

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				2 *****
				3 *
				4 * EXPERIMENTAL pending further PoP definition
				5 *
				6 * Zvector E6 instruction tests for VRR-a encoded:
				7 *
				8 * E655 VCNF - VECTOR FP CONVERT TO NNP
				9 * E656 VCLFNH - VECTOR FP CONVERT AND LENGTHEN FROM NNP HIGH
				10 * E65D VCFN - VECTOR FP CONVERT FROM NNP
				11 * E65E VCLFNL - VECTOR FP CONVERT AND LENGTHEN FROM NNP LOW
				12 *
				13 * and partial testing of
				14 *
				15 * E675 VCRNF - VECTOR FP CONVERT AND ROUND TO NNP
				16 *
				17 * during cross check tests for VCLFNH and VCLFNL
				18 *
				19 * James Wekel August 2024
				20 *****
				22 *****
				23 *
				24 * basic instruction tests
				25 *
				26 *****
				27 * This program tests EXPERIMENTAL functioning of the z/arch E6 VRR-a
				28 * Neural-network-processing-assist facility vector instructions.
				29 * These test are EXPERIMENTAL pending further PoP definition of
				30 * NNP-data-type-1.
				31 *
				32 * If requested and if VXC == 0 after test instruction execution,
				33 * a cross check test is performed. A cross check uses the result
				34 * of the instruction test to recreate the test source.
				35 *
				36 * Exceptions (including trapable IEEE exceptions) are not tested.
				37 *
				38 * PLEASE NOTE that the tests are very SIMPLE TESTS designed to catch
				39 * obvious coding errors. None of the tests are thorough. They are
				40 * NOT designed to test all aspects of any of the instructions.
				41 *
				42 *****
				43 *
				44 * *Testcase zvector-e6-20-NNPconvert
				45 * *
				46 * * EXPERIMENTAL pending further PoP definition
				47 * *
				48 * * Zvector E6 instruction tests for VRR-a encoded:
				49 * *
				50 * * E655 VCNF - VECTOR FP CONVERT TO NNP
				51 * * E656 VCLFNH - VECTOR FP CONVERT AND LENGTHEN FROM NNP HIGH
				52 * * E65D VCFN - VECTOR FP CONVERT FROM NNP
				53 * * E65E VCLFNL - VECTOR FP CONVERT AND LENGTHEN FROM NNP LOW
				54 * *
				55 * * # -----
				56 * * # This tests only the basic function of the instruction.

```

57 *      *      #      Exceptions are NOT tested.
58 *      *      #      -----
59 *      *
60 *      mainsize      2
61 *      numcpu        1
62 *      sysclear
63 *      archlvl       z/Arch
64 *
65 *      loadcore       "$(testpath)/zvector-e6-20-NNPconvert.core" 0x0
66 *
67 *      diag8cmd       enable      # (needed for messages to Hercules console)
68 *      runtest 2
69 *      diag8cmd       disable     # (reset back to default)
70 *
71 *      *Done
72 *
73 *****

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LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				75 *****
				76 * FCHECK Macro - Is a Facility Bit set?
				77 *
				78 * If the facility bit is NOT set, an message is issued and
				79 * the test is skipped.
				80 *
				81 * Fcheck uses R0, R1 and R2
				82 *
				83 * eg. FCHECK 134,'vector-packed-decimal'
				84 *****
				85 MACRO
				86 FCHECK &BITNO,&NOTSETMSG
				87 .* &BITNO : facility bit number to check
				88 .* &NOTSETMSG : 'facility name'
				89 LCLA &FBBYTE Facility bit in Byte
				90 LCLA &FBBIT Facility bit within Byte
				91
				92 LCLA &L(8)
				93 &L(1) SetA 128,64,32,16,8,4,2,1 bit positions within byte
				94
				95 &FBBYTE SETA &BITNO/8
				96 &FBBIT SETA &L((&BITNO-(&FBBYTE*8))+1)
				97 .* MNOTE 0,'checking Bit=&BITNO: FBBYTE=&FBBYTE, FBBIT=&FBBIT'
				98
				99 B X&SYSNDX
				100 * Fcheck data area
				101 * skip messgae
				102 SKT&SYSNDX DC C' Skipping tests: '
				103 DC C&NOTSETMSG
				104 DC C' facility (bit &BITNO) is not installed.'
				105 SKL&SYSNDX EQU *-SKT&SYSNDX
				106 * facility bits
				107 DS FD gap
				108 FB&SYSNDX DS 4FD
				109 DS FD gap
				110 *
				111 X&SYSNDX EQU *
				112 LA R0,((X&SYSNDX-FB&SYSNDX)/8)-1
				113 STFLE FB&SYSNDX get facility bits
				114
				115 XGR R0,R0
				116 IC R0,FB&SYSNDX+&FBBYTE get fbit byte
				117 N R0,=F'&FBBIT' is bit set?
				118 BNZ XC&SYSNDX
				119 *
				120 * facility bit not set, issue message and exit
				121 *
				122 LA R0,SKL&SYSNDX message length
				123 LA R1,SKT&SYSNDX message address
				124 BAL R2,MSG
				125
				126 B EOJ
				127 XC&SYSNDX EQU *
				128 MEND

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				149
				150 *****
				151 * The actual "ZVE6TST" program itself...
				152 *****
				153 *
				154 * Architecture Mode: z/Arch
				155 * Register Usage:
				156 *
				157 * R0 (work)
				158 * R1-4 (work)
				159 * R5 Testing control table - current test base
				160 * R6 (work)
				161 * R7 msg return
				162 * R8 First base register
				163 * R9 Second base register
				164 * R10 Third base register
				165 * R11 E6TEST call return
				166 * R12 E6TESTS register
				167 * R13 (work)
				168 * R14 Subroutine call
				169 * R15 Secondary Subroutine call or work
				170 *
				171 *****
00000200		00000200		173 USING BEGIN,R8 FIRST Base Register
00000200		00001200		174 USING BEGIN+4096,R9 SECOND Base Register
00000200		00002200		175 USING BEGIN+8192,R10 THIRD Base Register
				176
00000200	0580			177 BEGIN BALR R8,0 Initialize FIRST base register
00000202	0680			178 BCTR R8,0 Initialize FIRST base register
00000204	0680			179 BCTR R8,0 Initialize FIRST base register
				180
00000206	4190 8800		00000800	181 LA R9,2048(,R8) Initialize SECOND base register
0000020A	4190 9800		00000800	182 LA R9,2048(,R9) Initialize SECOND base register
				183
0000020E	41A0 9800		00000800	184 LA R10,2048(,R9) Initialize THIRD base register
00000212	41A0 A800		00000800	185 LA R10,2048(,R10) Initialize THIRD base register
				186
00000216	B600 859C		0000079C	187 STCTL R0,R0,CTLR0 Store CR0 to enable AFP
0000021A	9604 859D		0000079D	188 OI CTLR0+1,X'04' Turn on AFP bit
0000021E	9602 859D		0000079D	189 OI CTLR0+1,X'02' Turn on Vector bit
00000222	B700 859C		0000079C	190 LCTL R0,R0,CTLR0 Reload updated CR0
				191
				192 *****
				193 * Is Neural-network-processing-assist facility 2 installed (bit 165)
				194 *****
				195
00000226	47F0 80C0		000002C0	196 FCHECK 165,'Neural-network-processing-assist'
				197+ B X0001
				198+*
				199+* Fcheck data area
				200+SKT0001 DC C' Skipping tests: '
0000022A	40404040 40404040			201+ DC C'Neural-network-processing-assist'
00000244	D585A499 81936095			202+ DC C' facility (bit 165) is not installed.'
00000264	40868183 899389A3	0000005F	00000001	203+SKL0001 EQU *-SKT0001
				204+* facility bits

LOC	OBJECT	CODE	ADDR1	ADDR2	STMT	
					225 *****	
					226 *	
					227 *****	
					228	
000002E8	58C0	85DC		000007DC	229 L R12,=A(E6TESTS)	get table of test addresses
					230	
			000002EC	00000001	231 NEXTE6 EQU *	
000002EC	5850	C000		00000000	232 L R5,0(0,R12)	get test address
000002F0	1255				233 LTR R5,R5	have a test?
000002F2	4780	846C		0000066C	234 BZ ENDTEST	done?
					235	
000002F6			00000000		236 USING E6TEST,R5	
					237	
000002F6	4800	5004		00000004	238 LH R0,TNUM	save current test number
000002FA	5000	8E04		00001004	239 ST R0,TESTING	for easy reference
					240	
000002FE	58B0	5000		00000000	241 L R11,TSUB	get address of test routine
00000302	05BB				242 BALR R11,R11	do test
					243	
00000304	45F0	813C		0000033C	244 BAL R15,XCHECK	
					245 *	
					246 * validate FPC first	
					247 *	
00000308	D500	500A	5041	00000041	248 CLC FLG(1),FPC_R+1	expected FPC flags?
0000030E	4780	8116		00000316	249 BE CHECKVXC	
00000312	4570	8310		00000510	250 BAL R7,FPCMSG	no, issue failed message
					251	
			00000316	00000001	252 CHECKVXC EQU *	
00000316	D500	500B	5042	00000042	253 CLC VXC(1),FPC_R+2	expected VXC?
0000031C	4780	8124		00000324	254 BE CHECKRES	
00000320	4570	8382		00000582	255 BAL R7,VXCMSG	no, issue failed message
					256	
			00000324	00000001	257 CHECKRES EQU *	
					258	
					259 * then validate results, if not inexact	
					260	
					261 * XGR R1,R1	
					262 * IC R1,FPC_R+1	FPC flags
					263 * N R1,=XL4'00000008'	check inexact flag
					264 * LTR R1,R1	
					265 * BNZ DONEXT	
					266	
00000324	E310	501C	0014	0000001C	267 LGF R1,READDR	expected result address
0000032A	D50F	5028	1000	00000000	268 CLC V10OUTPUT,0(R1)	
00000330	4770	83F4		000005F4	269 BNE FAILMSG	no, issue failed message
					270	
			00000334	00000001	271 DONEXT EQU *	
00000334	41C0	C004		00000004	272 LA R12,4(0,R12)	next test address
00000338	47F0	80EC		000002EC	273 B NEXTE6	

LOC	OBJECT	CODE	ADDR1	ADDR2	STMT			
					331			
000003DA	9500	507A		0000007A	332	CLI	FPC_XC+2,0	
000003DE	477F	0000		00000000	333	BNZ	0(R15)	some VXC error , so exit
					334			
000003E2	E310	500C	0014	0000000C	335	LGF	R1,V2ADDR	expected source address
000003E8	D507	5060	1000	00000000	336	CLC	XCOUTPUT(8),0(R1)	
000003EE	4770	825C		0000045C	337	BNE	XCFAILMSG	
000003F2	07FF				338	BR	R15	return from xcheck
					339			
					340	* cross check: VCFN - VECTOR FP CONVERT FROM NNP		
					341	*		
000003F4					342	XCVCFN	DS 0F	
000003F4	B29D	85A4		000007A4	343	LFPC	FPCINIT	initialize FPC
					344			
000003F8	E700	5028	0806	00000028	345	VL	V16,V10OUTPUT	
000003FE	E6F0	0001	0455		346	VCNF	V15,V16,0,1	
00000404	B29C	5078		00000078	347	STFPC	FPC_XC	save FPC
00000408	E7F0	5060	000E	00000060	348	VST	V15,XCOUTPUT	
					349			
0000040E	9500	507A		0000007A	350	CLI	FPC_XC+2,0	
00000412	477F	0000		00000000	351	BNZ	0(R15)	some VXC error , so exit
					352			
00000416	E310	500C	0014	0000000C	353	LGF	R1,V2ADDR	expected source address
0000041C	D50F	5060	1000	00000000	354	CLC	XCOUTPUT,0(R1)	
00000422	4770	825C		0000045C	355	BNE	XCFAILMSG	
00000426	07FF				356	BR	R15	return from xcheck
					357			
					358	* cross check: VCLFNL - VECTOR FP CONVERT AND LENGTHEN FROM NNP LOW		
					359	*		
00000428					360	XCVCLFNL	DS 0F	
00000428	B29D	85A4		000007A4	361	LFPC	FPCINIT	initialize FPC
					362			
0000042C	E700	5028	0806	00000028	363	VL	V16,V10OUTPUT	
00000432	E6F0	0002	0675		364	VCRNF	V15,V16,V16,0,2	
00000438	B29C	5078		00000078	365	STFPC	FPC_XC	save FPC
0000043C	E7F0	5060	000E	00000060	366	VST	V15,XCOUTPUT	
					367			
00000442	9500	507A		0000007A	368	CLI	FPC_XC+2,0	
00000446	477F	0000		00000000	369	BNZ	0(R15)	some VXC error , so exit
					370			
0000044A	E310	500C	0014	0000000C	371	LGF	R1,V2ADDR	expected source address
00000450	D507	5060	1008	00000008	372	CLC	XCOUTPUT(8),8(R1)	low part of vector
00000456	4770	825C		0000045C	373	BNE	XCFAILMSG	
0000045A	07FF				374	BR	R15	return from xcheck
					375			
					376			
					377	* xcheck failed message		
0000045C					378	XCFAILMSG	DS 0H	
0000045C	4820	5004		00000004	379	LH	R2,TNUM	get test number and convert
00000460	4E20	8F90		00001190	380	CVD	R2,DECNUM	
00000464	D211	8F7A	8F64	0000117A	381	MVC	PRT3,EDIT	
0000046A	DE11	8F7A	8F90	0000117A	382	ED	PRT3,DECNUM	
00000470	D202	8F23	8F87	00001123	383	MVC	XCPTNUM(3),PRT3+13	fill in message with test #
					384			
00000476	D207	8F45	5010	00001145	385	MVC	XCPNAME,OPNAME	fill in message with instruction
					386			

LOC	OBJECT CODE			ADDR1	ADDR2	STMT
						453 *****
						454 * VXC not as expected:
						455 * issue message with test number, instruction under test,
						456 * expected FPC and received FPC
						457 *****
00000582	5800	85E0		00000582	00000001	458 VXCMSG EQU *
00000586	5000	8E00			000007E0	459 L R0,=F'1' set failed test indicator
					00001000	460 ST R0,FAILED
						461
0000058A	4820	5004			00000004	462 LH R2,TNUM get test number and convert
0000058E	4E20	8F90			00001190	463 CVD R2,DECNUM
00000592	D211	8F7A 8F64	0000117A	00001164		464 MVC PRT3,EDIT
00000598	DE11	8F7A 8F90	0000117A	00001190		465 ED PRT3,DECNUM
0000059E	D202	8E78 8F87	00001078	00001187		466 MVC VXCPNUM(3),PRT3+13 fill in message with test #
						467
000005A4	D207	8E9C 5010	0000109C	00000010		468 MVC VXCPNAME,OPNAME fill in message with instruction
						469 *
000005AA	B982	0022				470 XGR R2,R2
000005AE	4320	500B		0000000B		471 IC R2,VXC get expected flags and convert
000005B2	4E20	8F90		00001190		472 CVD R2,DECNUM
000005B6	D211	8F7A 8F64	0000117A	00001164		473 MVC PRT3,EDIT
000005BC	DE11	8F7A 8F90	0000117A	00001190		474 ED PRT3,DECNUM
000005C2	D202	8EB5 8F87	000010B5	00001187		475 MVC VXCPEXP(3),PRT3+13 fill in message with m3 field
						476 *
000005C8	B982	0022				477 XGR R2,R2
000005CC	4320	5042		00000042		478 IC R2,FPC_R+2 get received flags and convert
000005D0	4E20	8F90		00001190		479 CVD R2,DECNUM
000005D4	D211	8F7A 8F64	0000117A	00001164		480 MVC PRT3,EDIT
000005DA	DE11	8F7A 8F90	0000117A	00001190		481 ED PRT3,DECNUM
000005E0	D202	8ECA 8F87	000010CA	00001187		482 MVC VXCPGOT(3),PRT3+13 fill in message with FPC got
						483 *
000005E6	4100	0063		00000063		484 LA R0,VXCPLNG message length
000005EA	4110	8E6B		0000106B		485 LA R1,VXCPLINE message address
000005EE	45F0	847A		0000067A		486 BAL R15,RPTERROR
000005F2	07F7					487 BR R7
						488

LOC	OBJECT CODE			ADDR1	ADDR2	STMT
						490 *****
						491 * result not as expected:
						492 * issue message with test number, instruction under test
						493 * and instruction m4
						494 *****
000005F4	4820	5004		000005F4	00000001	495 FAILMSG EQU *
000005F8	4E20	8F90			00000004	496 LH R2,TNUM get test number and convert
000005FC	D211	8F7A 8F64		0000117A	00001164	497 CVD R2,DECNUM
00000602	DE11	8F7A 8F90		0000117A	00001190	498 MVC PRT3,EDIT
00000608	D202	8EDB 8F87		000010DB	00001187	499 ED PRT3,DECNUM
						500 MVC PRTNUM(3),PRT3+13 fill in message with test #
						501
0000060E	D207	8EF6 5010		000010F6	00000010	502 MVC PRTNAME,OPNAME fill in message with instruction
						503 *
00000614	B982	0022				504 XGR R2,R2
00000618	4320	5008			00000008	505 IC R2,M3 get m3 and convert
0000061C	4E20	8F90			00001190	506 CVD R2,DECNUM
00000620	D211	8F7A 8F64		0000117A	00001164	507 MVC PRT3,EDIT
00000626	DE11	8F7A 8F90		0000117A	00001190	508 ED PRT3,DECNUM
0000062C	D201	8F07 8F88		00001107	00001188	509 MVC PRTM3(2),PRT3+14 fill in message with m3 field
						510 *
00000632	B982	0022				511 XGR R2,R2
00000636	4320	5009			00000009	512 IC R2,M4 get m4 and convert
0000063A	4E20	8F90			00001190	513 CVD R2,DECNUM
0000063E	D211	8F7A 8F64		0000117A	00001164	514 MVC PRT3,EDIT
00000644	DE11	8F7A 8F90		0000117A	00001190	515 ED PRT3,DECNUM
0000064A	D201	8F13 8F88		00001113	00001188	516 MVC PRTM4(2),PRT3+14 fill in message with m4 field
						517 *
00000650	4100	0048			00000048	518 LA R0,PRTLNG message length
00000654	4110	8ECE			000010CE	519 LA R1,PRTLINE message address
00000658	45F0	847A			0000067A	520 BAL R15,RPTERROR
						522 *****
						523 * continue after a failed test
						524 *****
0000065C	5800	85E0		0000065C	00000001	525 FAILCONT EQU *
00000660	5000	8E00			000007E0	526 L R0,=F'1' set failed test indicator
					00001000	527 ST R0,FAILED
						528
00000664	41C0	C004			00000004	529 LA R12,4(0,R12) next test address
00000668	47F0	80EC			000002EC	530 B NEXTE6
						532 *****
						533 * end of testing; set ending psw
						534 *****
0000066C	5810	8E00		0000066C	00000001	535 ENDTEST EQU *
00000670	1211				00001000	536 L R1,FAILED did a test fail?
						537 LTR R1,R1
00000672	4780	8580			00000780	538 BZ EOJ No, exit
00000676	47F0	8598			00000798	539 B FAILTEST Yes, exit with BAD PSW
						540

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				605 *****
				606 * Normal completion or Abnormal termination PSWs
				607 *****
00000770	00020001 80000000			609 EOJPSW DC 0D'0',X'0002000180000000',AD(0)
00000780	B2B2 8570		00000770	611 EOJ LPSWE EOJPSW Normal completion
00000788	00020001 80000000			613 FAILPSW DC 0D'0',X'0002000180000000',AD(X'BAD')
00000798	B2B2 8588		00000788	615 FAILTEST LPSWE FAILPSW Abnormal termination
				617 *****
				618 * Working Storage
				619 *****
0000079C	00000000			621 CTLR0 DS F CR0
000007A0	00000000			622 DS F
000007A4	00000000			623 FPCINIT DC XL4'00000000' FPC before test
000007A8				625
000007A8	E2D2C9D7 40E7C340			626 LTORG , Literals pool
000007B0	40404040 40404040			627 =CL8'SKIP XC ' ,
000007B8	E5C3D5C6 40404040			628 =CL8'
000007C0	E5C3D3C6 D5C84040			629 =CL8'VCNF'
000007C8	E5C3C6D5 40404040			630 =CL8'VCLFNH'
000007D0	E5C3D3C6 D5D34040			631 =CL8'VCFN'
000007D8	00000004			632 =CL8'VCLFNL'
000007DC	0000548C			633 =F'4'
000007E0	00000001			634 =A(E6TESTS)
000007E4	0000			635 =F'1'
000007E6	0064			636 =H'0'
000007E8	E2			637 =AL2(L'MSGMSG)
				638 =CL1'S'
				639
				640 * some constants
				641
	00000400 00000001			642 K EQU 1024 One KB
	00001000 00000001			643 PAGE EQU (4*K) Size of one page
	00010000 00000001			644 K64 EQU (64*K) 64 KB

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				651 *=====
				652 *
				653 * NOTE: start data on an address that is easy to display
				654 * within Hercules
				655 *
				656 *=====
000007E9		000007E9	00001000	657
00001000	00000000			658 ORG ZVE6TST+X'1000'
00001004	00000000			659 FAILED DC F'0' some test failed?
				660 TESTING DC F'0' current test #
				662
				663 *****
				664 * TEST failed : FPC message
				665 *****
				666 *
				667 * failed message and associated editing
				668 *
00001008	40404040	4040E385		669 FPCPLINE DC C' Test # '
00001015	A7A7A7			670 FPCPNUM DC C'xxx'
00001018	40A69996	958740C6		671 DC c' wrong FPC flags for instruction '
00001039	A7A7A7A7	A7A7A7A7		672 FPCPNAME DC CL8'xxxxxxxxx'
00001041	4085A797	8583A385		673 DC C' expected: flags='
00001052	A7A7A7			674 FPCPEXP DC C'xxx'
00001055	6B			675 DC C','
00001056	40998583	8589A585		676 DC C' received: flags='
00001067	A7A7A7			677 FPCPGOT DC C'xxx'
0000106A	4B			678 DC C'.'
		00000063	00000001	679 FPCPLNG EQU *-FPCPLINE
				681
				682 *****
				683 * TEST failed : VXC message
				684 *****
				685 *
				686 * failed message and associated editing
				687 *
0000106B	40404040	4040E385		688 VXCPLINE DC C' Test # '
00001078	A7A7A7			689 VXCPNUM DC C'xxx'
0000107B	40A69996	958740C6		690 DC c' wrong FPC VXC for instruction '
0000109C	A7A7A7A7	A7A7A7A7		691 VXCPNAME DC CL8'xxxxxxxxx'
000010A4	4085A797	8583A385		692 DC C' expected: VXC='
000010B5	A7A7A7			693 VXCPEXP DC C'xxx'
000010B8	6B			694 DC C','
000010B9	40998583	8589A585		695 DC C' received: VXC='
000010CA	A7A7A7			696 VXCPGOT DC C'xxx'
000010CD	4B			697 DC C'.'
		00000063	00000001	698 VXCPLNG EQU *-VXCPLINE

LOC	OBJECT	CODE	ADDR1	ADDR2	STMT
					700 *****
					701 * TEST failed : result messgae
					702 *****
					703 *
					704 * failed message and associated editting
					705 *
000010CE	40404040	4040E385			706 PRTLNE DC C' Test # '
000010DB	A7A7A7				707 PRTNUM DC C'xxx'
000010DE	40868189	93858440			708 DC c' failed for instruction '
000010F6	A7A7A7A7	A7A7A7A7			709 PRTNAME DC CL8'xxxxxxxxx'
000010FE	40A689A3	884094F3			710 DC C' with m3='
00001107	A7A7				711 PRTM3 DC C'xx'
00001109	6B				712 DC C','
0000110A	40A689A3	884094F4			713 DC C' with m4='
00001113	A7A7				714 PRTM4 DC C'xx'
00001115	4B				715 DC C'.'
			00000048	00000001	716 PRTLNG EQU *-PRTLNE
					718 *****
					719 * TEST failed : XCHECK
					720 *****
					721 *
					722 * XCHECK failed message
					723 *
00001116	40404040	4040E385			724 XCPLINE DC C' Test # '
00001123	A7A7A7				725 XCPTNUM DC C'xxx'
00001126	40E7C3C8	C5C3D240			726 DC c' XCHECK failed for instruction '
00001145	A7A7A7A7	A7A7A7A7			727 XCPNAME DC CL8'xxxxxxxxx'
0000114D	40A689A3	884094F3			728 DC C' with m3='
00001156	A7A7				729 XCPM3 DC C'xx'
00001158	40A689A3	884094F4			730 DC C' with m4='
00001161	A7A7				731 XCPM4 DC C'xx'
00001163	4B				732 DC C'.'
			0000004E	00000001	733 XCPLNG EQU *-XCPLINE

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				753 *****
				754 * E6TEST DSECT
				755 *****
				757 E6TEST DSECT ,
00000000	00000000			758 TSUB DC A(0) pointer to test
00000004	0000			759 TNUM DC H'00' Test Number
00000006	00			760 DC X'00'
00000007	40			761 XCSKIP DC CL1' ' Y = skip cross check
00000008	00			762 M3 DC HL1'00' m3 used
00000009	00			763 M4 DC HL1'00' m4 used
0000000A	00			764 FLG DC X'00' expected FPC flags
0000000B	00			765 VXC DC X'00' VXC expected
0000000C	00000000			766 V2ADDR DC A(0) address of v2: 16-byte packed decimal
00000010	40404040	40404040		767 OPNAME DC CL8' ' E6 name
00000018	00000000			768 RELEN DC A(0) result length
0000001C	00000000			769 READDR DC A(0) expected result address
00000020	00000000	00000000		770 DS FD gap
00000028	00000000	00000000		771 V10OUTPUT DS XL16 V1 Output
00000038	00000000	00000000		772 DS FD gap
00000040	00000000			773 FPC_R DS F FPC after instruction
00000048	00000000	00000000		774 DS FD gap
00000050	40404040	40404040		775 SKIPXC DC CL8' ' was cross check skipped?
00000058	00000000	00000000		776 DS FD gap
00000060	00000000	00000000		777 XCOUTPUT DS XL16 Cross check Output
00000070	00000000	00000000		778 DS FD gap
00000078	00000000			779 FPC_XC DS F FPC after cross check
00000080	00000000	00000000		780 DS FD gap
00000088	00000000			781 WK1 DS F debug area
0000008C	00000000			782 WK2 DS F
00000090				783 DS 0F
				784 **
				785 * test routine will be here (from VRR-a macro)
				787 ZVE6TST CSECT ,
000011E0		00000000	000055D7	788 DS 0F
				790 *****
				791 * Macros to help build test tables
				792 *****
				794 *
				795 * macro to generate individual test
				796 *
				797 MACRO
				798 VRR_A &INST,&M3,&M4,&FLAGS,&VXC,&SKIP
				799 .* &INST - VRR-a instruction under test
				800 .* &m3 - m3 field
				801 .* &m4 - m4 field
				802 .* &flags - expected FPC flags (HEX)
				803 .* &VXC - expected VXC (HEX)

LOC	OBJECT CODE	ADDR1	ADDR2	STMT	
				804 .*	&SKIP - S = skip cross check
				805	GBLA &TNUM
				806 &TNUM	SETA &TNUM+1
				807	
				808	DS 0FD
				809	USING *,R5
				810	
				811 T&TNUM	DC A(X&TNUM)
				812	DC H'&TNUM'
				813	DC X'00'
				814	DC CL1'&SKIP'
				815	DC HL1'&M3'
				816	DC HL1'&M4'
				817 FLG&TNUM	DC X'&FLAGS'
				818 VXC&TNUM	DC X'&VXC'
				819 V2_&TNUM	DC A(RE&TNUM+16)
				820	DC CL8'&INST'
				821	DC A(16)
				822	DC A(RE&TNUM)
				823	DS FD
				824 V10&TNUM	DS XL16
				825	DS FD
				826 FPC_R_&TNUM	DS F
				827	DS FD
				828	DC CL8' '
				829	DS FD
				830 XCO&TNUM	DS XL16
				831	DS FD
				832 FPC_XC_&TNUM	DS F
				833	DS FD
				834	DS F
				835	DS F
				836 .*	
				837 *	
				838 X&TNUM	DS 0F
				839	LFPC FPCINIT
				840 *	LFPC =XL4'08000000'
				841	
				842	LGF R2,V2_&TNUM
				843	VL V22,0(R2)
				844	
				845	&INST V22,V22,&M3,&M4
				846	
				847	STFPC FPC_R_&TNUM
				848	VST V22,V10&TNUM
				849	
				850	BR R11
				851	
				852 RE&TNUM	DS 0F
				853	DROP R5
				854	
				855	MEND
				857 *	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
-----	-------------	-------	-------	------

```
858 * macro to generate table of pointers to individual tests
```

859 *

860 MACRO

861 PTTABLE

862 GBLA & TNUM

863 LCLA &CUR

864 &CUR SETA 1

865 . *

866 TTABLE DS 0F

867 .LOOP ANOP

868 . *

869	DC	A(T&CUR)	TEST &CUR
-----	----	----------	-----------

870 . *

871 &CUR SETA &CUR+1

```
872      AIF    (&CUR LE &TNUM).LOOP
```

873 *

874	DC	A(0)	END OF TABLE
-----	----	------	--------------

875	DC	A(0)
-----	----	------

876 . *

877 **MEND**

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				879 *****
				880 * E6 VRR-a tests
				881 *****
				882 PRINT DATA
				883 *
				884 * E655 VCNF - VECTOR FP CONVERT TO NNP
				885 * E656 VCLFNH - VECTOR FP CONVERT AND LENGTHEN FROM NNP HIGH
				886 * E65D VCFN - VECTOR FP CONVERT FROM NNP
				887 * E65E VCLFNL - VECTOR FP CONVERT AND LENGTHEN FROM NNP LOW
				888 *
				889 * -----
				890 * VRR-a instruction, m3, m4, flags, VXC, SKIP
				891 * followed by
				892 * followed by
				893 * v1 - 16 byte expected result
				894 * v2 - 16 byte source
				895 * -----
				896 * VCNF - VECTOR FP CONVERT TO NNP
				897 * -----
				898 * tinyb -> dlfloat (with cross check dlfloat -> tinyb)
				899 *
				900 * some of these tests use numbers from PoP SA22-7832-13,
				901 * Figure 9-2. Examples of Floating-Point Numbers (page 9-6)
				902 *
				903 * +0 simple instruction test and 'test' test
				904 VRR_A VCNF,0,1,00,00,N
000011E0				905+ DS 0FD
000011E0		000011E0		906+ USING *,R5 base for test data and test routine
000011E0	00001270			907+T1 DC A(X1) address of test routine
000011E4	0001			908+ DC H'1' test number
000011E6	00			909+ DC X'00'
000011E7	D5			910+ DC CL1'N' Y = skip cross check
000011E8	00			911+ DC HL1'0' m3
000011E9	01			912+ DC HL1'1' m4
000011EA	00			913+FLG1 DC X'00' expected FPC flags
000011EB	00			914+VXC1 DC X'00' expected VXC
000011EC	000012A4			915+V2_1 DC A(RE1+16) address of v2: 16-byte packed decimal
000011F0	E5C3D5C6 40404040			916+ DC CL8'VCNF' instruction name
000011F8	00000010			917+ DC A(16) result length
000011FC	00001294			918+ DC A(RE1) address of expected resul
00001200	00000000 00000000			919+ DS FD gap
00001208	00000000 00000000			920+V101 DS XL16 V1 output
00001210	00000000 00000000			
00001218	00000000 00000000			921+ DS FD gap
00001220	00000000			922+FPC_R_1 DS F FPC after instruction
00001228	00000000 00000000			923+ DS FD gap
00001230	40404040 40404040			924+ DC CL8' ' was cross check skipped?
00001238	00000000 00000000			925+ DS FD gap
00001240	00000000 00000000			926+XC01 DS XL16 Cross check Output
00001248	00000000 00000000			
00001250	00000000 00000000			927+ DS FD gap
00001258	00000000			928+FPC_XC_1 DS F FPC after cross check
00001260	00000000 00000000			929+ DS FD gap
00001268	00000000			930+ DS F debug area
0000126C	00000000			931+ DS F
				932+*

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001270				933+X1	DS	0F	
00001270	B29D 85A4		000007A4	934+	LFPC	FPCINIT	initialize FPC
				935+*	LFPC	=XL4'08000000'	set trap exception
00001274	E320 500C 0014		000011EC	936+	LGF	R2,V2_1	get v2
0000127A	E762 0000 0806		00000000	937+	VL	V22,0(R2)	
00001280	E666 0001 0C55			938+	VCNF	V22,V22,0,1	test instruction (dest is source)
00001286	B29C 9020		00001220	939+	STFPC	FPC_R_1	save FPC
0000128A	E760 9008 080E		00001208	940+	VST	V22,V101	save instruction result
00001290	07FB			941+	BR	R11	return
00001294				942+RE1	DS	0F	expected 16 byte result
00001294				943+	DROP	R5	
00001294	00000000 00000000			944	DC	XL16'00000000000000000000000000000000'	
0000129C	00000000 00000000						
000012A4	00000000 00000000			945	DC	XL16'00000000000000000000000000000000'	
000012AC	00000000 00000000						
				946			
				947 * 1.0, -1, -4096			
				948	VRR_A	VCNF,0,1,00,00,N	
000012B8				949+	DS	0FD	
000012B8		000012B8		950+	USING	*,R5	base for test data and test routine
000012B8	00001348			951+T2	DC	A(X2)	address of test routine
000012BC	0002			952+	DC	H'2'	test number
000012BE	00			953+	DC	X'00'	
000012BF	D5			954+	DC	CL1'N'	Y = skip cross check
000012C0	00			955+	DC	HL1'0'	m3
000012C1	01			956+	DC	HL1'1'	m4
000012C2	00			957+FLG2	DC	X'00'	expected FPC flags
000012C3	00			958+VXC2	DC	X'00'	expected VXC
000012C4	0000137C			959+V2_2	DC	A(RE2+16)	address of v2: 16-byte packed decimal
000012C8	E5C3D5C6 40404040			960+	DC	CL8'VCNF'	instruction name
000012D0	00000010			961+	DC	A(16)	result length
000012D4	0000136C			962+	DC	A(RE2)	address of expected resul
000012D8	00000000 00000000			963+	DS	FD	gap
000012E0	00000000 00000000			964+V102	DS	XL16	V1 output
000012E8	00000000 00000000						
000012F0	00000000 00000000			965+	DS	FD	gap
000012F8	00000000			966+FPC_R_2	DS	F	FPC after instruction
00001300	00000000 00000000			967+	DS	FD	gap
00001308	40404040 40404040			968+	DC	CL8' '	was cross check skipped?
00001310	00000000 00000000			969+	DS	FD	gap
00001318	00000000 00000000			970+XC02	DS	XL16	Cross check Output
00001320	00000000 00000000						
00001328	00000000 00000000			971+	DS	FD	gap
00001330	00000000			972+FPC_XC_2	DS	F	FPC after cross check
00001338	00000000 00000000			973+	DS	FD	gap
00001340	00000000			974+	DS	F	debug area
00001344	00000000			975+	DS	F	
				976+*			
00001348				977+X2	DS	0F	
00001348	B29D 85A4		000007A4	978+	LFPC	FPCINIT	initialize FPC
				979+*	LFPC	=XL4'08000000'	set trap exception
0000134C	E320 500C 0014		000012C4	980+	LGF	R2,V2_2	get v2
00001352	E762 0000 0806		00000000	981+	VL	V22,0(R2)	
00001358	E666 0001 0C55			982+	VCNF	V22,V22,0,1	test instruction (dest is source)
0000135E	B29C 5040		000012F8	983+	STFPC	FPC_R_2	save FPC
00001362	E760 5028 080E		000012E0	984+	VST	V22,V102	save instruction result

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001368	07FB			985+	BR	R11	return
0000136C				986+RE2	DS	0F	expected 16 byte result
0000136C				987+	DROP	R5	
0000136C	3E000000 BE000000			988	DC	XL16'	3E000000BE00000000000000000000D800'
00001374	00000000 0000D800						
0000137C	3C000000 BC000000			989	DC	XL16'	3C000000BC0000000000000000000F000'
00001384	00000000 0000F000						
				990 * 0.5			
				991	VRR_A	VCNF,0,1,00,00,N	
00001390				992+	DS	0FD	
00001390		00001390		993+	USING	*,R5	base for test data and test routine
00001390	00001420			994+T3	DC	A(X3)	address of test routine
00001394	0003			995+	DC	H'3'	test number
00001396	00			996+	DC	X'00'	
00001397	D5			997+	DC	CL1'N'	Y = skip cross check
00001398	00			998+	DC	HL1'0'	m3
00001399	01			999+	DC	HL1'1'	m4
0000139A	00			1000+FLG3	DC	X'00'	expected FPC flags
0000139B	00			1001+VXC3	DC	X'00'	expected VXC
0000139C	00001454			1002+V2_3	DC	A(RE3+16)	address of v2: 16-byte packed decimal
000013A0	E5C3D5C6 40404040			1003+	DC	CL8'VCNF'	instruction name
000013A8	00000010			1004+	DC	A(16)	result length
000013AC	00001444			1005+	DC	A(RE3)	address of expected resul
000013B0	00000000 00000000			1006+	DS	FD	gap
000013B8	00000000 00000000			1007+V103	DS	XL16	V1 output
000013C0	00000000 00000000						
000013C8	00000000 00000000			1008+	DS	FD	gap
000013D0	00000000			1009+FPC_R_3	DS	F	FPC after instruction
000013D8	00000000 00000000			1010+	DS	FD	gap
000013E0	40404040 40404040			1011+	DC	CL8' '	was cross check skipped?
000013E8	00000000 00000000			1012+	DS	FD	gap
000013F0	00000000 00000000			1013+XC03	DS	XL16	Cross check Output
000013F8	00000000 00000000						
00001400	00000000 00000000			1014+	DS	FD	gap
00001408	00000000			1015+FPC_XC_3	DS	F	FPC after cross check
00001410	00000000 00000000			1016+	DS	FD	gap
00001418	00000000			1017+	DS	F	debug area
0000141C	00000000			1018+	DS	F	
				1019+*			
00001420				1020+X3	DS	0F	
00001420	B29D 85A4		000007A4	1021+	LFPC	FPCINIT	initialize FPC
				1022+*	LFPC	=XL4'08000000'	set trap exception
00001424	E320 500C 0014		0000139C	1023+	LGF	R2,V2_3	get v2
0000142A	E762 0000 0806		00000000	1024+	VL	V22,0(R2)	
00001430	E666 0001 0C55			1025+	VCNF	V22,V22,0,1	test instruction (dest is source)
00001436	B29C 5040		000013D0	1026+	STFPC	FPC_R_3	save FPC
0000143A	E760 5028 080E		000013B8	1027+	VST	V22,V103	save instruction result
00001440	07FB			1028+	BR	R11	return
00001444				1029+RE3	DS	0F	expected 16 byte result
00001444				1030+	DROP	R5	
00001444	3C000000 00000000			1031	DC	XL16'	3C0000000000000000000000000D800'
0000144C	00000000 0000D800						
00001454	38000000 00000000			1032	DC	XL16'	380000000000000000000000000F000'
0000145C	00000000 0000F000						
				1033			
				1034 * 1/64			

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
				1035	VRR_A	VCNF,0,1,00,00,N
00001468				1036+	DS	0FD
00001468		00001468		1037+	USING	*,R5
00001468	000014F8			1038+T4	DC	A(X4)
0000146C	0004			1039+	DC	H'4'
0000146E	00			1040+	DC	X'00'
0000146F	D5			1041+	DC	CL1'N'
00001470	00			1042+	DC	HL1'0'
00001471	01			1043+	DC	HL1'1'
00001472	00			1044+FLG4	DC	X'00'
00001473	00			1045+VXC4	DC	X'00'
00001474	0000152C			1046+V2_4	DC	A(RE4+16)
00001478	E5C3D5C6 40404040			1047+	DC	CL8'VCNF'
00001480	00000010			1048+	DC	A(16)
00001484	0000151C			1049+	DC	A(RE4)
00001488	00000000 00000000			1050+	DS	FD
00001490	00000000 00000000			1051+V104	DS	XL16
00001498	00000000 00000000					
000014A0	00000000 00000000			1052+	DS	FD
000014A8	00000000			1053+FPC_R_4	DS	F
000014B0	00000000 00000000			1054+	DS	FD
000014B8	40404040 40404040			1055+	DC	CL8' '
000014C0	00000000 00000000			1056+	DS	FD
000014C8	00000000 00000000			1057+XC04	DS	XL16
000014D0	00000000 00000000					
000014D8	00000000 00000000			1058+	DS	FD
000014E0	00000000			1059+FPC_XC_4	DS	F
000014E8	00000000 00000000			1060+	DS	FD
000014F0	00000000			1061+	DS	F
000014F4	00000000			1062+	DS	F
				1063+*		
000014F8				1064+X4	DS	0F
000014F8	B29D 85A4		000007A4	1065+	LFPC	FPCINIT
				1066+*	LFPC	=XL4'08000000'
000014FC	E320 500C 0014		00001474	1067+	LGF	R2,V2_4
00001502	E762 0000 0806		00000000	1068+	VL	V22,0(R2)
00001508	E666 0001 0C55			1069+	VCNF	V22,V22,0,1
0000150E	B29C 5040		000014A8	1070+	STFPC	FPC_R_4
00001512	E760 5028 080E		00001490	1071+	VST	V22,V104
00001518	07FB			1072+	BR	R11
0000151C				1073+RE4	DS	0F
0000151C				1074+	DROP	R5
0000151C	32000000 00000000			1075	DC	XL16'3200000000000000000000000000D800'
00001524	00000000 0000D800					
0000152C	24000000 00000000			1076	DC	XL16'2400000000000000000000000000F000'
00001534	00000000 0000F000					
				1077		
				1078 * +0, -0		
				1079	VRR_A	VCNF,0,1,00,00,N
00001540				1080+	DS	0FD
00001540		00001540		1081+	USING	*,R5
00001540	000015D0			1082+T5	DC	A(X5)
00001544	0005			1083+	DC	H'5'
00001546	00			1084+	DC	X'00'
00001547	D5			1085+	DC	CL1'N'
00001548	00			1086+	DC	HL1'0'

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001549	01			1087+	DC	HL1'1'	m4
0000154A	00			1088+FLG5	DC	X'00'	expected FPC flags
0000154B	00			1089+VXC5	DC	X'00'	expected VXC
0000154C	00001604			1090+V2_5	DC	A(RE5+16)	address of v2: 16-byte packed decimal
00001550	E5C3D5C6 40404040			1091+	DC	CL8'VCNF'	instruction name
00001558	00000010			1092+	DC	A(16)	result length
0000155C	000015F4			1093+	DC	A(RE5)	address of expected resul
00001560	00000000 00000000			1094+	DS	FD	gap
00001568	00000000 00000000			1095+V105	DS	XL16	V1 output
00001570	00000000 00000000						
00001578	00000000 00000000			1096+	DS	FD	gap
00001580	00000000			1097+FPC_R_5	DS	F	FPC after instruction
00001588	00000000 00000000			1098+	DS	FD	gap
00001590	40404040 40404040			1099+	DC	CL8' '	was cross check skipped?
00001598	00000000 00000000			1100+	DS	FD	gap
000015A0	00000000 00000000			1101+XC05	DS	XL16	Cross check Output
000015A8	00000000 00000000						
000015B0	00000000 00000000			1102+	DS	FD	gap
000015B8	00000000			1103+FPC_XC_5	DS	F	FPC after cross check
000015C0	00000000 00000000			1104+	DS	FD	gap
000015C8	00000000			1105+	DS	F	debug area
000015CC	00000000			1106+	DS	F	
				1107+*			
000015D0				1108+X5	DS	0F	
000015D0	B29D 85A4		000007A4	1109+	LFPC	FPCINIT	initialize FPC
				1110+*	LFPC	=XL4'08000000'	set trap exception
000015D4	E320 500C 0014		0000154C	1111+	LGF	R2,V2_5	get v2
000015DA	E762 0000 0806		00000000	1112+	VL	V22,0(R2)	
000015E0	E666 0001 0C55			1113+	VCNF	V22,V22,0,1	test instruction (dest is source)
000015E6	B29C 5040		00001580	1114+	STFPC	FPC_R_5	save FPC
000015EA	E760 5028 080E		00001568	1115+	VST	V22,V105	save instruction result
000015F0	07FB			1116+	BR	R11	return
000015F4				1117+RE5	DS	0F	expected 16 byte result
000015F4				1118+	DROP	R5	
000015F4	00008000 00000000			1119	DC	XL16'0000800000000000000000000000D800'	
000015FC	00000000 0000D800						
00001604	00008000 00000000			1120	DC	XL16'0000800000000000000000000000F000'	
0000160C	00000000 0000F000						
				1121			
				1122 * -15			
				1123	VRR_A	VCNF,0,1,00,00,N	
00001618				1124+	DS	0FD	
00001618		00001618		1125+	USING	*,R5	base for test data and test routine
00001618	000016A8			1126+T6	DC	A(X6)	address of test routine
0000161C	0006			1127+	DC	H'6'	test number
0000161E	00			1128+	DC	X'00'	
0000161F	D5			1129+	DC	CL1'N'	Y = skip cross check
00001620	00			1130+	DC	HL1'0'	m3
00001621	01			1131+	DC	HL1'1'	m4
00001622	00			1132+FLG6	DC	X'00'	expected FPC flags
00001623	00			1133+VXC6	DC	X'00'	expected VXC
00001624	000016DC			1134+V2_6	DC	A(RE6+16)	address of v2: 16-byte packed decimal
00001628	E5C3D5C6 40404040			1135+	DC	CL8'VCNF'	instruction name
00001630	00000010			1136+	DC	A(16)	result length
00001634	000016CC			1137+	DC	A(RE6)	address of expected resul
00001638	00000000 00000000			1138+	DS	FD	gap

LOC	OBJECT CODE		ADDR1	ADDR2	STMT			
00001640	00000000	00000000			1139+V106	DS	XL16	V1 output
00001648	00000000	00000000						
00001650	00000000	00000000			1140+	DS	FD	gap
00001658	00000000				1141+FPC_R_6	DS	F	FPC after instruction
00001660	00000000	00000000			1142+	DS	FD	gap
00001668	40404040	40404040			1143+	DC	CL8' '	was cross check skipped?
00001670	00000000	00000000			1144+	DS	FD	gap
00001678	00000000	00000000			1145+XC06	DS	XL16	Cross check Output
00001680	00000000	00000000						
00001688	00000000	00000000			1146+	DS	FD	gap
00001690	00000000				1147+FPC_XC_6	DS	F	FPC after cross check
00001698	00000000	00000000			1148+	DS	FD	gap
000016A0	00000000				1149+	DS	F	debug area
000016A4	00000000				1150+	DS	F	
					1151+*			
000016A8					1152+X6	DS	0F	
000016A8	B29D 85A4		000007A4		1153+	LFPC	FPCINIT	initialize FPC
					1154+*	LFPC	=XL4'08000000'	set trap exception
000016AC	E320 500C 0014		00001624		1155+	LGF	R2,V2_6	get v2
000016B2	E762 0000 0806		00000000		1156+	VL	V22,0(R2)	
000016B8	E666 0001 0C55				1157+	VCNF	V22,V22,0,1	test instruction (dest is source)
000016BE	B29C 5040		00001658		1158+	STFPC	FPC_R_6	save FPC
000016C2	E760 5028 080E		00001640		1159+	VST	V22,V106	save instruction result
000016C8	07FB				1160+	BR	R11	return
000016CC					1161+RE6	DS	0F	expected 16 byte result
000016CC					1162+	DROP	R5	
000016CC	C5C00000 00000000				1163	DC	XL16'C5C0000000000000000000000000D800'	
000016D4	00000000 0000D800							
000016DC	CB800000 00000000				1164	DC	XL16'CB80000000000000000000000000F000'	
000016E4	00000000 0000F000							
					1165			
					1166 * 20/7 (about)			
					1167	VRR_A	VCNF,0,1,00,00,N	
000016F0					1168+	DS	0FD	
000016F0			000016F0		1169+	USING	*,R5	base for test data and test routine
000016F0	00001780				1170+T7	DC	A(X7)	address of test routine
000016F4	0007				1171+	DC	H'7'	test number
000016F6	00				1172+	DC	X'00'	
000016F7	D5				1173+	DC	CL1'N'	Y = skip cross check
000016F8	00				1174+	DC	HL1'0'	m3
000016F9	01				1175+	DC	HL1'1'	m4
000016FA	00				1176+FLG7	DC	X'00'	expected FPC flags
000016FB	00				1177+VXC7	DC	X'00'	expected VXC
000016FC	000017B4				1178+V2_7	DC	A(RE7+16)	address of v2: 16-byte packed decimal
00001700	E5C3D5C6 40404040				1179+	DC	CL8'VCNF'	instruction name
00001708	00000010				1180+	DC	A(16)	result length
0000170C	000017A4				1181+	DC	A(RE7)	address of expected resul
00001710	00000000 00000000				1182+	DS	FD	gap
00001718	00000000 00000000				1183+V107	DS	XL16	V1 output
00001720	00000000 00000000							
00001728	00000000 00000000				1184+	DS	FD	gap
00001730	00000000				1185+FPC_R_7	DS	F	FPC after instruction
00001738	00000000 00000000				1186+	DS	FD	gap
00001740	40404040 40404040				1187+	DC	CL8' '	was cross check skipped?
00001748	00000000 00000000				1188+	DS	FD	gap
00001750	00000000 00000000				1189+XC07	DS	XL16	Cross check Output

LOC	OBJECT CODE	ADDR1	ADDR2	STMT	
00001758	00000000 00000000				
00001760	00000000 00000000			1190+	DS FD gap
00001768	00000000			1191+FPC_XC_7	DS F FPC after cross check
00001770	00000000 00000000			1192+	DS FD gap
00001778	00000000			1193+	DS F debug area
0000177C	00000000			1194+	DS F
				1195+*	
00001780				1196+X7	DS 0F
00001780	B29D 85A4		000007A4	1197+	LFPC FPCINIT initialize FPC
				1198+*	LFPC =XL4'08000000' set trap exception
00001784	E320 500C 0014		000016FC	1199+	LGF R2,V2_7 get v2
0000178A	E762 0000 0806		00000000	1200+	VL V22,0(R2)
00001790	E666 0001 0C55			1201+	VCNF V22,V22,0,1 test instruction (dest is source)
00001796	B29C 5040		00001730	1202+	STFPC FPC_R_7 save FPC
0000179A	E760 5028 080E		00001718	1203+	VST V22,V107 save instruction result
000017A0	07FB			1204+	BR R11 return
000017A4				1205+RE7	DS 0F expected 16 byte result
000017A4				1206+	DROP R5
000017A4	40DB0000 00000000			1207	DC XL16'40DB000000000000000000000000D800'
000017AC	00000000 0000D800				
000017B4	41B60000 00000000			1208	DC XL16'41B6000000000000000000000000F000'
000017BC	00000000 0000F000				
				1209	
				1210	* additional tests
				1211	* max tiny: (1 - 2^-11) * 2^16 (65504)
				1212	VRR_A VCNF,0,1,00,00,S skip xc - rounded
000017C8				1213+	DS 0FD
000017C8		000017C8		1214+	USING *,R5 base for test data and test routine
000017C8	00001858			1215+T8	DC A(X8) address of test routine
000017CC	0008			1216+	DC H'8' test number
000017CE	00			1217+	DC X'00'
000017CF	E2			1218+	DC CL1'S' Y = skip cross check
000017D0	00			1219+	DC HL1'0' m3
000017D1	01			1220+	DC HL1'1' m4
000017D2	00			1221+FLG8	DC X'00' expected FPC flags
000017D3	00			1222+VXC8	DC X'00' expected VXC
000017D4	0000188C			1223+V2_8	DC A(RE8+16) address of v2: 16-byte packed decimal
000017D8	E5C3D5C6 40404040			1224+	DC CL8'VCNF' instruction name
000017E0	00000010			1225+	DC A(16) result length
000017E4	0000187C			1226+	DC A(RE8) address of expected resul
000017E8	00000000 00000000			1227+	DS FD gap
000017F0	00000000 00000000			1228+V108	DS XL16 V1 output
000017F8	00000000 00000000				
00001800	00000000 00000000			1229+	DS FD gap
00001808	00000000			1230+FPC_R_8	DS F FPC after instruction
00001810	00000000 00000000			1231+	DS FD gap
00001818	40404040 40404040			1232+	DC CL8' ' was cross check skipped?
00001820	00000000 00000000			1233+	DS FD gap
00001828	00000000 00000000			1234+XC08	DS XL16 Cross check Output
00001830	00000000 00000000				
00001838	00000000 00000000			1235+	DS FD gap
00001840	00000000			1236+FPC_XC_8	DS F FPC after cross check
00001848	00000000 00000000			1237+	DS FD gap
00001850	00000000			1238+	DS F debug area
00001854	00000000			1239+	DS F
				1240+*	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001858				1241+X8	DS	0F	
00001858	B29D 85A4		000007A4	1242+	LFPC	FPCINIT	initialize FPC
				1243+*	LFPC	=XL4'08000000'	set trap exception
0000185C	E320 500C 0014		000017D4	1244+	LGF	R2,V2_8	get v2
00001862	E762 0000 0806		00000000	1245+	VL	V22,0(R2)	
00001868	E666 0001 0C55			1246+	VCNF	V22,V22,0,1	test instruction (dest is source)
0000186E	B29C 5040		00001808	1247+	STFPC	FPC_R_8	save FPC
00001872	E760 5028 080E		000017F0	1248+	VST	V22,V108	save instruction result
00001878	07FB			1249+	BR	R11	return
0000187C				1250+RE8	DS	0F	expected 16 byte result
0000187C				1251+	DROP	R5	
0000187C	5E000000 00000000			1252	DC	XL16'5E00000000000000000000000000D800'	
00001884	00000000 0000D800						
0000188C	7BFF0000 00000000			1253	DC	XL16'7BFF000000000000000000000000F000'	
00001894	00000000 0000F000						
				1254			
				1255 * min tiny (normal): 2^-14 (0.00006103515625)			
				1256	VRR_A	VCNF,0,1,00,00,N	
000018A0				1257+	DS	0FD	
000018A0		000018A0		1258+	USING	*,R5	base for test data and test routine
000018A0	00001930			1259+T9	DC	A(X9)	address of test routine
000018A4	0009			1260+	DC	H'9'	test number
000018A6	00			1261+	DC	X'00'	
000018A7	D5			1262+	DC	CL1'N'	Y = skip cross check
000018A8	00			1263+	DC	HL1'0'	m3
000018A9	01			1264+	DC	HL1'1'	m4
000018AA	00			1265+FLG9	DC	X'00'	expected FPC flags
000018AB	00			1266+VXC9	DC	X'00'	expected VXC
000018AC	00001964			1267+V2_9	DC	A(RE9+16)	address of v2: 16-byte packed decimal
000018B0	E5C3D5C6 40404040			1268+	DC	CL8'VCNF'	instruction name
000018B8	00000010			1269+	DC	A(16)	result length
000018BC	00001954			1270+	DC	A(RE9)	address of expected resul
000018C0	00000000 00000000			1271+	DS	FD	gap
000018C8	00000000 00000000			1272+V109	DS	XL16	V1 output
000018D0	00000000 00000000						
000018D8	00000000 00000000			1273+	DS	FD	gap
000018E0	00000000			1274+FPC_R_9	DS	F	FPC after instruction
000018E8	00000000 00000000			1275+	DS	FD	gap
000018F0	40404040 40404040			1276+	DC	CL8' '	was cross check skipped?
000018F8	00000000 00000000			1277+	DS	FD	gap
00001900	00000000 00000000			1278+XC09	DS	XL16	Cross check Output
00001908	00000000 00000000						
00001910	00000000 00000000			1279+	DS	FD	gap
00001918	00000000			1280+FPC_XC_9	DS	F	FPC after cross check
00001920	00000000 00000000			1281+	DS	FD	gap
00001928	00000000			1282+	DS	F	debug area
0000192C	00000000			1283+	DS	F	
				1284+*			
00001930				1285+X9	DS	0F	
00001930	B29D 85A4		000007A4	1286+	LFPC	FPCINIT	initialize FPC
				1287+*	LFPC	=XL4'08000000'	set trap exception
00001934	E320 500C 0014		000018AC	1288+	LGF	R2,V2_9	get v2
0000193A	E762 0000 0806		00000000	1289+	VL	V22,0(R2)	
00001940	E666 0001 0C55			1290+	VCNF	V22,V22,0,1	test instruction (dest is source)
00001946	B29C 5040		000018E0	1291+	STFPC	FPC_R_9	save FPC
0000194A	E760 5028 080E		000018C8	1292+	VST	V22,V109	save instruction result

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00001950	07FB			1293+	BR R11	return
00001954				1294+RE9	DS 0F	expected 16 byte result
00001954				1295+	DROP R5	
00001954	22000000 00000000			1296	DC XL16'	2200000000000000000000000000D800'
0000195C	00000000 0000D800					
00001964	04000000 00000000			1297	DC XL16'	04000000000000000000000000F000'
0000196C	00000000 0000F000					
				1298		
				1299	* min tiny (subnormal): 2^-24 (0.000000059604644775390625)	
				1300	VRR_A VCNF,0,1,00,00,N	
00001978				1301+	DS 0FD	
00001978		00001978		1302+	USING *,R5	base for test data and test routine
00001978	00001A08			1303+T10	DC A(X10)	address of test routine
0000197C	000A			1304+	DC H'10'	test number
0000197E	00			1305+	DC X'00'	
0000197F	D5			1306+	DC CL1'N'	Y = skip cross check
00001980	00			1307+	DC HL1'0'	m3
00001981	01			1308+	DC HL1'1'	m4
00001982	00			1309+FLG10	DC X'00'	expected FPC flags
00001983	00			1310+VXC10	DC X'00'	expected VXC
00001984	00001A3C			1311+V2_10	DC A(RE10+16)	address of v2: 16-byte packed decimal
00001988	E5C3D5C6 40404040			1312+	DC CL8'VCNF'	instruction name
00001990	00000010			1313+	DC A(16)	result length
00001994	00001A2C			1314+	DC A(RE10)	address of expected resul
00001998	00000000 00000000			1315+	DS FD	gap
000019A0	00000000 00000000			1316+V1010	DS XL16	V1 output
000019A8	00000000 00000000					
000019B0	00000000 00000000			1317+	DS FD	gap
000019B8	00000000			1318+FPC_R_10	DS F	FPC after instruction
000019C0	00000000 00000000			1319+	DS FD	gap
000019C8	40404040 40404040			1320+	DC CL8' '	was cross check skipped?
000019D0	00000000 00000000			1321+	DS FD	gap
000019D8	00000000 00000000			1322+XC010	DS XL16	Cross check Output
000019E0	00000000 00000000					
000019E8	00000000 00000000			1323+	DS FD	gap
000019F0	00000000			1324+FPC_XC_10	DS F	FPC after cross check
000019F8	00000000 00000000			1325+	DS FD	gap
00001A00	00000000			1326+	DS F	debug area
00001A04	00000000			1327+	DS F	
				1328+*		
00001A08				1329+X10	DS 0F	
00001A08	B29D 85A4		000007A4	1330+	LFPC FPCINIT	initialize FPC
				1331+*	LFPC =XL4'08000000'	set trap exception
00001A0C	E320 500C 0014		00001984	1332+	LGF R2,V2_10	get v2
00001A12	E762 0000 0806		00000000	1333+	VL V22,0(R2)	
00001A18	E666 0001 0C55			1334+	VCNF V22,V22,0,1	test instruction (dest is source)
00001A1E	B29C 5040		000019B8	1335+	STFPC FPC_R_10	save FPC
00001A22	E760 5028 080E		000019A0	1336+	VST V22,V1010	save instruction result
00001A28	07FB			1337+	BR R11	return
00001A2C				1338+RE10	DS 0F	expected 16 byte result
00001A2C				1339+	DROP R5	
00001A2C	0E000000 00000000			1340	DC XL16'0E0000000000000000000000D800'	
00001A34	00000000 0000D800					
00001A3C	00010000 00000000			1341	DC XL16'000100000000000000000000F000'	
00001A44	00000000 0000F000					
				1342		

LOC	OBJECT CODE	ADDR1	ADDR2	STMT	
				1343 * NAN,	
				1344	VRR_A VCNF,0,1,80,01,S skip xcheck - includes NAN
00001A50				1345+	DS 0FD
00001A50		00001A50		1346+	USING *,R5 base for test data and test routine
00001A50	00001AE0			1347+T11	DC A(X11) address of test routine
00001A54	000B			1348+	DC H'11' test number
00001A56	00			1349+	DC X'00'
00001A57	E2			1350+	DC CL1'S' Y = skip cross check
00001A58	00			1351+	DC HL1'0' m3
00001A59	01			1352+	DC HL1'1' m4
00001A5A	80			1353+FLG11	DC X'80' expected FPC flags
00001A5B	01			1354+VXC11	DC X'01' expected VXC
00001A5C	00001B14			1355+V2_11	DC A(RE11+16) address of v2: 16-byte packed decimal
00001A60	E5C3D5C6 40404040			1356+	DC CL8'VCNF' instruction name
00001A68	00000010			1357+	DC A(16) result length
00001A6C	00001B04			1358+	DC A(RE11) address of expected resul
00001A70	00000000 00000000			1359+	DS FD gap
00001A78	00000000 00000000			1360+V1011	DS XL16 V1 output
00001A80	00000000 00000000				
00001A88	00000000 00000000			1361+	DS FD gap
00001A90	00000000			1362+FPC_R_11	DS F FPC after instruction
00001A98	00000000 00000000			1363+	DS FD gap
00001AA0	40404040 40404040			1364+	DC CL8' ' was cross check skipped?
00001AA8	00000000 00000000			1365+	DS FD gap
00001AB0	00000000 00000000			1366+XC011	DS XL16 Cross check Output
00001AB8	00000000 00000000				
00001AC0	00000000 00000000			1367+	DS FD gap
00001AC8	00000000			1368+FPC_XC_11	DS F FPC after cross check
00001AD0	00000000 00000000			1369+	DS FD gap
00001AD8	00000000			1370+	DS F debug area
00001ADC	00000000			1371+	DS F
				1372+*	
00001AE0				1373+X11	DS 0F
00001AE0	B29D 85A4		000007A4	1374+	LFPC FPCINIT initialize FPC
				1375+*	LFPC =XL4'08000000' set trap exception
00001AE4	E320 500C 0014		00001A5C	1376+	LGF R2,V2_11 get v2
00001AEA	E762 0000 0806		00000000	1377+	VL V22,0(R2)
00001AF0	E666 0001 0C55			1378+	VCNF V22,V22,0,1 test instruction (dest is source)
00001AF6	B29C 5040		00001A90	1379+	STFPC FPC_R_11 save FPC
00001AFA	E760 5028 080E		00001A78	1380+	VST V22,V1011 save instruction result
00001B00	07FB			1381+	BR R11 return
00001B04				1382+RE11	DS 0F expected 16 byte result
00001B04				1383+	DROP R5
00001B04	FFFF0000 00000000			1384	DC XL16'FFFF000000000000000000000000D800'
00001B0C	00000000 0000D800				
00001B14	FFFF0000 00000000			1385	DC XL16'FFFF000000000000000000000000F000'
00001B1C	00000000 0000F000				
				1386	
				1387 * inexact - bad m3	
				1388	VRR_A VCNF,5,1,08,05,S skip xc - inexact
00001B28				1389+	DS 0FD
00001B28		00001B28		1390+	USING *,R5 base for test data and test routine
00001B28	00001BB8			1391+T12	DC A(X12) address of test routine
00001B2C	000C			1392+	DC H'12' test number
00001B2E	00			1393+	DC X'00'
00001B2F	E2			1394+	DC CL1'S' Y = skip cross check

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001B30	05			1395+	DC	HL1'5'	m3
00001B31	01			1396+	DC	HL1'1'	m4
00001B32	08			1397+FLG12	DC	X'08'	expected FPC flags
00001B33	05			1398+VXC12	DC	X'05'	expected VXC
00001B34	00001BEC			1399+V2_12	DC	A(RE12+16)	address of v2: 16-byte packed decimal
00001B38	E5C3D5C6 40404040			1400+	DC	CL8'VCNF'	instruction name
00001B40	00000010			1401+	DC	A(16)	result length
00001B44	00001BDC			1402+	DC	A(RE12)	address of expected resul
00001B48	00000000 00000000			1403+	DS	FD	gap
00001B50	00000000 00000000			1404+V1012	DS	XL16	V1 output
00001B58	00000000 00000000						
00001B60	00000000 00000000			1405+	DS	FD	gap
00001B68	00000000			1406+FPC_R_12	DS	F	FPC after instruction
00001B70	00000000 00000000			1407+	DS	FD	gap
00001B78	40404040 40404040			1408+	DC	CL8' '	was cross check skipped?
00001B80	00000000 00000000			1409+	DS	FD	gap
00001B88	00000000 00000000			1410+XC012	DS	XL16	Cross check Output
00001B90	00000000 00000000						
00001B98	00000000 00000000			1411+	DS	FD	gap
00001BA0	00000000			1412+FPC_XC_12	DS	F	FPC after cross check
00001BA8	00000000 00000000			1413+	DS	FD	gap
00001BB0	00000000			1414+	DS	F	debug area
00001BB4	00000000			1415+	DS	F	
				1416+*			
00001BB8				1417+X12	DS	0F	
00001BB8	B29D 85A4		000007A4	1418+	LFPC	FPCINIT	initialize FPC
				1419+*	LFPC	=XL4'08000000'	set trap exception
00001BBC	E320 500C 0014		00001B34	1420+	LGF	R2,V2_12	get v2
00001BC2	E762 0000 0806		00000000	1421+	VL	V22,0(R2)	
00001BC8	E666 0001 5C55			1422+	VCNF	V22,V22,5,1	test instruction (dest is source)
00001BCE	B29C 5040		00001B68	1423+	STFPC	FPC_R_12	save FPC
00001BD2	E760 5028 080E		00001B50	1424+	VST	V22,V1012	save instruction result
00001BD8	07FB			1425+	BR	R11	return
00001BDC				1426+RE12	DS	0F	expected 16 byte result
00001BDC				1427+	DROP	R5	
00001BDC	00000000 00000000			1428	DC	XL16'00000000000000000000000000000000'	
00001BE4	00000000 00000000						
00001BEC	00010000 00000000			1429	DC	XL16'0001000000000000000000000000F000'	
00001BF4	00000000 0000F000						
				1430			
				1431 * inexact - bad m4			
				1432	VRR_A	VCNF,0,5,08,05,S	skip xc - inexact
00001C00				1433+	DS	0FD	
00001C00		00001C00		1434+	USING	*,R5	base for test data and test routine
00001C00	00001C90			1435+T13	DC	A(X13)	address of test routine
00001C04	000D			1436+	DC	H'13'	test number
00001C06	00			1437+	DC	X'00'	
00001C07	E2			1438+	DC	CL1'S'	Y = skip cross check
00001C08	00			1439+	DC	HL1'0'	m3
00001C09	05			1440+	DC	HL1'5'	m4
00001C0A	08			1441+FLG13	DC	X'08'	expected FPC flags
00001C0B	05			1442+VXC13	DC	X'05'	expected VXC
00001C0C	00001CC4			1443+V2_13	DC	A(RE13+16)	address of v2: 16-byte packed decimal
00001C10	E5C3D5C6 40404040			1444+	DC	CL8'VCNF'	instruction name
00001C18	00000010			1445+	DC	A(16)	result length
00001C1C	00001CB4			1446+	DC	A(RE13)	address of expected resul

LOC	OBJECT CODE			ADDR1	ADDR2	STMT			
00001C20	00000000	00000000				1447+	DS	FD	gap
00001C28	00000000	00000000				1448+V1013	DS	XL16	V1 output
00001C30	00000000	00000000							
00001C38	00000000	00000000				1449+	DS	FD	gap
00001C40	00000000					1450+FPC_R_13	DS	F	FPC after instruction
00001C48	00000000	00000000				1451+	DS	FD	gap
00001C50	40404040	40404040				1452+	DC	CL8' '	was cross check skipped?
00001C58	00000000	00000000				1453+	DS	FD	gap
00001C60	00000000	00000000				1454+XC013	DS	XL16	Cross check Output
00001C68	00000000	00000000							
00001C70	00000000	00000000				1455+	DS	FD	gap
00001C78	00000000					1456+FPC_XC_13	DS	F	FPC after cross check
00001C80	00000000	00000000				1457+	DS	FD	gap
00001C88	00000000					1458+	DS	F	debug area
00001C8C	00000000					1459+	DS	F	
						1460+*			
00001C90						1461+X13	DS	0F	
00001C90	B29D 85A4			000007A4		1462+	LFPC	FPCINIT	initialize FPC
						1463+*	LFPC	=XL4'08000000'	set trap exception
00001C94	E320 500C 0014			00001C0C		1464+	LGF	R2,V2_13	get v2
00001C9A	E762 0000 0806			00000000		1465+	VL	V22,0(R2)	
00001CA0	E666 0005 0C55					1466+	VCNF	V22,V22,0,5	test instruction (dest is source)
00001CA6	B29C 5040			00001C40		1467+	STFPC	FPC_R_13	save FPC
00001CAA	E760 5028 080E			00001C28		1468+	VST	V22,V1013	save instruction result
00001CB0	07FB					1469+	BR	R11	return
00001CB4						1470+RE13	DS	0F	expected 16 byte result
00001CB4						1471+	DROP	R5	
00001CB4	00000000	00000000				1472	DC	XL16'00000000000000000000000000000000'	
00001CBC	00000000	00000000							
00001CC4	00010000	00000000				1473	DC	XL16'0001000000000000000000000000F000'	
00001CCC	00000000	0000F000							
						1474			
						1475	* more tests		
						1476	* +3.140625, -3.140625		
						1477	VRR_A	VCNF,0,1,00,00,N	
00001CD8						1478+	DS	0FD	
00001CD8			00001CD8			1479+	USING	*,R5	base for test data and test routine
00001CD8	00001D68					1480+T14	DC	A(X14)	address of test routine
00001CDC	000E					1481+	DC	H'14'	test number
00001CDE	00					1482+	DC	X'00'	
00001CDF	D5					1483+	DC	CL1'N'	Y = skip cross check
00001CE0	00					1484+	DC	HL1'0'	m3
00001CE1	01					1485+	DC	HL1'1'	m4
00001CE2	00					1486+FLG14	DC	X'00'	expected FPC flags
00001CE3	00					1487+VXC14	DC	X'00'	expected VXC
00001CE4	00001D9C					1488+V2_14	DC	A(RE14+16)	address of v2: 16-byte packed decimal
00001CE8	E5C3D5C6 40404040					1489+	DC	CL8'VCNF'	instruction name
00001CF0	00000010					1490+	DC	A(16)	result length
00001CF4	00001D8C					1491+	DC	A(RE14)	address of expected resul
00001CF8	00000000	00000000				1492+	DS	FD	gap
00001D00	00000000	00000000				1493+V1014	DS	XL16	V1 output
00001D08	00000000	00000000							
00001D10	00000000	00000000				1494+	DS	FD	gap
00001D18	00000000					1495+FPC_R_14	DS	F	FPC after instruction
00001D20	00000000	00000000				1496+	DS	FD	gap
00001D28	40404040	40404040				1497+	DC	CL8' '	was cross check skipped?

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001D30	00000000 00000000			1498+	DS	FD	gap
00001D38	00000000 00000000			1499+XC014	DS	XL16	Cross check Output
00001D40	00000000 00000000						
00001D48	00000000 00000000			1500+	DS	FD	gap
00001D50	00000000			1501+FPC_XC_14	DS	F	FPC after cross check
00001D58	00000000 00000000			1502+	DS	FD	gap
00001D60	00000000			1503+	DS	F	debug area
00001D64	00000000			1504+	DS	F	
				1505+*			
00001D68				1506+X14	DS	0F	
00001D68	B29D 85A4		000007A4	1507+	LFPC	FPCINIT	initialize FPC
				1508+*	LFPC	=XL4'08000000'	set trap exception
00001D6C	E320 500C 0014		00001CE4	1509+	LGF	R2,V2_14	get v2
00001D72	E762 0000 0806		00000000	1510+	VL	V22,0(R2)	
00001D78	E666 0001 0C55			1511+	VCNF	V22,V22,0,1	test instruction (dest is source)
00001D7E	B29C 5040		00001D18	1512+	STFPC	FPC_R_14	save FPC
00001D82	E760 5028 080E		00001D00	1513+	VST	V22,V1014	save instruction result
00001D88	07FB			1514+	BR	R11	return
00001D8C				1515+RE14	DS	0F	expected 16 byte result
00001D8C				1516+	DROP	R5	
00001D8C	42240000 C2240000			1517	DC	XL16'42240000C2240000000000000000D800'	
00001D94	00000000 0000D800						
00001D9C	44480000 C4480000			1518	DC	XL16'44480000C4480000000000000000F000'	
00001DA4	00000000 0000F000						
				1519			
				1520 * +27.15625, -27.15625			skip xc - lost digits
				1521	VRR_A	VCNF,0,1,00,00,S	
00001DB0				1522+	DS	0FD	
00001DB0		00001DB0		1523+	USING	*,R5	base for test data and test routine
00001DB0	00001E40			1524+T15	DC	A(X15)	address of test routine
00001DB4	000F			1525+	DC	H'15'	test number
00001DB6	00			1526+	DC	X'00'	
00001DB7	E2			1527+	DC	CL1'S'	Y = skip cross check
00001DB8	00			1528+	DC	HL1'0'	m3
00001DB9	01			1529+	DC	HL1'1'	m4
00001DBA	00			1530+FLG15	DC	X'00'	expected FPC flags
00001DBB	00			1531+VXC15	DC	X'00'	expected VXC
00001DBC	00001E74			1532+V2_15	DC	A(RE15+16)	address of v2: 16-byte packed decimal
00001DC0	E5C3D5C6 40404040			1533+	DC	CL8'VCNF'	instruction name
00001DC8	00000010			1534+	DC	A(16)	result length
00001DCC	00001E64			1535+	DC	A(RE15)	address of expected resul
00001DD0	00000000 00000000			1536+	DS	FD	gap
00001DD8	00000000 00000000			1537+V1015	DS	XL16	V1 output
00001DE0	00000000 00000000						
00001DE8	00000000 00000000			1538+	DS	FD	gap
00001DF0	00000000			1539+FPC_R_15	DS	F	FPC after instruction
00001DF8	00000000 00000000			1540+	DS	FD	gap
00001E00	40404040 40404040			1541+	DC	CL8' '	was cross check skipped?
00001E08	00000000 00000000			1542+	DS	FD	gap
00001E10	00000000 00000000			1543+XC015	DS	XL16	Cross check Output
00001E18	00000000 00000000						
00001E20	00000000 00000000			1544+	DS	FD	gap
00001E28	00000000			1545+FPC_XC_15	DS	F	FPC after cross check
00001E30	00000000 00000000			1546+	DS	FD	gap
00001E38	00000000			1547+	DS	F	debug area
00001E3C	00000000			1548+	DS	F	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001E40				1549+*			
00001E40	B29D 85A4		000007A4	1550+X15	DS	0F	
				1551+	LFPC	FPCINIT	initialize FPC
				1552+*	LFPC	=XL4'08000000'	set trap exception
00001E44	E320 500C 0014		00001DBC	1553+	LGF	R2,V2_15	get v2
00001E4A	E762 0000 0806		00000000	1554+	VL	V22,0(R2)	
00001E50	E666 0001 0C55			1555+	VCNF	V22,V22,0,1	test instruction (dest is source)
00001E56	B29C 5040		00001DF0	1556+	STFPC	FPC_R_15	save FPC
00001E5A	E760 5028 080E		00001DD8	1557+	VST	V22,V1015	save instruction result
00001E60	07FB			1558+	BR	R11	return
00001E64				1559+RE15	DS	0F	expected 16 byte result
00001E64				1560+	DROP	R5	
00001E64	2F650000 AF650000			1561	DC	XL16'2F650000AF6500000000000000000D800'	
00001E6C	00000000 0000D800						
00001E74	1EC90000 9EC90000			1562	DC	XL16'1EC900009EC90000000000000000F000'	
00001E7C	00000000 0000F000						
				1563			
				1564 * +65504, -65504			
				1565	VRR_A	VCNF,0,1,00,00,N	
00001E88				1566+	DS	0FD	
00001E88		00001E88		1567+	USING	*,R5	base for test data and test routine
00001E88	00001F18			1568+T16	DC	A(X16)	address of test routine
00001E8C	0010			1569+	DC	H'16'	test number
00001E8E	00			1570+	DC	X'00'	
00001E8F	D5			1571+	DC	CL1'N'	Y = skip cross check
00001E90	00			1572+	DC	HL1'0'	m3
00001E91	01			1573+	DC	HL1'1'	m4
00001E92	00			1574+FLG16	DC	X'00'	expected FPC flags
00001E93	00			1575+VXC16	DC	X'00'	expected VXC
00001E94	00001F4C			1576+V2_16	DC	A(RE16+16)	address of v2: 16-byte packed decimal
00001E98	E5C3D5C6 40404040			1577+	DC	CL8'VCNF'	instruction name
00001EA0	00000010			1578+	DC	A(16)	result length
00001EA4	00001F3C			1579+	DC	A(RE16)	address of expected resul
00001EA8	00000000 00000000			1580+	DS	FD	gap
00001EB0	00000000 00000000			1581+V1016	DS	XL16	V1 output
00001EB8	00000000 00000000						
00001EC0	00000000 00000000			1582+	DS	FD	gap
00001EC8	00000000			1583+FPC_R_16	DS	F	FPC after instruction
00001ED0	00000000 00000000			1584+	DS	FD	gap
00001ED8	40404040 40404040			1585+	DC	CL8' '	was cross check skipped?
00001EE0	00000000 00000000			1586+	DS	FD	gap
00001EE8	00000000 00000000			1587+XC016	DS	XL16	Cross check Output
00001EF0	00000000 00000000						
00001EF8	00000000 00000000			1588+	DS	FD	gap
00001F00	00000000			1589+FPC_XC_16	DS	F	FPC after cross check
00001F08	00000000 00000000			1590+	DS	FD	gap
00001F10	00000000			1591+	DS	F	debug area
00001F14	00000000			1592+	DS	F	
				1593+*			
00001F18				1594+X16	DS	0F	
00001F18	B29D 85A4		000007A4	1595+	LFPC	FPCINIT	initialize FPC
				1596+*	LFPC	=XL4'08000000'	set trap exception
00001F1C	E320 500C 0014		00001E94	1597+	LGF	R2,V2_16	get v2
00001F22	E762 0000 0806		00000000	1598+	VL	V22,0(R2)	
00001F28	E666 0001 0C55			1599+	VCNF	V22,V22,0,1	test instruction (dest is source)
00001F2E	B29C 5040		00001EC8	1600+	STFPC	FPC_R_16	save FPC

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001F32	E760 5028 080E		00001EB0	1601+	VST	V22,V1016	save instruction result
00001F38	07FB			1602+	BR	R11	return
00001F3C				1603+RE16	DS	0F	expected 16 byte result
00001F3C				1604+	DROP	R5	
00001F3C	5E000000 DE000000			1605	DC	XL16'5E000000DE0000000000000000D800'	
00001F44	00000000 0000D800						
00001F4C	7BFF0000 FBFF0000			1606	DC	XL16'7BFF0000FBFF000000000000000F000'	
00001F54	00000000 0000F000						
				1607			
				1608 * +Normal, -Normal			
				1609	VRR_A	VCNF,0,1,00,00,N	
00001F60				1610+	DS	0FD	
00001F60		00001F60		1611+	USING	*,R5	base for test data and test routine
00001F60	00001FF0			1612+T17	DC	A(X17)	address of test routine
00001F64	0011			1613+	DC	H'17'	test number
00001F66	00			1614+	DC	X'00'	
00001F67	D5			1615+	DC	CL1'N'	Y = skip cross check
00001F68	00			1616+	DC	HL1'0'	m3
00001F69	01			1617+	DC	HL1'1'	m4
00001F6A	00			1618+FLG17	DC	X'00'	expected FPC flags
00001F6B	00			1619+VXC17	DC	X'00'	expected VXC
00001F6C	00002024			1620+V2_17	DC	A(RE17+16)	address of v2: 16-byte packed decimal
00001F70	E5C3D5C6 40404040			1621+	DC	CL8'VCNF'	instruction name
00001F78	00000010			1622+	DC	A(16)	result length
00001F7C	00002014			1623+	DC	A(RE17)	address of expected resul
00001F80	00000000 00000000			1624+	DS	FD	gap
00001F88	00000000 00000000			1625+V1017	DS	XL16	V1 output
00001F90	00000000 00000000						
00001F98	00000000 00000000			1626+	DS	FD	gap
00001FA0	00000000			1627+FPC_R_17	DS	F	FPC after instruction
00001FA8	00000000 00000000			1628+	DS	FD	gap
00001FB0	40404040 40404040			1629+	DC	CL8' '	was cross check skipped?
00001FB8	00000000 00000000			1630+	DS	FD	gap
00001FC0	00000000 00000000			1631+XC017	DS	XL16	Cross check Output
00001FC8	00000000 00000000						
00001FD0	00000000 00000000			1632+	DS	FD	gap
00001FD8	00000000			1633+FPC_XC_17	DS	F	FPC after cross check
00001FE0	00000000 00000000			1634+	DS	FD	gap
00001FE8	00000000			1635+	DS	F	debug area
00001FEC	00000000			1636+	DS	F	
				1637+*			
00001FF0				1638+X17	DS	0F	
00001FF0	B29D 85A4		000007A4	1639+	LFPC	FPCINIT	initialize FPC
				1640+*	LFPC	=XL4'08000000'	set trap exception
00001FF4	E320 500C 0014		00001F6C	1641+	LGF	R2,V2_17	get v2
00001FFA	E762 0000 0806		00000000	1642+	VL	V22,0(R2)	
00002000	E666 0001 0C55			1643+	VCNF	V22,V22,0,1	test instruction (dest is source)
00002006	B29C 5040		00001FA0	1644+	STFPC	FPC_R_17	save FPC
0000200A	E760 5028 080E		00001F88	1645+	VST	V22,V1017	save instruction result
00002010	07FB			1646+	BR	R11	return
00002014				1647+RE17	DS	0F	expected 16 byte result
00002014				1648+	DROP	R5	
00002014	40800000 C0800000			1649	DC	XL16'40800000C0800000000000000D800'	
0000201C	00000000 0000D800						
00002024	41000000 C1000000		1650	DC	XL16'41000000C10000000000000000F000'		
0000202C	00000000 0000F000						

LOC	OBJECT CODE	ADDR1	ADDR2	STMT	
				1651	
				1652 * +Subnormal, -Subnormal	
				1653 VRR_A VCNF,0,1,00,00,N	
00002038				1654+ DS 0FD	
00002038		00002038		1655+ USING *,R5	base for test data and test routine
00002038	000020C8			1656+T18 DC A(X18)	address of test routine
0000203C	0012			1657+ DC H'18'	test number
0000203E	00			1658+ DC X'00'	
0000203F	D5			1659+ DC CL1'N'	Y = skip cross check
00002040	00			1660+ DC HL1'0'	m3
00002041	01			1661+ DC HL1'1'	m4
00002042	00			1662+FLG18 DC X'00'	expected FPC flags
00002043	00			1663+VXC18 DC X'00'	expected VXC
00002044	000020FC			1664+V2_18 DC A(RE18+16)	address of v2: 16-byte packed decimal
00002048	E5C3D5C6 40404040			1665+ DC CL8'VCNF'	instruction name
00002050	00000010			1666+ DC A(16)	result length
00002054	000020EC			1667+ DC A(RE18)	address of expected resul
00002058	00000000 00000000			1668+ DS FD	gap
00002060	00000000 00000000			1669+V1018 DS XL16	V1 output
00002068	00000000 00000000				
00002070	00000000 00000000			1670+ DS FD	gap
00002078	00000000			1671+FPC_R_18 DS F	FPC after instruction
00002080	00000000 00000000			1672+ DS FD	gap
00002088	40404040 40404040			1673+ DC CL8' '	was cross check skipped?
00002090	00000000 00000000			1674+ DS FD	gap
00002098	00000000 00000000			1675+XC018 DS XL16	Cross check Output
000020A0	00000000 00000000				
000020A8	00000000 00000000			1676+ DS FD	gap
000020B0	00000000			1677+FPC_XC_18 DS F	FPC after cross check
000020B8	00000000 00000000			1678+ DS FD	gap
000020C0	00000000			1679+ DS F	debug area
000020C4	00000000			1680+ DS F	
				1681+*	
000020C8				1682+X18 DS 0F	
000020C8	B29D 85A4		000007A4	1683+ LFPC FPCINIT	initialize FPC
				1684+* LFPC =XL4'08000000'	set trap exception
000020CC	E320 500C 0014		00002044	1685+ LGF R2,V2_18	get v2
000020D2	E762 0000 0806		00000000	1686+ VL V22,0(R2)	
000020D8	E666 0001 0C55			1687+ VCNF V22,V22,0,1	test instruction (dest is source)
000020DE	B29C 5040		00002078	1688+ STFPC FPC_R_18	save FPC
000020E2	E760 5028 080E		00002060	1689+ VST V22,V1018	save instruction result
000020E8	07FB			1690+ BR R11	return
000020EC				1691+RE18 DS 0F	expected 16 byte result
000020EC				1692+ DROP R5	
000020EC	21FF0000 A1FF0000			1693 DC XL16'21FF0000A1FF0000000000000000D800'	
000020F4	00000000 0000D800				
000020FC	03FF0000 83FF0000			1694 DC XL16'03FF000083FF0000000000000000F000'	
00002104	00000000 0000F000				
				1695	
				1696 * +Infinity, -Infinity	skip xc - infinity
				1697 VRR_A VCNF,0,1,20,23,S	
00002110				1698+ DS 0FD	
00002110		00002110		1699+ USING *,R5	base for test data and test routine
00002110	000021A0			1700+T19 DC A(X19)	address of test routine
00002114	0013			1701+ DC H'19'	test number
00002116	00			1702+ DC X'00'	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00002117	E2			1703+	DC	CL1'S'
00002118	00			1704+	DC	HL1'0'
00002119	01			1705+	DC	HL1'1'
0000211A	20			1706+FLG19	DC	X'20'
0000211B	23			1707+VXC19	DC	X'23'
0000211C	000021D4			1708+V2_19	DC	A(RE19+16)
00002120	E5C3D5C6 40404040			1709+	DC	CL8'VCNF'
00002128	00000010			1710+	DC	A(16)
0000212C	000021C4			1711+	DC	A(RE19)
00002130	00000000 00000000			1712+	DS	FD
00002138	00000000 00000000			1713+V1019	DS	XL16
00002140	00000000 00000000					
00002148	00000000 00000000			1714+	DS	FD
00002150	00000000			1715+FPC_R_19	DS	F
00002158	00000000 00000000			1716+	DS	FD
00002160	40404040 40404040			1717+	DC	CL8' '
00002168	00000000 00000000			1718+	DS	FD
00002170	00000000 00000000			1719+XC019	DS	XL16
00002178	00000000 00000000					
00002180	00000000 00000000			1720+	DS	FD
00002188	00000000			1721+FPC_XC_19	DS	F
00002190	00000000 00000000			1722+	DS	FD
00002198	00000000			1723+	DS	F
0000219C	00000000			1724+	DS	F
				1725+*		
000021A0				1726+X19	DS	0F
000021A0	B29D 85A4		000007A4	1727+	LFPC	FPCINIT
				1728+*	LFPC	=XL4'08000000'
000021A4	E320 500C 0014		0000211C	1729+	LGF	R2,V2_19
000021AA	E762 0000 0806		00000000	1730+	VL	V22,0(R2)
000021B0	E666 0001 0C55			1731+	VCNF	V22,V22,0,1
000021B6	B29C 5040		00002150	1732+	STFPC	FPC_R_19
000021BA	E760 5028 080E		00002138	1733+	VST	V22,V1019
000021C0	07FB			1734+	BR	R11
000021C4				1735+RE19	DS	0F
000021C4				1736+	DROP	R5
000021C4	7FFF0000 FFFF0000			1737	DC	XL16'7FFF0000FFFF0000000000000000D800'
000021CC	00000000 0000D800					
000021D4	7C000000 FC000000			1738	DC	XL16'7C000000FC00000000000000000F000'
000021DC	00000000 0000F000					
				1739		
				1740 * +QNaN, -QNaN		skip xc - NaN
				1741	VRR_A	VCNF,0,1,80,21,S
000021E8				1742+	DS	0FD
000021E8		000021E8		1743+	USING	*,R5
000021E8	00002278			1744+T20	DC	A(X20)
000021EC	0014			1745+	DC	H'20'
000021EE	00			1746+	DC	X'00'
000021EF	E2			1747+	DC	CL1'S'
000021F0	00			1748+	DC	HL1'0'
000021F1	01			1749+	DC	HL1'1'
000021F2	80			1750+FLG20	DC	X'80'
000021F3	21			1751+VXC20	DC	X'21'
000021F4	000022AC			1752+V2_20	DC	A(RE20+16)
000021F8	E5C3D5C6 40404040			1753+	DC	CL8'VCNF'
00002200	00000010			1754+	DC	A(16)

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00002204	0000229C			1755+	DC	A(RE20)
00002208	00000000 00000000			1756+	DS	FD
00002210	00000000 00000000			1757+V1020	DS	XL16
00002218	00000000 00000000					
00002220	00000000 00000000			1758+	DS	FD
00002228	00000000			1759+FPC_R_20	DS	F
00002230	00000000 00000000			1760+	DS	FD
00002238	40404040 40404040			1761+	DC	CL8' '
00002240	00000000 00000000			1762+	DS	FD
00002248	00000000 00000000			1763+XC020	DS	XL16
00002250	00000000 00000000					
00002258	00000000 00000000			1764+	DS	FD
00002260	00000000			1765+FPC_XC_20	DS	F
00002268	00000000 00000000			1766+	DS	FD
00002270	00000000			1767+	DS	F
00002274	00000000			1768+	DS	F
				1769+*		
00002278				1770+X20	DS	0F
00002278	B29D 85A4		000007A4	1771+	LFPC	FPCINIT
				1772+*	LFPC	=XL4'08000000'
0000227C	E320 500C 0014		000021F4	1773+	LGF	R2,V2_20
00002282	E762 0000 0806		00000000	1774+	VL	V22,0(R2)
00002288	E666 0001 0C55			1775+	VCNF	V22,V22,0,1
0000228E	B29C A028		00002228	1776+	STFPC	FPC_R_20
00002292	E760 A010 080E		00002210	1777+	VST	V22,V1020
00002298	07FB			1778+	BR	R11
0000229C				1779+RE20	DS	0F
0000229C				1780+	DROP	R5
0000229C	7FFF0000 FFFF0000			1781	DC	XL16'7FFF0000FFFF0000000000000000D800'
000022A4	00000000 0000D800					
000022AC	7E080000 FE080000			1782	DC	XL16'7E080000FE0800000000000000000F000'
000022B4	00000000 0000F000					
				1783		
				1784 * +SNaN, -SNaN		skip xc - NaN
				1785	VRR_A	VCNF,0,1,80,21,S
000022C0				1786+	DS	0FD
000022C0		000022C0		1787+	USING	*,R5
000022C0	00002350			1788+T21	DC	A(X21)
000022C4	0015			1789+	DC	H'21'
000022C6	00			1790+	DC	X'00'
000022C7	E2			1791+	DC	CL1'S'
000022C8	00			1792+	DC	HL1'0'
000022C9	01			1793+	DC	HL1'1'
000022CA	80			1794+FLG21	DC	X'80'
000022CB	21			1795+VXC21	DC	X'21'
000022CC	00002384			1796+V2_21	DC	A(RE21+16)
000022D0	E5C3D5C6 40404040			1797+	DC	CL8'VCNF'
000022D8	00000010			1798+	DC	A(16)
000022DC	00002374			1799+	DC	A(RE21)
000022E0	00000000 00000000			1800+	DS	FD
000022E8	00000000 00000000			1801+V1021	DS	XL16
000022F0	00000000 00000000					
000022F8	00000000 00000000			1802+	DS	FD
00002300	00000000			1803+FPC_R_21	DS	F
00002308	00000000 00000000			1804+	DS	FD
00002310	40404040 40404040			1805+	DC	CL8' '

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00002318	00000000	00000000		1806+	DS	FD	gap
00002320	00000000	00000000		1807+XC021	DS	XL16	Cross check Output
00002328	00000000	00000000					
00002330	00000000	00000000		1808+	DS	FD	gap
00002338	00000000			1809+FPC_XC_21	DS	F	FPC after cross check
00002340	00000000	00000000		1810+	DS	FD	gap
00002348	00000000			1811+	DS	F	debug area
0000234C	00000000			1812+	DS	F	
				1813+*			
00002350				1814+X21	DS	0F	
00002350	B29D 85A4		000007A4	1815+	LFPC	FPCINIT	initialize FPC
				1816+*	LFPC	=XL4'08000000'	set trap exception
00002354	E320 500C 0014		000022CC	1817+	LGF	R2,V2_21	get v2
0000235A	E762 0000 0806		00000000	1818+	VL	V22,0(R2)	
00002360	E666 0001 0C55			1819+	VCNF	V22,V22,0,1	test instruction (dest is source)
00002366	B29C 5040		00002300	1820+	STFPC	FPC_R_21	save FPC
0000236A	E760 5028 080E		000022E8	1821+	VST	V22,V1021	save instruction result
00002370	07FB			1822+	BR	R11	return
00002374				1823+RE21	DS	0F	expected 16 byte result
00002374				1824+	DROP	R5	
00002374	7FFF0000 FFFF0000			1825	DC	XL16'7FFF0000FFFF0000000000000000D800'	
0000237C	00000000 0000D800						
00002384	7C080000 FC080000			1826	DC	XL16'7C080000FC0800000000000000000F000'	
0000238C	00000000 0000F000						
				1827			
				1828 *	multiple exception types		
				1829 *	+Infinity, -QNaN		
				1830	VRR_A	VCNF,0,1,A0,21,S	skip xc - infinity
00002398				1831+	DS	0FD	
00002398			00002398	1832+	USING	*,R5	base for test data and test routine
00002398	00002428			1833+T22	DC	A(X22)	address of test routine
0000239C	0016			1834+	DC	H'22'	test number
0000239E	00			1835+	DC	X'00'	
0000239F	E2			1836+	DC	CL1'S'	Y = skip cross check
000023A0	00			1837+	DC	HL1'0'	m3
000023A1	01			1838+	DC	HL1'1'	m4
000023A2	A0			1839+FLG22	DC	X'A0'	expected FPC flags
000023A3	21			1840+VXC22	DC	X'21'	expected VXC
000023A4	0000245C			1841+V2_22	DC	A(RE22+16)	address of v2: 16-byte packed decimal
000023A8	E5C3D5C6 40404040			1842+	DC	CL8'VCNF'	instruction name
000023B0	00000010			1843+	DC	A(16)	result length
000023B4	0000244C			1844+	DC	A(RE22)	address of expected resul
000023B8	00000000 00000000			1845+	DS	FD	gap
000023C0	00000000 00000000			1846+V1022	DS	XL16	V1 output
000023C8	00000000 00000000						
000023D0	00000000 00000000			1847+	DS	FD	gap
000023D8	00000000			1848+FPC_R_22	DS	F	FPC after instruction
000023E0	00000000 00000000			1849+	DS	FD	gap
000023E8	40404040 40404040			1850+	DC	CL8' '	was cross check skipped?
000023F0	00000000 00000000			1851+	DS	FD	gap
000023F8	00000000 00000000			1852+XC022	DS	XL16	Cross check Output
00002400	00000000 00000000						
00002408	00000000 00000000			1853+	DS	FD	gap
00002410	00000000			1854+FPC_XC_22	DS	F	FPC after cross check
00002418	00000000 00000000			1855+	DS	FD	gap
00002420	00000000			1856+	DS	F	debug area

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00002424	00000000			1857+	DS	F	
				1858+*			
00002428				1859+X22	DS	0F	
00002428	B29D 85A4		000007A4	1860+	LFPC	FPCINIT	initialize FPC
				1861+*	LFPC	=XL4'08000000'	set trap exception
0000242C	E320 500C 0014		000023A4	1862+	LGF	R2,V2_22	get v2
00002432	E762 0000 0806		00000000	1863+	VL	V22,0(R2)	
00002438	E666 0001 0C55			1864+	VCNF	V22,V22,0,1	test instruction (dest is source)
0000243E	B29C 5040		000023D8	1865+	STFPC	FPC_R_22	save FPC
00002442	E760 5028 080E		000023C0	1866+	VST	V22,V1022	save instruction result
00002448	07FB			1867+	BR	R11	return
0000244C				1868+RE22	DS	0F	expected 16 byte result
0000244C				1869+	DROP	R5	
0000244C	7FFF0000 FFFF0000			1870	DC	XL16'7FFF0000FFFF0000000000000000D800'	
00002454	00000000 0000D800						
0000245C	7C000000 FE080000			1871	DC	XL16'7C000000FE0800000000000000000F000'	
00002464	00000000 0000F000						
				1872			
				1873 * +Infinity,	-QNaN, -SNaN		skip xc - infinity
				1874	VRR_A	VCNF,0,1,A0,41,S	
00002470				1875+	DS	0FD	
00002470		00002470		1876+	USING	*,R5	base for test data and test routine
00002470	00002500			1877+T23	DC	A(X23)	address of test routine
00002474	0017			1878+	DC	H'23'	test number
00002476	00			1879+	DC	X'00'	
00002477	E2			1880+	DC	CL1'S'	Y = skip cross check
00002478	00			1881+	DC	HL1'0'	m3
00002479	01			1882+	DC	HL1'1'	m4
0000247A	A0			1883+FLG23	DC	X'A0'	expected FPC flags
0000247B	41			1884+VXC23	DC	X'41'	expected VXC
0000247C	00002534			1885+V2_23	DC	A(RE23+16)	address of v2: 16-byte packed decimal
00002480	E5C3D5C6 40404040			1886+	DC	CL8'VCNF'	instruction name
00002488	00000010			1887+	DC	A(16)	result length
0000248C	00002524			1888+	DC	A(RE23)	address of expected resul
00002490	00000000 00000000			1889+	DS	FD	gap
00002498	00000000 00000000			1890+V1023	DS	XL16	V1 output
000024A0	00000000 00000000						
000024A8	00000000 00000000			1891+	DS	FD	gap
000024B0	00000000			1892+FPC_R_23	DS	F	FPC after instruction
000024B8	00000000 00000000			1893+	DS	FD	gap
000024C0	40404040 40404040			1894+	DC	CL8' '	was cross check skipped?
000024C8	00000000 00000000			1895+	DS	FD	gap
000024D0	00000000 00000000			1896+XC023	DS	XL16	Cross check Output
000024D8	00000000 00000000						
000024E0	00000000 00000000			1897+	DS	FD	gap
000024E8	00000000			1898+FPC_XC_23	DS	F	FPC after cross check
000024F0	00000000 00000000			1899+	DS	FD	gap
000024F8	00000000			1900+	DS	F	debug area
000024FC	00000000			1901+	DS	F	
				1902+*			
00002500				1903+X23	DS	0F	
00002500	B29D 85A4		000007A4	1904+	LFPC	FPCINIT	initialize FPC
				1905+*	LFPC	=XL4'08000000'	set trap exception
00002504	E320 500C 0014		0000247C	1906+	LGF	R2,V2_23	get v2
0000250A	E762 0000 0806		00000000	1907+	VL	V22,0(R2)	
00002510	E666 0001 0C55			1908+	VCNF	V22,V22,0,1	test instruction (dest is source)

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00002516	B29C 5040		000024B0	1909+	STFPC	FPC_R_23	save FPC
0000251A	E760 5028 080E		00002498	1910+	VST	V22,V1023	save instruction result
00002520	07FB			1911+	BR	R11	return
00002524				1912+RE23	DS	0F	expected 16 byte result
00002524				1913+	DROP	R5	
00002524	7FFF0000 FFFF0000			1914	DC	XL16'7FFF0000FFFF0000FFFF00000000D800'	
0000252C	FFFF0000 0000D800						
00002534	7C000000 FE080000			1915	DC	XL16'7C000000FE080000FC0800000000F000'	
0000253C	FC080000 0000F000						
				1916			
				1917 *			
				1918 * VCLFNH	-	VECTOR FP CONVERT AND LENGTHEN FROM NNP HIGH	
				1919 *			
				1920 *		dlfloat -> short float (with cross check: short float -> dlfloor)	
				1921 *			
				1922			
				1923 * +0, -0		simple instruction and 'test' test	
				1924	VRR_A	VCLFNH,2,0,00,00,N	
00002548				1925+	DS	0FD	
00002548		00002548		1926+	USING	*,R5	base for test data and test routine
00002548	000025D8			1927+T24	DC	A(X24)	address of test routine
0000254C	0018			1928+	DC	H'24'	test number
0000254E	00			1929+	DC	X'00'	
0000254F	D5			1930+	DC	CL1'N'	Y = skip cross check
00002550	02			1931+	DC	HL1'2'	m3
00002551	00			1932+	DC	HL1'0'	m4
00002552	00			1933+FLG24	DC	X'00'	expected FPC flags
00002553	00			1934+VXC24	DC	X'00'	expected VXC
00002554	0000260C			1935+V2_24	DC	A(RE24+16)	address of v2: 16-byte packed decimal
00002558	E5C3D3C6 D5C84040			1936+	DC	CL8'VCLFNH'	instruction name
00002560	00000010			1937+	DC	A(16)	result length
00002564	000025FC			1938+	DC	A(RE24)	address of expected resul
00002568	00000000 00000000			1939+	DS	FD	gap
00002570	00000000 00000000			1940+V1024	DS	XL16	V1 output
00002578	00000000 00000000						
00002580	00000000 00000000			1941+	DS	FD	gap
00002588	00000000			1942+FPC_R_24	DS	F	FPC after instruction
00002590	00000000 00000000			1943+	DS	FD	gap
00002598	40404040 40404040			1944+	DC	CL8' '	was cross check skipped?
000025A0	00000000 00000000			1945+	DS	FD	gap
000025A8	00000000 00000000			1946+XC024	DS	XL16	Cross check Output
000025B0	00000000 00000000						
000025B8	00000000 00000000			1947+	DS	FD	gap
000025C0	00000000			1948+FPC_XC_24	DS	F	FPC after cross check
000025C8	00000000 00000000			1949+	DS	FD	gap
000025D0	00000000			1950+	DS	F	debug area
000025D4	00000000			1951+	DS	F	
				1952+*			
000025D8				1953+X24	DS	0F	
000025D8	B29D 85A4		000007A4	1954+	LFPC	FPCINIT	initialize FPC
				1955+*	LFPC	=XL4'08000000'	set trap exception
000025DC	E320 500C 0014		00002554	1956+	LGF	R2,V2_24	get v2
000025E2	E762 0000 0806		00000000	1957+	VL	V22,0(R2)	
000025E8	E666 0000 2C56			1958+	VCLFNH	V22,V22,2,0	test instruction (dest is source)
000025EE	B29C 5040		00002588	1959+	STFPC	FPC_R_24	save FPC
000025F2	E760 5028 080E		00002570	1960+	VST	V22,V1024	save instruction result

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
000025F8	07FB			1961+	BR R11	return
000025FC				1962+RE24	DS 0F	expected 16 byte result
000025FC				1963+	DROP R5	
000025FC	00000000 80000000			1964	DC XL16'	00000000080000000000000000000000'
00002604	00000000 00000000					
0000260C	00008000 00000000			1965	DC XL16'	00008000000000000000000000000000'
00002614	00000000 00000000					
				1966		
				1967 * +1, -1		
				1968	VRR_A VCLFNH, 2, 0, 00, 00, N	
00002620				1969+	DS 0FD	
00002620		00002620		1970+	USING *, R5	base for test data and test routine
00002620	000026B0			1971+T25	DC A(X25)	address of test routine
00002624	0019			1972+	DC H'25'	test number
00002626	00			1973+	DC X'00'	
00002627	D5			1974+	DC CL1'N'	Y = skip cross check
00002628	02			1975+	DC HL1'2'	m3
00002629	00			1976+	DC HL1'0'	m4
0000262A	00			1977+FLG25	DC X'00'	expected FPC flags
0000262B	00			1978+VXC25	DC X'00'	expected VXC
0000262C	000026E4			1979+V2_25	DC A(RE25+16)	address of v2: 16-byte packed decimal
00002630	E5C3D3C6 D5C84040			1980+	DC CL8'VCLFNH'	instruction name
00002638	00000010			1981+	DC A(16)	result length
0000263C	000026D4			1982+	DC A(RE25)	address of expected resul
00002640	00000000 00000000			1983+	DS FD	gap
00002648	00000000 00000000			1984+V1025	DS XL16	V1 output
00002650	00000000 00000000					
00002658	00000000 00000000			1985+	DS FD	gap
00002660	00000000			1986+FPC_R_25	DS F	FPC after instruction
00002668	00000000 00000000			1987+	DS FD	gap
00002670	40404040 40404040			1988+	DC CL8' '	was cross check skipped?
00002678	00000000 00000000			1989+	DS FD	gap
00002680	00000000 00000000			1990+XC025	DS XL16	Cross check Output
00002688	00000000 00000000					
00002690	00000000 00000000			1991+	DS FD	gap
00002698	00000000			1992+FPC_XC_25	DS F	FPC after cross check
000026A0	00000000 00000000			1993+	DS FD	gap
000026A8	00000000			1994+	DS F	debug area
000026AC	00000000			1995+	DS F	
				1996+*		
000026B0				1997+X25	DS 0F	
000026B0	B29D 85A4		000007A4	1998+	LFPC FPCINIT	initialize FPC
				1999+*	LFPC =XL4'08000000'	set trap exception
000026B4	E320 500C 0014		0000262C	2000+	LGF R2, V2_25	get v2
000026BA	E762 0000 0806		00000000	2001+	VL V22, 0(R2)	
000026C0	E666 0000 2C56			2002+	VCLFNH V22, V22, 2, 0	test instruction (dest is source)
000026C6	B29C 5040		00002660	2003+	STFPC FPC_R_25	save FPC
000026CA	E760 5028 080E		00002648	2004+	VST V22, V1025	save instruction result
000026D0	07FB			2005+	BR R11	return
000026D4				2006+RE25	DS 0F	expected 16 byte result
000026D4				2007+	DROP R5	
000026D4	3F800000 BF800000			2008	DC XL16'	3F800000BF8000000000000000000000'
000026DC	00000000 00000000					
000026E4	3E00BE00 00000000			2009	DC XL16'	3E00BE00000000000000000000000000'
000026EC	00000000 00000000					
				2010		

LOC	OBJECT CODE			ADDR1	ADDR2	STMT			
000029E0	00000000	00000000				2166+XC029	DS	XL16	Cross check Output
000029E8	00000000	00000000							
000029F0	00000000	00000000				2167+	DS	FD	gap
000029F8	00000000					2168+FPC_XC_29	DS	F	FPC after cross check
00002A00	00000000	00000000				2169+	DS	FD	gap
00002A08	00000000					2170+	DS	F	debug area
00002A0C	00000000					2171+	DS	F	
						2172+*			
00002A10						2173+X29	DS	0F	
00002A10	B29D	85A4			000007A4	2174+	LFPC	FPCINIT	initialize FPC
						2175+*	LFPC	=XL4'08000000'	set trap exception
00002A14	E320	500C	0014		0000298C	2176+	LGF	R2,V2_29	get v2
00002A1A	E762	0000	0806		00000000	2177+	VL	V22,0(R2)	
00002A20	E666	0000	2C56			2178+	VCLFNH	V22,V22,2,0	test instruction (dest is source)
00002A26	B29C	5040			000029C0	2179+	STFPC	FPC_R_29	save FPC
00002A2A	E760	5028	080E		000029A8	2180+	VST	V22,V1029	save instruction result
00002A30	07FB					2181+	BR	R11	return
00002A34						2182+RE29	DS	0F	expected 16 byte result
00002A34						2183+	DROP	R5	
00002A34	4036C000	C036C000				2184	DC	XL16'4036C000C036C000000000000000000000000'	
00002A3C	00000000	00000000							
00002A44	40DBC0DB	00000000				2185	DC	XL16'40DBC0DB000000000000000000000000000000'	
00002A4C	00000000	00000000							
						2186			
						2187	* additional tests		
						2188	* max tiny: (1 - 2^-11) * 2^16 (65504)		
						2189	VRR_A	VCLFNH,2,0,00,00,N	
00002A58						2190+	DS	0FD	
00002A58				00002A58		2191+	USING	*,R5	base for test data and test routine
00002A58	00002AE8					2192+T30	DC	A(X30)	address of test routine
00002A5C	001E					2193+	DC	H'30'	test number
00002A5E	00					2194+	DC	X'00'	
00002A5F	D5					2195+	DC	CL1'N'	Y = skip cross check
00002A60	02					2196+	DC	HL1'2'	m3
00002A61	00					2197+	DC	HL1'0'	m4
00002A62	00					2198+FLG30	DC	X'00'	expected FPC flags
00002A63	00					2199+VXC30	DC	X'00'	expected VXC
00002A64	00002B1C					2200+V2_30	DC	A(RE30+16)	address of v2: 16-byte packed decimal
00002A68	E5C3D3C6	D5C84040				2201+	DC	CL8'VCLFNH'	instruction name
00002A70	00000010					2202+	DC	A(16)	result length
00002A74	00002B0C					2203+	DC	A(RE30)	address of expected resul
00002A78	00000000	00000000				2204+	DS	FD	gap
00002A80	00000000	00000000				2205+V1030	DS	XL16	V1 output
00002A88	00000000	00000000							
00002A90	00000000	00000000				2206+	DS	FD	gap
00002A98	00000000					2207+FPC_R_30	DS	F	FPC after instruction
00002AA0	00000000	00000000				2208+	DS	FD	gap
00002AA8	40404040	40404040				2209+	DC	CL8' '	was cross check skipped?
00002AB0	00000000	00000000				2210+	DS	FD	gap
00002AB8	00000000	00000000				2211+XC030	DS	XL16	Cross check Output
00002AC0	00000000	00000000							
00002AC8	00000000	00000000				2212+	DS	FD	gap
00002AD0	00000000					2213+FPC_XC_30	DS	F	FPC after cross check
00002AD8	00000000	00000000				2214+	DS	FD	gap
00002AE0	00000000					2215+	DS	F	debug area
00002AE4	00000000					2216+	DS	F	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00002AE8				2217+*			
00002AE8	B29D 85A4		000007A4	2218+X30	DS	0F	
				2219+	LFPC	FPCINIT	initialize FPC
				2220+*	LFPC	=XL4'08000000'	set trap exception
00002AEC	E320 500C 0014		00002A64	2221+	LGF	R2,V2_30	get v2
00002AF2	E762 0000 0806		00000000	2222+	VL	V22,0(R2)	
00002AF8	E666 0000 2C56			2223+	VCLFNH	V22,V22,2,0	test instruction (dest is source)
00002AFE	B29C 5040		00002A98	2224+	STFPC	FPC_R_30	save FPC
00002B02	E760 5028 080E		00002A80	2225+	VST	V22,V1030	save instruction result
00002B08	07FB			2226+	BR	R11	return
00002B0C				2227+RE30	DS	0F	expected 16 byte result
00002B0C				2228+	DROP	R5	
00002B0C	47800000 00000000			2229	DC	XL16'47800000000000000000000000000000'	
00002B14	00000000 00000000						
00002B1C	5E000000 00000000			2230	DC	XL16'5E000000000000000000000000000000'	
00002B24	00000000 00000000						
				2231			
				2232 * max dfloat: 2^(33)-2ulp			
				2233	VRR_A	VCLFNH,2,0,00,00,N	
00002B30			00002B30	2234+	DS	0FD	
00002B30	00002BC0			2235+	USING	*,R5	base for test data and test routine
00002B30	001F			2236+T31	DC	A(X31)	address of test routine
00002B34	00			2237+	DC	H'31'	test number
00002B36	00			2238+	DC	X'00'	
00002B37	D5			2239+	DC	CL1'N'	Y = skip cross check
00002B38	02			2240+	DC	HL1'2'	m3
00002B39	00			2241+	DC	HL1'0'	m4
00002B3A	00			2242+FLG31	DC	X'00'	expected FPC flags
00002B3B	00			2243+VXC31	DC	X'00'	expected VXC
00002B3C	00002BF4			2244+V2_31	DC	A(RE31+16)	address of v2: 16-byte packed decimal
00002B40	E5C3D3C6 D5C84040			2245+	DC	CL8'VCLFNH'	instruction name
00002B48	00000010			2246+	DC	A(16)	result length
00002B4C	00002BE4			2247+	DC	A(RE31)	address of expected resul
00002B50	00000000 00000000			2248+	DS	FD	gap
00002B58	00000000 00000000			2249+V1031	DS	XL16	V1 output
00002B60	00000000 00000000						
00002B68	00000000 00000000			2250+	DS	FD	gap
00002B70	00000000			2251+FPC_R_31	DS	F	FPC after instruction
00002B78	00000000 00000000			2252+	DS	FD	gap
00002B80	40404040 40404040			2253+	DC	CL8' '	was cross check skipped?
00002B88	00000000 00000000			2254+	DS	FD	gap
00002B90	00000000 00000000			2255+XC031	DS	XL16	Cross check Output
00002B98	00000000 00000000						
00002BA0	00000000 00000000			2256+	DS	FD	gap
00002BA8	00000000			2257+FPC_XC_31	DS	F	FPC after cross check
00002BB0	00000000 00000000			2258+	DS	FD	gap
00002BB8	00000000			2259+	DS	F	debug area
00002BBC	00000000			2260+	DS	F	
				2261+*			
00002BC0				2262+X31	DS	0F	
00002BC0	B29D 85A4		000007A4	2263+	LFPC	FPCINIT	initialize FPC
				2264+*	LFPC	=XL4'08000000'	set trap exception
00002BC4	E320 500C 0014		00002B3C	2265+	LGF	R2,V2_31	get v2
00002BCA	E762 0000 0806		00000000	2266+	VL	V22,0(R2)	
00002BD0	E666 0000 2C56			2267+	VCLFNH	V22,V22,2,0	test instruction (dest is source)
00002BD6	B29C 5040		00002B70	2268+	STFPC	FPC_R_31	save FPC

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00002BDA	E760 5028 080E		00002B58	2269+	VST	V22,V1031	save instruction result
00002BE0	07FB			2270+	BR	R11	return
00002BE4				2271+RE31	DS	0F	expected 16 byte result
00002BE4				2272+	DROP	R5	
00002BE4	4FFF8000 00000000			2273	DC	XL16'4FFF8000000000000000000000000000'	
00002BEC	00000000 00000000						
00002BF4	7FFE0000 00000000			2274	DC	XL16'7FFE0000000000000000000000000000'	
00002BFC	00000000 00000000						
				2275			
				2276 * min	dlfloat:	2^(-31)*+ulp	
				2277	VRR_A	VCLFNH,2,0,00,00,N	
00002C08				2278+	DS	0FD	
00002C08		00002C08		2279+	USING	*,R5	base for test data and test routine
00002C08	00002C98			2280+T32	DC	A(X32)	address of test routine
00002C0C	0020			2281+	DC	H'32'	test number
00002C0E	00			2282+	DC	X'00'	
00002C0F	D5			2283+	DC	CL1'N'	Y = skip cross check
00002C10	02			2284+	DC	HL1'2'	m3
00002C11	00			2285+	DC	HL1'0'	m4
00002C12	00			2286+FLG32	DC	X'00'	expected FPC flags
00002C13	00			2287+VXC32	DC	X'00'	expected VXC
00002C14	00002CCC			2288+V2_32	DC	A(RE32+16)	address of v2: 16-byte packed decimal
00002C18	E5C3D3C6 D5C84040			2289+	DC	CL8'VCLFNH'	instruction name
00002C20	00000010			2290+	DC	A(16)	result length
00002C24	00002CBC			2291+	DC	A(RE32)	address of expected resul
00002C28	00000000 00000000			2292+	DS	FD	gap
00002C30	00000000 00000000			2293+V1032	DS	XL16	V1 output
00002C38	00000000 00000000						
00002C40	00000000 00000000			2294+	DS	FD	gap
00002C48	00000000			2295+FPC_R_32	DS	F	FPC after instruction
00002C50	00000000 00000000			2296+	DS	FD	gap
00002C58	40404040 40404040			2297+	DC	CL8' '	was cross check skipped?
00002C60	00000000 00000000			2298+	DS	FD	gap
00002C68	00000000 00000000			2299+XC032	DS	XL16	Cross check Output
00002C70	00000000 00000000						
00002C78	00000000 00000000			2300+	DS	FD	gap
00002C80	00000000			2301+FPC_XC_32	DS	F	FPC after cross check
00002C88	00000000 00000000			2302+	DS	FD	gap
00002C90	00000000			2303+	DS	F	debug area
00002C94	00000000			2304+	DS	F	
				2305+*			
00002C98				2306+X32	DS	0F	
00002C98	B29D 85A4		000007A4	2307+	LFPC	FPCINIT	initialize FPC
				2308+*	LFPC	=XL4'08000000'	set trap exception
00002C9C	E320 500C 0014		00002C14	2309+	LGF	R2,V2_32	get v2
00002CA2	E762 0000 0806		00000000	2310+	VL	V22,0(R2)	
00002CA8	E666 0000 2C56			2311+	VCLFNH	V22,V22,2,0	test instruction (dest is source)
00002CAE	B29C 5040		00002C48	2312+	STFPC	FPC_R_32	save FPC
00002CB2	E760 5028 080E		00002C30	2313+	VST	V22,V1032	save instruction result
00002CB8	07FB			2314+	BR	R11	return
00002CBC				2315+RE32	DS	0F	expected 16 byte result
00002CBC				2316+	DROP	R5	
00002CBC	30004000 00000000			2317	DC	XL16'30004000000000000000000000000000'	
00002CC4	00000000 00000000						
00002CCC	00010000 00000000			2318	DC	XL16'00010000000000000000000000000000'	
00002CD4	00000000 00000000						

LOC	OBJECT CODE	ADDR1	ADDR2	STMT	
				2319	
				2320 * +Infinity, -Infinity	
00002CE0				2321 VRR_A VCLFNH,2,0,80,21,S	skip xcheck - infinity
00002CE0		00002CE0		2322+ DS 0FD	
00002CE0	00002D70			2323+ USING *,R5	base for test data and test routine
00002CE4	0021			2324+T33 DC A(X33)	address of test routine
00002CE6	00			2325+ DC H'33'	test number
00002CE7	E2			2326+ DC X'00'	
00002CE8	02			2327+ DC CL1'S'	Y = skip cross check
00002CE9	00			2328+ DC HL1'2'	m3
00002CEA	80			2329+ DC HL1'0'	m4
00002CEB	21			2330+FLG33 DC X'80'	expected FPC flags
00002CEC	00002DA4			2331+VXC33 DC X'21'	expected VXC
00002CF0	E5C3D3C6 D5C84040			2332+V2_33 DC A(RE33+16)	address of v2: 16-byte packed decimal
00002CF8	00000010			2333+ DC CL8'VCLFNH'	instruction name
00002CFC	00002D94			2334+ DC A(16)	result length
00002D00	00000000 00000000			2335+ DC A(RE33)	address of expected resul
00002D08	00000000 00000000			2336+ DS FD	gap
00002D10	00000000 00000000			2337+V1033 DS XL16	V1 output
00002D18	00000000 00000000			2338+ DS FD	gap
00002D20	00000000			2339+FPC_R_33 DS F	FPC after instruction
00002D28	00000000 00000000			2340+ DS FD	gap
00002D30	40404040 40404040			2341+ DC CL8' '	was cross check skipped?
00002D38	00000000 00000000			2342+ DS FD	gap
00002D40	00000000 00000000			2343+XC033 DS XL16	Cross check Output
00002D48	00000000 00000000				
00002D50	00000000 00000000			2344+ DS FD	gap
00002D58	00000000			2345+FPC_XC_33 DS F	FPC after cross check
00002D60	00000000 00000000			2346+ DS FD	gap
00002D68	00000000			2347+ DS F	debug area
00002D6C	00000000			2348+ DS F	
				2349+*	
00002D70				2350+X33 DS 0F	
00002D70	B29D 85A4		000007A4	2351+ LFPC FPCINIT	initialize FPC
				2352+* LFPC =XL4'08000000'	set trap exception
00002D74	E320 500C 0014		00002CEC	2353+ LGF R2,V2_33	get v2
00002D7A	E762 0000 0806		00000000	2354+ VL V22,0(R2)	
00002D80	E666 0000 2C56			2355+ VCLFNH V22,V22,2,0	test instruction (dest is source)
00002D86	B29C 5040		00002D20	2356+ STFPC FPC_R_33	save FPC
00002D8A	E760 5028 080E		00002D08	2357+ VST V22,V1033	save instruction result
00002D90	07FB			2358+ BR R11	return
00002D94				2359+RE33 DS 0F	expected 16 byte result
00002D94				2360+ DROP R5	
00002D94	7FC00000 00000000			2361 DC XL16'7FC0000000000000FFC000000000000'	
00002D9C	FFC00000 00000000				
00002DA4	7FFF0000 FFFF0000			2362 DC XL16'7FFF0000FFFF00000000000000000000'	
00002DAC	00000000 00000000				
				2363	
				2364 * inexact - bad m3	
00002DB8				2365 VRR_A VCLFNH,5,0,08,05,S	skip xc - inexact
00002DB8		00002DB8		2366+ DS 0FD	
00002DB8	00002E48			2367+ USING *,R5	base for test data and test routine
00002DBC	0022			2368+T34 DC A(X34)	address of test routine
00002DBE	00			2369+ DC H'34'	test number
				2370+ DC X'00'	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00002DBF	E2			2371+	DC	CL1'S'
00002DC0	05			2372+	DC	HL1'5'
00002DC1	00			2373+	DC	HL1'0'
00002DC2	08			2374+FLG34	DC	X'08'
00002DC3	05			2375+VXC34	DC	X'05'
00002DC4	00002E7C			2376+V2_34	DC	A(RE34+16)
00002DC8	E5C3D3C6 D5C84040			2377+	DC	CL8'VCLFNH'
00002DD0	00000010			2378+	DC	A(16)
00002DD4	00002E6C			2379+	DC	A(RE34)
00002DD8	00000000 00000000			2380+	DS	FD
00002DE0	00000000 00000000			2381+V1034	DS	XL16
00002DE8	00000000 00000000					
00002DF0	00000000 00000000			2382+	DS	FD
00002DF8	00000000			2383+FPC_R_34	DS	F
00002E00	00000000 00000000			2384+	DS	FD
00002E08	40404040 40404040			2385+	DC	CL8' '
00002E10	00000000 00000000			2386+	DS	FD
00002E18	00000000 00000000			2387+XC034	DS	XL16
00002E20	00000000 00000000					
00002E28	00000000 00000000			2388+	DS	FD
00002E30	00000000			2389+FPC_XC_34	DS	F
00002E38	00000000 00000000			2390+	DS	FD
00002E40	00000000			2391+	DS	F
00002E44	00000000			2392+	DS	F
				2393+*		
00002E48				2394+X34	DS	0F
00002E48	B29D 85A4		000007A4	2395+	LFPC	FPCINIT
				2396+*	LFPC	=XL4'08000000'
00002E4C	E320 500C 0014		00002DC4	2397+	LGF	R2,V2_34
00002E52	E762 0000 0806		00000000	2398+	VL	V22,0(R2)
00002E58	E666 0000 5C56			2399+	VCLFNH	V22,V22,5,0
00002E5E	B29C 5040		00002DF8	2400+	STFPC	FPC_R_34
00002E62	E760 5028 080E		00002DE0	2401+	VST	V22,V1034
00002E68	07FB			2402+	BR	R11
00002E6C				2403+RE34	DS	0F
00002E6C				2404+	DROP	R5
00002E6C	00000000 00000000			2405	DC	XL16'00000000000000000000000000000000'
00002E74	00000000 00000000					
00002E7C	00010000 00000000			2406	DC	XL16'0001000000000000000000000000F000'
00002E84	00000000 0000F000					
				2407		
				2408	* inexact - bad m4	
				2409	VRR_A	VCLFNH,2,5,08,05,S
				2410+	DS	0FD
00002E90		00002E90		2411+	USING	*,R5
00002E90	00002F20			2412+T35	DC	A(X35)
00002E94	0023			2413+	DC	H'35'
00002E96	00			2414+	DC	X'00'
00002E97	E2			2415+	DC	CL1'S'
00002E98	02			2416+	DC	HL1'2'
00002E99	05			2417+	DC	HL1'5'
00002E9A	08			2418+FLG35	DC	X'08'
00002E9B	05			2419+VXC35	DC	X'05'
00002E9C	00002F54			2420+V2_35	DC	A(RE35+16)
00002EA0	E5C3D3C6 D5C84040			2421+	DC	CL8'VCLFNH'
00002EA8	00000010			2422+	DC	A(16)

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00002EAC	00002F44			2423+	DC	A(RE35)
00002EB0	00000000 00000000			2424+	DS	FD
00002EB8	00000000 00000000			2425+V1035	DS	XL16
00002EC0	00000000 00000000					
00002EC8	00000000 00000000			2426+	DS	FD
00002ED0	00000000			2427+FPC_R_35	DS	F
00002ED8	00000000 00000000			2428+	DS	FD
00002EE0	40404040 40404040			2429+	DC	CL8' '
00002EE8	00000000 00000000			2430+	DS	FD
00002EF0	00000000 00000000			2431+XC035	DS	XL16
00002EF8	00000000 00000000					
00002F00	00000000 00000000			2432+	DS	FD
00002F08	00000000			2433+FPC_XC_35	DS	F
00002F10	00000000 00000000			2434+	DS	FD
00002F18	00000000			2435+	DS	F
00002F1C	00000000			2436+	DS	F
				2437+*		
00002F20				2438+X35	DS	0F
00002F20	B29D 85A4		000007A4	2439+	LFPC	FPCINIT
				2440+*	LFPC	=XL4'08000000'
00002F24	E320 500C 0014		00002E9C	2441+	LGF	R2,V2_35
00002F2A	E762 0000 0806		00000000	2442+	VL	V22,0(R2)
00002F30	E666 0005 2C56			2443+	VCLFNH	V22,V22,2,5
00002F36	B29C 5040		00002ED0	2444+	STFPC	FPC_R_35
00002F3A	E760 5028 080E		00002EB8	2445+	VST	V22,V1035
00002F40	07FB			2446+	BR	R11
00002F44				2447+RE35	DS	0F
00002F44				2448+	DROP	R5
00002F44	00000000 00000000			2449	DC	XL16'00000000000000000000000000000000'
00002F4C	00000000 00000000					
00002F54	00010000 00000000			2450	DC	XL16'0001000000000000000000000000F000'
00002F5C	00000000 0000F000					
				2451		
				2452 * more tests		
				2453 * +3.140625, -3.140625		
				2454	VRR_A	VCLFNH,2,0,00,00,N
00002F68				2455+	DS	0FD
00002F68		00002F68		2456+	USING	*,R5
00002F68	00002FF8			2457+T36	DC	A(X36)
00002F6C	0024			2458+	DC	H'36'
00002F6E	00			2459+	DC	X'00'
00002F6F	D5			2460+	DC	CL1'N'
00002F70	02			2461+	DC	HL1'2'
00002F71	00			2462+	DC	HL1'0'
00002F72	00			2463+FLG36	DC	X'00'
00002F73	00			2464+VXC36	DC	X'00'
00002F74	0000302C			2465+V2_36	DC	A(RE36+16)
00002F78	E5C3D3C6 D5C84040			2466+	DC	CL8'VCLFNH'
00002F80	00000010			2467+	DC	A(16)
00002F84	0000301C			2468+	DC	A(RE36)
00002F88	00000000 00000000			2469+	DS	FD
00002F90	00000000 00000000			2470+V1036	DS	XL16
00002F98	00000000 00000000					
00002FA0	00000000 00000000			2471+	DS	FD
00002FA8	00000000			2472+FPC_R_36	DS	F
00002FB0	00000000 00000000			2473+	DS	FD

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00002FB8	40404040 40404040			2474+	DC	CL8' ' was cross check skipped?
00002FC0	00000000 00000000			2475+	DS	FD gap
00002FC8	00000000 00000000			2476+XC036	DS	XL16 Cross check Output
00002FD0	00000000 00000000					
00002FD8	00000000 00000000			2477+	DS	FD gap
00002FE0	00000000			2478+FPC_XC_36	DS	F FPC after cross check
00002FE8	00000000 00000000			2479+	DS	FD gap
00002FF0	00000000			2480+	DS	F debug area
00002FF4	00000000			2481+	DS	F
				2482+*		
00002FF8				2483+X36	DS	0F
00002FF8	B29D 85A4		000007A4	2484+	LFPC	FPCINIT initialize FPC
				2485+*	LFPC	=XL4'08000000' set trap exception
00002FFC	E320 500C 0014		00002F74	2486+	LGF	R2,V2_36 get v2
00003002	E762 0000 0806		00000000	2487+	VL	V22,0(R2)
00003008	E666 0000 2C56			2488+	VCLFNH	V22,V22,2,0 test instruction (dest is source)
0000300E	B29C 5040		00002FA8	2489+	STFPC	FPC_R_36 save FPC
00003012	E760 5028 080E		00002F90	2490+	VST	V22,V1036 save instruction result
00003018	07FB			2491+	BR	R11 return
0000301C				2492+RE36	DS	0F expected 16 byte result
0000301C				2493+	DROP	R5
0000301C	40890000 00000000			2494	DC	XL16'4089000000000000C089000000000000'
00003024	C0890000 00000000					
0000302C	42240000 C2240000			2495	DC	XL16'42240000C22400000000000000000000'
00003034	00000000 00000000					
				2496		
				2497		
				2498	*	+27.15625, -27.15625
				2499	VRR_A	VCLFNH,2,0,00,00,N
00003040				2500+	DS	0FD
00003040		00003040		2501+	USING	*,R5 base for test data and test routine
00003040	000030D0			2502+T37	DC	A(X37) address of test routine
00003044	0025			2503+	DC	H'37' test number
00003046	00			2504+	DC	X'00'
00003047	D5			2505+	DC	CL1'N' Y = skip cross check
00003048	02			2506+	DC	HL1'2' m3
00003049	00			2507+	DC	HL1'0' m4
0000304A	00			2508+FLG37	DC	X'00' expected FPC flags
0000304B	00			2509+VXC37	DC	X'00' expected VXC
0000304C	00003104			2510+V2_37	DC	A(RE37+16) address of v2: 16-byte packed decimal
00003050	E5C3D3C6 D5C84040			2511+	DC	CL8'VCLFNH' instruction name
00003058	00000010			2512+	DC	A(16) result length
0000305C	000030F4			2513+	DC	A(RE37) address of expected resul
00003060	00000000 00000000			2514+	DS	FD gap
00003068	00000000 00000000			2515+V1037	DS	XL16 V1 output
00003070	00000000 00000000					
00003078	00000000 00000000			2516+	DS	FD gap
00003080	00000000			2517+FPC_R_37	DS	F FPC after instruction
00003088	00000000 00000000			2518+	DS	FD gap
00003090	40404040 40404040			2519+	DC	CL8' ' was cross check skipped?
00003098	00000000 00000000			2520+	DS	FD gap
000030A0	00000000 00000000			2521+XC037	DS	XL16 Cross check Output
000030A8	00000000 00000000					
000030B0	00000000 00000000			2522+	DS	FD gap
000030B8	00000000			2523+FPC_XC_37	DS	F FPC after cross check
000030C0	00000000 00000000			2524+	DS	FD gap

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000030C8	00000000			2525+	DS	F	debug area
000030CC	00000000			2526+	DS	F	
				2527+*			
000030D0				2528+X37	DS	0F	
000030D0	B29D 85A4		000007A4	2529+	LFPC	FPCINIT	initialize FPC
				2530+*	LFPC	=XL4'08000000'	set trap exception
000030D4	E320 500C 0014		0000304C	2531+	LGF	R2,V2_37	get v2
000030DA	E762 0000 0806		00000000	2532+	VL	V22,0(R2)	
000030E0	E666 0000 2C56			2533+	VCLFNH	V22,V22,2,0	test instruction (dest is source)
000030E6	B29C 5040		00003080	2534+	STFPC	FPC_R_37	save FPC
000030EA	E760 5028 080E		00003068	2535+	VST	V22,V1037	save instruction result
000030F0	07FB			2536+	BR	R11	return
000030F4				2537+RE37	DS	0F	expected 16 byte result
000030F4				2538+	DROP	R5	
000030F4	3BD94000 00000000			2539	DC	XL16'3BD9400000000000BBD9400000000000'	
000030FC	BBD94000 00000000						
00003104	2F650000 AF650000			2540	DC	XL16'2F650000AF6500000000000000000000000'	
0000310C	00000000 00000000						
				2541			
				2542 * +65504, -65504			
				2543	VRR_A	VCLFNH,2,0,00,00,N	
00003118				2544+	DS	0FD	
00003118		00003118		2545+	USING	*,R5	base for test data and test routine
00003118	000031A8			2546+T38	DC	A(X38)	address of test routine
0000311C	0026			2547+	DC	H'38'	test number
0000311E	00			2548+	DC	X'00'	
0000311F	D5			2549+	DC	CL1'N'	Y = skip cross check
00003120	02			2550+	DC	HL1'2'	m3
00003121	00			2551+	DC	HL1'0'	m4
00003122	00			2552+FLG38	DC	X'00'	expected FPC flags
00003123	00			2553+VXC38	DC	X'00'	expected VXC
00003124	000031DC			2554+V2_38	DC	A(RE38+16)	address of v2: 16-byte packed decimal
00003128	E5C3D3C6 D5C84040			2555+	DC	CL8'VCLFNH'	instruction name
00003130	00000010			2556+	DC	A(16)	result length
00003134	000031CC			2557+	DC	A(RE38)	address of expected resul
00003138	00000000 00000000			2558+	DS	FD	gap
00003140	00000000 00000000			2559+V1038	DS	XL16	V1 output
00003148	00000000 00000000						
00003150	00000000 00000000			2560+	DS	FD	gap
00003158	00000000			2561+FPC_R_38	DS	F	FPC after instruction
00003160	00000000 00000000			2562+	DS	FD	gap
00003168	40404040 40404040			2563+	DC	CL8' '	was cross check skipped?
00003170	00000000 00000000			2564+	DS	FD	gap
00003178	00000000 00000000			2565+XC038	DS	XL16	Cross check Output
00003180	00000000 00000000						
00003188	00000000 00000000			2566+	DS	FD	gap
00003190	00000000			2567+FPC_XC_38	DS	F	FPC after cross check
00003198	00000000 00000000			2568+	DS	FD	gap
000031A0	00000000			2569+	DS	F	debug area
000031A4	00000000			2570+	DS	F	
				2571+*			
000031A8				2572+X38	DS	0F	
000031A8	B29D 85A4		000007A4	2573+	LFPC	FPCINIT	initialize FPC
				2574+*	LFPC	=XL4'08000000'	set trap exception
000031AC	E320 500C 0014		00003124	2575+	LGF	R2,V2_38	get v2
000031B2	E762 0000 0806		00000000	2576+	VL	V22,0(R2)	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
000031B8	E666 0000 2C56			2577+	VCLFNH V22,V22,2,0	test instruction (dest is source)
000031BE	B29C 5040		00003158	2578+	STFPC FPC_R_38	save FPC
000031C2	E760 5028 080E		00003140	2579+	VST V22,V1038	save instruction result
000031C8	07FB			2580+	BR R11	return
000031CC				2581+RE38	DS 0F	expected 16 byte result
000031CC				2582+	DROP R5	
000031CC	47800000 00000000			2583	DC XL16'4780000000000000C780000000000000'	
000031D4	C7800000 00000000					
000031DC	5E000000 DE000000			2584	DC XL16'5E000000DE0000000000000000000000'	
000031E4	00000000 00000000					
				2585		
				2586 * +Normal, -Normal (2.5)		
				2587	VRR_A VCLFNH,2,0,00,00,N	
000031F0				2588+	DS 0FD	
000031F0		000031F0		2589+	USING *,R5	base for test data and test routine
000031F0	00003280			2590+T39	DC A(X39)	address of test routine
000031F4	0027			2591+	DC H'39'	test number
000031F6	00			2592+	DC X'00'	
000031F7	D5			2593+	DC CL1'N'	Y = skip cross check
000031F8	02			2594+	DC HL1'2'	m3
000031F9	00			2595+	DC HL1'0'	m4
000031FA	00			2596+FLG39	DC X'00'	expected FPC flags
000031FB	00			2597+VXC39	DC X'00'	expected VXC
000031FC	000032B4			2598+V2_39	DC A(RE39+16)	address of v2: 16-byte packed decimal
00003200	E5C3D3C6 D5C84040			2599+	DC CL8'VCLFNH'	instruction name
00003208	00000010			2600+	DC A(16)	result length
0000320C	000032A4			2601+	DC A(RE39)	address of expected resul
00003210	00000000 00000000			2602+	DS FD	gap
00003218	00000000 00000000			2603+V1039	DS XL16	V1 output
00003220	00000000 00000000					
00003228	00000000 00000000			2604+	DS FD	gap
00003230	00000000			2605+FPC_R_39	DS F	FPC after instruction
00003238	00000000 00000000			2606+	DS FD	gap
00003240	40404040 40404040			2607+	DC CL8' '	was cross check skipped?
00003248	00000000 00000000			2608+	DS FD	gap
00003250	00000000 00000000			2609+XC039	DS XL16	Cross check Output
00003258	00000000 00000000					
00003260	00000000 00000000			2610+	DS FD	gap
00003268	00000000			2611+FPC_XC_39	DS F	FPC after cross check
00003270	00000000 00000000			2612+	DS FD	gap
00003278	00000000			2613+	DS F	debug area
0000327C	00000000			2614+	DS F	
				2615+*		
00003280				2616+X39	DS 0F	
00003280	B29D 85A4		000007A4	2617+	LFPC FPCINIT	initialize FPC
				2618+*	LFPC =XL4'08000000'	set trap exception
00003284	E320 500C 0014		000031FC	2619+	LGF R2,V2_39	get v2
0000328A	E762 0000 0806		00000000	2620+	VL V22,0(R2)	
00003290	E666 0000 2C56			2621+	VCLFNH V22,V22,2,0	test instruction (dest is source)
00003296	B29C 5040		00003230	2622+	STFPC FPC_R_39	save FPC
0000329A	E760 5028 080E		00003218	2623+	VST V22,V1039	save instruction result
000032A0	07FB			2624+	BR R11	return
000032A4				2625+RE39	DS 0F	expected 16 byte result
000032A4				2626+	DROP R5	
000032A4	40200000 00000000			2627	DC XL16'4020000000000000C020000000000000'	
000032AC	C0200000 00000000					

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
000032B4	40800000 C0800000			2628	DC	XL16'40800000C08000000000000000000000'
000032BC	00000000 00000000					
				2629		
				2630	* +0.00006097, -0.00006097 (about)	
				2631	VRR_A	VCLFNH,2,0,00,00,N
000032C8				2632+	DS	0FD
000032C8		000032C8		2633+	USING	*,R5
000032C8	00003358			2634+T40	DC	A(X40)
000032CC	0028			2635+	DC	H'40'
000032CE	00			2636+	DC	X'00'
000032CF	D5			2637+	DC	CL1'N'
000032D0	02			2638+	DC	HL1'2'
000032D1	00			2639+	DC	HL1'0'
000032D2	00			2640+FLG40	DC	X'00'
000032D3	00			2641+VXC40	DC	X'00'
000032D4	0000338C			2642+V2_40	DC	A(RE40+16)
000032D8	E5C3D3C6 D5C84040			2643+	DC	CL8'VCLFNH'
000032E0	00000010			2644+	DC	A(16)
000032E4	0000337C			2645+	DC	A(RE40)
000032E8	00000000 00000000			2646+	DS	FD
000032F0	00000000 00000000			2647+V1040	DS	XL16
000032F8	00000000 00000000					
00003300	00000000 00000000			2648+	DS	FD
00003308	00000000			2649+FPC_R_40	DS	F
00003310	00000000 00000000			2650+	DS	FD
00003318	40404040 40404040			2651+	DC	CL8' '
00003320	00000000 00000000			2652+	DS	FD
00003328	00000000 00000000			2653+XC040	DS	XL16
00003330	00000000 00000000					
00003338	00000000 00000000			2654+	DS	FD
00003340	00000000			2655+FPC_XC_40	DS	F
00003348	00000000 00000000			2656+	DS	FD
00003350	00000000			2657+	DS	F
00003354	00000000			2658+	DS	F
				2659+*		
00003358				2660+X40	DS	0F
00003358	B29D 85A4		000007A4	2661+	LFPC	FPCINIT
				2662+*	LFPC	=XL4'08000000'
0000335C	E320 500C 0014		000032D4	2663+	LGF	R2,V2_40
00003362	E762 0000 0806		00000000	2664+	VL	V22,0(R2)
00003368	E666 0000 2C56			2665+	VCLFNH	V22,V22,2,0
0000336E	B29C 5040		00003308	2666+	STFPC	FPC_R_40
00003372	E760 5028 080E		000032F0	2667+	VST	V22,V1040
00003378	07FB			2668+	BR	R11
0000337C				2669+RE40	DS	0F
0000337C				2670+	DROP	R5
0000337C	387FC000 00000000			2671	DC	XL16'387FC00000000000B87FC00000000000'
00003384	B87FC000 00000000					
0000338C	21FF0000 A1FF0000			2672	DC	XL16'21FF0000A1FF00000000000000000000'
00003394	00000000 00000000					
				2673		
				2674	*-----	
				2675	* VCFN - VECTOR FP CONVERT FROM NNP	
				2676	*-----	
				2677	* dlfloat -> tiny float (with cross check: tiny float -> dlfloat)	
				2678	*	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT	
				2679	
				2680 * +0, -0	simple instruction and 'test' test
				2681	VRR_A VCFN,1,0,00,00,N
000033A0				2682+	DS 0FD
000033A0		000033A0		2683+	USING *,R5
000033A0	00003430			2684+T41	DC A(X41)
000033A4	0029			2685+	DC H'41'
000033A6	00			2686+	DC X'00'
000033A7	D5			2687+	DC CL1'N'
000033A8	01			2688+	DC HL1'1'
000033A9	00			2689+	DC HL1'0'
000033AA	00			2690+FLG41	DC X'00'
000033AB	00			2691+VXC41	DC X'00'
000033AC	00003464			2692+V2_41	DC A(RE41+16)
000033B0	E5C3C6D5 40404040			2693+	DC CL8'VCFN'
000033B8	00000010			2694+	DC A(16)
000033BC	00003454			2695+	DC A(RE41)
000033C0	00000000 00000000			2696+	DS FD
000033C8	00000000 00000000			2697+V1041	DS XL16
000033D0	00000000 00000000				
000033D8	00000000 00000000			2698+	DS FD
000033E0	00000000			2699+FPC_R_41	DS F
000033E8	00000000 00000000			2700+	DS FD
000033F0	40404040 40404040			2701+	DC CL8' '
000033F8	00000000 00000000			2702+	DS FD
00003400	00000000 00000000			2703+XC041	DS XL16
00003408	00000000 00000000				
00003410	00000000 00000000			2704+	DS FD
00003418	00000000			2705+FPC_XC_41	DS F
00003420	00000000 00000000			2706+	DS FD
00003428	00000000			2707+	DS F
0000342C	00000000			2708+	DS F
				2709+*	
00003430				2710+X41	DS 0F
00003430	B29D 85A4		000007A4	2711+	LFPC FPCINIT
				2712+*	LFPC =XL4'08000000'
00003434	E320 500C 0014		000033AC	2713+	LGF R2,V2_41
0000343A	E762 0000 0806		00000000	2714+	VL V22,0(R2)
00003440	E666 0000 1C5D			2715+	VCFN V22,V22,1,0
00003446	B29C 5040		000033E0	2716+	STFPC FPC_R_41
0000344A	E760 5028 080E		000033C8	2717+	VST V22,V1041
00003450	07FB			2718+	BR R11
00003454				2719+RE41	DS 0F
00003454				2720+	DROP R5
00003454	00008000 00000000			2721	DC XL16'00008000000000000000000000000000'
0000345C	00000000 00000000				
00003464	00008000 00000000			2722	DC XL16'00008000000000000000000000000000'
0000346C	00000000 00000000				
				2723	
				2724 * 1.0, -1	
				2725	VRR_A VCFN,1,0,00,00,N
00003478				2726+	DS 0FD
00003478		00003478		2727+	USING *,R5
00003478	00003508			2728+T42	DC A(X42)
0000347C	002A			2729+	DC H'42'
0000347E	00			2730+	DC X'00'

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
0000347F	D5			2731+	DC	CL1'N' Y = skip cross check
00003480	01			2732+	DC	HL1'1' m3
00003481	00			2733+	DC	HL1'0' m4
00003482	00			2734+FLG42	DC	X'00' expected FPC flags
00003483	00			2735+VXC42	DC	X'00' expected VXC
00003484	0000353C			2736+V2_42	DC	A(RE42+16) address of v2: 16-byte packed decimal
00003488	E5C3C6D5 40404040			2737+	DC	CL8'VCFN' instruction name
00003490	00000010			2738+	DC	A(16) result length
00003494	0000352C			2739+	DC	A(RE42) address of expected resul
00003498	00000000 00000000			2740+	DS	FD gap
000034A0	00000000 00000000			2741+V1042	DS	XL16 V1 output
000034A8	00000000 00000000					
000034B0	00000000 00000000			2742+	DS	FD gap
000034B8	00000000			2743+FPC_R_42	DS	F FPC after instruction
000034C0	00000000 00000000			2744+	DS	FD gap
000034C8	40404040 40404040			2745+	DC	CL8' ' was cross check skipped?
000034D0	00000000 00000000			2746+	DS	FD gap
000034D8	00000000 00000000			2747+XC042	DS	XL16 Cross check Output
000034E0	00000000 00000000					
000034E8	00000000 00000000			2748+	DS	FD gap
000034F0	00000000			2749+FPC_XC_42	DS	F FPC after cross check
000034F8	00000000 00000000			2750+	DS	FD gap
00003500	00000000			2751+	DS	F debug area
00003504	00000000			2752+	DS	F
				2753+*		
00003508				2754+X42	DS	0F
00003508	B29D 85A4		000007A4	2755+	LFPC	FPCINIT initialize FPC
				2756+*	LFPC	=XL4'08000000' set trap exception
0000350C	E320 500C 0014		00003484	2757+	LGF	R2,V2_42 get v2
00003512	E762 0000 0806		00000000	2758+	VL	V22,0(R2)
00003518	E666 0000 1C5D			2759+	VCFN	V22,V22,1,0 test instruction (dest is source)
0000351E	B29C 5040		000034B8	2760+	STFPC	FPC_R_42 save FPC
00003522	E760 5028 080E		000034A0	2761+	VST	V22,V1042 save instruction result
00003528	07FB			2762+	BR	R11 return
0000352C				2763+RE42	DS	0F expected 16 byte result
0000352C				2764+	DROP	R5
0000352C	3C000000 BC000000			2765	DC	XL16'3C000000BC00000000000000000000000F000'
00003534	00000000 0000F000					
0000353C	3E000000 BE000000			2766	DC	XL16'3E000000BE0000000000000000000000D800'
00003544	00000000 0000D800					
				2767		
				2768 * 0.5		
				2769	VRR_A	VCFN,1,0,00,00,N
00003550				2770+	DS	0FD
00003550		00003550		2771+	USING	*,R5 base for test data and test routine
00003550	000035E0			2772+T43	DC	A(X43) address of test routine
00003554	002B			2773+	DC	H'43' test number
00003556	00			2774+	DC	X'00'
00003557	D5			2775+	DC	CL1'N' Y = skip cross check
00003558	01			2776+	DC	HL1'1' m3
00003559	00			2777+	DC	HL1'0' m4
0000355A	00			2778+FLG43	DC	X'00' expected FPC flags
0000355B	00			2779+VXC43	DC	X'00' expected VXC
0000355C	00003614			2780+V2_43	DC	A(RE43+16) address of v2: 16-byte packed decimal
00003560	E5C3C6D5 40404040			2781+	DC	CL8'VCFN' instruction name
00003568	00000010			2782+	DC	A(16) result length

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
0000356C	00003604			2783+	DC	A(RE43)
00003570	00000000 00000000			2784+	DS	FD
00003578	00000000 00000000			2785+V1043	DS	XL16
00003580	00000000 00000000					
00003588	00000000 00000000			2786+	DS	FD
00003590	00000000			2787+FPC_R_43	DS	F
00003598	00000000 00000000			2788+	DS	FD
000035A0	40404040 40404040			2789+	DC	CL8' '
000035A8	00000000 00000000			2790+	DS	FD
000035B0	00000000 00000000			2791+XC043	DS	XL16
000035B8	00000000 00000000					
000035C0	00000000 00000000			2792+	DS	FD
000035C8	00000000			2793+FPC_XC_43	DS	F
000035D0	00000000 00000000			2794+	DS	FD
000035D8	00000000			2795+	DS	F
000035DC	00000000			2796+	DS	F
				2797+*		
000035E0				2798+X43	DS	0F
000035E0	B29D 85A4		000007A4	2799+	LFPC	FPCINIT
				2800+*	LFPC	=XL4'08000000'
000035E4	E320 500C 0014		0000355C	2801+	LGF	R2,V2_43
000035EA	E762 0000 0806		00000000	2802+	VL	V22,0(R2)
000035F0	E666 0000 1C5D			2803+	VCFN	V22,V22,1,0
000035F6	B29C 5040		00003590	2804+	STFPC	FPC_R_43
000035FA	E760 5028 080E		00003578	2805+	VST	V22,V1043
00003600	07FB			2806+	BR	R11
00003604				2807+RE43	DS	0F
00003604				2808+	DROP	R5
00003604	38000000 00000000			2809	DC	XL16'3800000000000000000000000000F000'
0000360C	00000000 0000F000					
00003614	3C000000 00000000			2810	DC	XL16'3C000000000000000000000000D800'
0000361C	00000000 0000D800					
				2811		
				2812 * 1/64		
				2813	VRR_A	VCFN,1,0,00,00,N
00003628				2814+	DS	0FD
00003628		00003628		2815+	USING	*,R5
00003628	000036B8			2816+T44	DC	A(X44)
0000362C	002C			2817+	DC	H'44'
0000362E	00			2818+	DC	X'00'
0000362F	D5			2819+	DC	CL1'N'
00003630	01			2820+	DC	HL1'1'
00003631	00			2821+	DC	HL1'0'
00003632	00			2822+FLG44	DC	X'00'
00003633	00			2823+VXC44	DC	X'00'
00003634	000036EC			2824+V2_44	DC	A(RE44+16)
00003638	E5C3C6D5 40404040			2825+	DC	CL8'VCFN'
00003640	00000010			2826+	DC	A(16)
00003644	000036DC			2827+	DC	A(RE44)
00003648	00000000 00000000			2828+	DS	FD
00003650	00000000 00000000			2829+V1044	DS	XL16
00003658	00000000 00000000					
00003660	00000000 00000000			2830+	DS	FD
00003668	00000000			2831+FPC_R_44	DS	F
00003670	00000000 00000000			2832+	DS	FD
00003678	40404040 40404040			2833+	DC	CL8' '

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00003680	00000000 00000000			2834+	DS	FD	gap
00003688	00000000 00000000			2835+XC044	DS	XL16	Cross check Output
00003690	00000000 00000000						
00003698	00000000 00000000			2836+	DS	FD	gap
000036A0	00000000			2837+FPC_XC_44	DS	F	FPC after cross check
000036A8	00000000 00000000			2838+	DS	FD	gap
000036B0	00000000			2839+	DS	F	debug area
000036B4	00000000			2840+	DS	F	
				2841+*			
000036B8				2842+X44	DS	0F	
000036B8	B29D 85A4		000007A4	2843+	LFPC	FPCINIT	initialize FPC
				2844+*	LFPC	=XL4'08000000'	set trap exception
000036BC	E320 500C 0014		00003634	2845+	LGF	R2,V2_44	get v2
000036C2	E762 0000 0806		00000000	2846+	VL	V22,0(R2)	
000036C8	E666 0000 1C5D			2847+	VCFN	V22,V22,1,0	test instruction (dest is source)
000036CE	B29C 5040		00003668	2848+	STFPC	FPC_R_44	save FPC
000036D2	E760 5028 080E		00003650	2849+	VST	V22,V1044	save instruction result
000036D8	07FB			2850+	BR	R11	return
000036DC				2851+RE44	DS	0F	expected 16 byte result
000036DC				2852+	DROP	R5	
000036DC	24000000 00000000			2853	DC	XL16'24000000000000000000000000000000F000'	
000036E4	00000000 0000F000						
000036EC	32000000 00000000			2854	DC	XL16'32000000000000000000000000000000D800'	
000036F4	00000000 0000D800						
				2855			
				2856 * +0, -0			
				2857	VRR_A	VCFN,1,0,00,00,N	
00003700				2858+	DS	0FD	
00003700		00003700		2859+	USING	*,R5	base for test data and test routine
00003700	00003790			2860+T45	DC	A(X45)	address of test routine
00003704	002D			2861+	DC	H'45'	test number
00003706	00			2862+	DC	X'00'	
00003707	D5			2863+	DC	CL1'N'	Y = skip cross check
00003708	01			2864+	DC	HL1'1'	m3
00003709	00			2865+	DC	HL1'0'	m4
0000370A	00			2866+FLG45	DC	X'00'	expected FPC flags
0000370B	00			2867+VXC45	DC	X'00'	expected VXC
0000370C	000037C4			2868+V2_45	DC	A(RE45+16)	address of v2: 16-byte packed decimal
00003710	E5C3C6D5 40404040			2869+	DC	CL8'VCFN'	instruction name
00003718	00000010			2870+	DC	A(16)	result length
0000371C	000037B4			2871+	DC	A(RE45)	address of expected resul
00003720	00000000 00000000			2872+	DS	FD	gap
00003728	00000000 00000000			2873+V1045	DS	XL16	V1 output
00003730	00000000 00000000						
00003738	00000000 00000000			2874+	DS	FD	gap
00003740	00000000			2875+FPC_R_45	DS	F	FPC after instruction
00003748	00000000 00000000			2876+	DS	FD	gap
00003750	40404040 40404040			2877+	DC	CL8' '	was cross check skipped?
00003758	00000000 00000000			2878+	DS	FD	gap
00003760	00000000 00000000			2879+XC045	DS	XL16	Cross check Output
00003768	00000000 00000000						
00003770	00000000 00000000			2880+	DS	FD	gap
00003778	00000000			2881+FPC_XC_45	DS	F	FPC after cross check
00003780	00000000 00000000			2882+	DS	FD	gap
00003788	00000000			2883+	DS	F	debug area
0000378C	00000000			2884+	DS	F	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00003790				2885+*			
00003790	B29D 85A4		000007A4	2886+X45	DS	0F	
				2887+	LFPC	FPCINIT	initialize FPC
				2888+*	LFPC	=XL4'08000000'	set trap exception
00003794	E320 500C 0014		0000370C	2889+	LGF	R2,V2_45	get v2
0000379A	E762 0000 0806		00000000	2890+	VL	V22,0(R2)	
000037A0	E666 0000 1C5D			2891+	VCFN	V22,V22,1,0	test instruction (dest is source)
000037A6	B29C 5040		00003740	2892+	STFPC	FPC_R_45	save FPC
000037AA	E760 5028 080E		00003728	2893+	VST	V22,V1045	save instruction result
000037B0	07FB			2894+	BR	R11	return
000037B4				2895+RE45	DS	0F	expected 16 byte result
000037B4				2896+	DROP	R5	
000037B4	00008000 00000000			2897	DC	XL16'0000800000000000000000000000F000'	
000037BC	00000000 0000F000						
000037C4	00008000 00000000			2898	DC	XL16'0000800000000000000000000000D800'	
000037CC	00000000 0000D800						
				2899			
				2900 * -15			
				2901	VRR_A	VCFN,1,0,00,00,N	
000037D8				2902+	DS	0FD	
000037D8		000037D8		2903+	USING	*,R5	base for test data and test routine
000037D8	00003868			2904+T46	DC	A(X46)	address of test routine
000037DC	002E			2905+	DC	H'46'	test number
000037DE	00			2906+	DC	X'00'	
000037DF	D5			2907+	DC	CL1'N'	Y = skip cross check
000037E0	01			2908+	DC	HL1'1'	m3
000037E1	00			2909+	DC	HL1'0'	m4
000037E2	00			2910+FLG46	DC	X'00'	expected FPC flags
000037E3	00			2911+VXC46	DC	X'00'	expected VXC
000037E4	0000389C			2912+V2_46	DC	A(RE46+16)	address of v2: 16-byte packed decimal
000037E8	E5C3C6D5 40404040			2913+	DC	CL8'VCFN'	instruction name
000037F0	00000010			2914+	DC	A(16)	result length
000037F4	0000388C			2915+	DC	A(RE46)	address of expected resul
000037F8	00000000 00000000			2916+	DS	FD	gap
00003800	00000000 00000000			2917+V1046	DS	XL16	V1 output
00003808	00000000 00000000						
00003810	00000000 00000000			2918+	DS	FD	gap
00003818	00000000			2919+FPC_R_46	DS	F	FPC after instruction
00003820	00000000 00000000			2920+	DS	FD	gap
00003828	40404040 40404040			2921+	DC	CL8' '	was cross check skipped?
00003830	00000000 00000000			2922+	DS	FD	gap
00003838	00000000 00000000			2923+XC046	DS	XL16	Cross check Output
00003840	00000000 00000000						
00003848	00000000 00000000			2924+	DS	FD	gap
00003850	00000000			2925+FPC_XC_46	DS	F	FPC after cross check
00003858	00000000 00000000			2926+	DS	FD	gap
00003860	00000000			2927+	DS	F	debug area
00003864	00000000			2928+	DS	F	
				2929+*			
00003868				2930+X46	DS	0F	
00003868	B29D 85A4		000007A4	2931+	LFPC	FPCINIT	initialize FPC
				2932+*	LFPC	=XL4'08000000'	set trap exception
0000386C	E320 500C 0014		000037E4	2933+	LGF	R2,V2_46	get v2
00003872	E762 0000 0806		00000000	2934+	VL	V22,0(R2)	
00003878	E666 0000 1C5D			2935+	VCFN	V22,V22,1,0	test instruction (dest is source)
0000387E	B29C 5040		00003818	2936+	STFPC	FPC_R_46	save FPC

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00003882	E760 5028 080E		00003800	2937+	VST	V22,V1046	save instruction result
00003888	07FB			2938+	BR	R11	return
0000388C				2939+RE46	DS	0F	expected 16 byte result
0000388C				2940+	DROP	R5	
0000388C	CB800000 00000000			2941	DC	XL16'CB80000000000000000000000000F000'	
00003894	00000000 0000F000						
0000389C	C5C00000 00000000		2942		DC	XL16'C5C0000000000000000000000000D800'	
000038A4	00000000 0000D800						
				2943			
				2944 * 20/7 (about)			
				2945	VRR_A	VCFN,1,0,00,00,N	
000038B0				2946+	DS	0FD	
000038B0		000038B0		2947+	USING	*,R5	base for test data and test routine
000038B0	00003940			2948+T47	DC	A(X47)	address of test routine
000038B4	002F			2949+	DC	H'47'	test number
000038B6	00			2950+	DC	X'00'	
000038B7	D5			2951+	DC	CL1'N'	Y = skip cross check
000038B8	01			2952+	DC	HL1'1'	m3
000038B9	00			2953+	DC	HL1'0'	m4
000038BA	00			2954+FLG47	DC	X'00'	expected FPC flags
000038BB	00			2955+VXC47	DC	X'00'	expected VXC
000038BC	00003974			2956+V2_47	DC	A(RE47+16)	address of v2: 16-byte packed decimal
000038C0	E5C3C6D5 40404040			2957+	DC	CL8'VCFN'	instruction name
000038C8	00000010			2958+	DC	A(16)	result length
000038CC	00003964			2959+	DC	A(RE47)	address of expected resul
000038D0	00000000 00000000			2960+	DS	FD	gap
000038D8	00000000 00000000			2961+V1047	DS	XL16	V1 output
000038E0	00000000 00000000						
000038E8	00000000 00000000			2962+	DS	FD	gap
000038F0	00000000			2963+FPC_R_47	DS	F	FPC after instruction
000038F8	00000000 00000000			2964+	DS	FD	gap
00003900	40404040 40404040			2965+	DC	CL8' '	was cross check skipped?
00003908	00000000 00000000			2966+	DS	FD	gap
00003910	00000000 00000000			2967+XC047	DS	XL16	Cross check Output
00003918	00000000 00000000						
00003920	00000000 00000000			2968+	DS	FD	gap
00003928	00000000			2969+FPC_XC_47	DS	F	FPC after cross check
00003930	00000000 00000000			2970+	DS	FD	gap
00003938	00000000			2971+	DS	F	debug area
0000393C	00000000			2972+	DS	F	
				2973+*			
00003940				2974+X47	DS	0F	
00003940	B29D 85A4		000007A4	2975+	LFPC	FPCINIT	initialize FPC
				2976+*	LFPC	=XL4'08000000'	set trap exception
00003944	E320 500C 0014		000038BC	2977+	LGF	R2,V2_47	get v2
0000394A	E762 0000 0806		00000000	2978+	VL	V22,0(R2)	
00003950	E666 0000 1C5D			2979+	VCFN	V22,V22,1,0	test instruction (dest is source)
00003956	B29C 5040		000038F0	2980+	STFPC	FPC_R_47	save FPC
0000395A	E760 5028 080E		000038D8	2981+	VST	V22,V1047	save instruction result
00003960	07FB			2982+	BR	R11	return
00003964				2983+RE47	DS	0F	expected 16 byte result
00003964				2984+	DROP	R5	
00003964	41B60000 00000000			2985	DC	XL16'41B60000000000000000000000F000'	
0000396C	00000000 0000F000						
00003974	40DB0000 00000000		2986		DC	XL16'40DB0000000000000000000000D800'	
0000397C	00000000 0000D800						

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				2987
				2988 * additional tests
				2989 * max tiny: (1 - 2 ⁻¹¹) * 2 ¹⁶ (65504) +1 (overflow)
				2990 VRR_A VCFN,1,0,20,03,S
00003988				2991+ DS 0FD
00003988		00003988		2992+ USING *,R5 base for test data and test routine
00003988	00003A18			2993+T48 DC A(X48) address of test routine
0000398C	0030			2994+ DC H'48' test number
0000398E	00			2995+ DC X'00'
0000398F	E2			2996+ DC CL1'S' Y = skip cross check
00003990	01			2997+ DC HL1'1' m3
00003991	00			2998+ DC HL1'0' m4
00003992	20			2999+FLG48 DC X'20' expected FPC flags
00003993	03			3000+VXC48 DC X'03' expected VXC
00003994	00003A4C			3001+V2_48 DC A(RE48+16) address of v2: 16-byte packed decimal
00003998	E5C3C6D5 40404040			3002+ DC CL8'VCFN' instruction name
000039A0	00000010			3003+ DC A(16) result length
000039A4	00003A3C			3004+ DC A(RE48) address of expected resul
000039A8	00000000 00000000			3005+ DS FD gap
000039B0	00000000 00000000			3006+V1048 DS XL16 V1 output
000039B8	00000000 00000000			
000039C0	00000000 00000000			3007+ DS FD gap
000039C8	00000000			3008+FPC_R_48 DS F FPC after instruction
000039D0	00000000 00000000			3009+ DS FD gap
000039D8	40404040 40404040			3010+ DC CL8' ' was cross check skipped?
000039E0	00000000 00000000			3011+ DS FD gap
000039E8	00000000 00000000			3012+XC048 DS XL16 Cross check Output
000039F0	00000000 00000000			
000039F8	00000000 00000000			3013+ DS FD gap
00003A00	00000000			3014+FPC_XC_48 DS F FPC after cross check
00003A08	00000000 00000000			3015+ DS FD gap
00003A10	00000000			3016+ DS F debug area
00003A14	00000000			3017+ DS F
				3018+*
00003A18				3019+X48 DS 0F
00003A18	B29D 85A4		000007A4	3020+ LFPC FPCINIT initialize FPC
				3021+* LFPC =XL4'08000000' set trap exception
00003A1C	E320 500C 0014		00003994	3022+ LGF R2,V2_48 get v2
00003A22	E762 0000 0806		00000000	3023+ VL V22,0(R2)
00003A28	E666 0000 1C5D			3024+ VCFN V22,V22,1,0 test instruction (dest is source)
00003A2E	B29C 5040		000039C8	3025+ STFPC FPC_R_48 save FPC
00003A32	E760 5028 080E		000039B0	3026+ VST V22,V1048 save instruction result
00003A38	07FB			3027+ BR R11 return
00003A3C				3028+RE48 DS 0F expected 16 byte result
00003A3C				3029+ DROP R5
00003A3C	7C000000 00000000			3030 DC XL16'7C00000000000000000000000000F000'
00003A44	00000000 0000F000			
00003A4C	5E000000 00000000			3031 DC XL16'5E00000000000000000000000000D800'
00003A54	00000000 0000D800			
				3032
				3033 * min tiny (normal): 2 ⁻¹⁴ (0.00006103515625)
				3034 VRR_A VCFN,1,0,00,00,N
00003A60				3035+ DS 0FD
00003A60		00003A60		3036+ USING *,R5 base for test data and test routine
00003A60	00003AF0			3037+T49 DC A(X49) address of test routine
00003A64	0031			3038+ DC H'49' test number

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00003A66	00			3039+	DC	X'00'
00003A67	D5			3040+	DC	CL1'N'
00003A68	01			3041+	DC	HL1'1'
00003A69	00			3042+	DC	HL1'0'
00003A6A	00			3043+FLG49	DC	X'00'
00003A6B	00			3044+VXC49	DC	X'00'
00003A6C	00003B24			3045+V2_49	DC	A(RE49+16)
00003A70	E5C3C6D5 40404040			3046+	DC	CL8'VCFN'
00003A78	00000010			3047+	DC	A(16)
00003A7C	00003B14			3048+	DC	A(RE49)
00003A80	00000000 00000000			3049+	DS	FD
00003A88	00000000 00000000			3050+V1049	DS	XL16
00003A90	00000000 00000000					
00003A98	00000000 00000000			3051+	DS	FD
00003AA0	00000000			3052+FPC_R_49	DS	F
00003AA8	00000000 00000000			3053+	DS	FD
00003AB0	40404040 40404040			3054+	DC	CL8' '
00003AB8	00000000 00000000			3055+	DS	FD
00003AC0	00000000 00000000			3056+XC049	DS	XL16
00003AC8	00000000 00000000					
00003AD0	00000000 00000000			3057+	DS	FD
00003AD8	00000000			3058+FPC_XC_49	DS	F
00003AE0	00000000 00000000			3059+	DS	FD
00003AE8	00000000			3060+	DS	F
00003AEC	00000000			3061+	DS	F
				3062+*		
00003AF0				3063+X49	DS	0F
00003AF0	B29D 85A4		000007A4	3064+	LFPC	FPCINIT
				3065+*	LFPC	=XL4'08000000'
00003AF4	E320 500C 0014		00003A6C	3066+	LGF	R2,V2_49
00003AFA	E762 0000 0806		00000000	3067+	VL	V22,0(R2)
00003B00	E666 0000 1C5D			3068+	VCFN	V22,V22,1,0
00003B06	B29C 5040		00003AA0	3069+	STFPC	FPC_R_49
00003B0A	E760 5028 080E		00003A88	3070+	VST	V22,V1049
00003B10	07FB			3071+	BR	R11
00003B14				3072+RE49	DS	0F
00003B14				3073+	DROP	R5
00003B14	04000000 00000000			3074	DC	XL16'0400000000000000000000000000F000'
00003B1C	00000000 0000F000					
00003B24	22000000 00000000			3075	DC	XL16'2200000000000000000000000000D800'
00003B2C	00000000 0000D800					
				3076		
				3077 * min tiny (subnormal): 2^-24 (0.000000059604644775390625)		
				3078	VRR_A	VCFN,1,0,00,00,N
00003B38				3079+	DS	0FD
00003B38		00003B38		3080+	USING	*,R5
00003B38	00003BC8			3081+T50	DC	A(X50)
00003B3C	0032			3082+	DC	H'50'
00003B3E	00			3083+	DC	X'00'
00003B3F	D5			3084+	DC	CL1'N'
00003B40	01			3085+	DC	HL1'1'
00003B41	00			3086+	DC	HL1'0'
00003B42	00			3087+FLG50	DC	X'00'
00003B43	00			3088+VXC50	DC	X'00'
00003B44	00003BFC			3089+V2_50	DC	A(RE50+16)
00003B48	E5C3C6D5 40404040			3090+	DC	CL8'VCFN'

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00003B50	00000010			3091+	DC	A(16)	result length
00003B54	00003BEC			3092+	DC	A(RE50)	address of expected resul
00003B58	00000000 00000000			3093+	DS	FD	gap
00003B60	00000000 00000000			3094+V1050	DS	XL16	V1 output
00003B68	00000000 00000000						
00003B70	00000000 00000000			3095+	DS	FD	gap
00003B78	00000000			3096+FPC_R_50	DS	F	FPC after instruction
00003B80	00000000 00000000			3097+	DS	FD	gap
00003B88	40404040 40404040			3098+	DC	CL8' '	was cross check skipped?
00003B90	00000000 00000000			3099+	DS	FD	gap
00003B98	00000000 00000000			3100+XC050	DS	XL16	Cross check Output
00003BA0	00000000 00000000						
00003BA8	00000000 00000000			3101+	DS	FD	gap
00003BB0	00000000			3102+FPC_XC_50	DS	F	FPC after cross check
00003BB8	00000000 00000000			3103+	DS	FD	gap
00003BC0	00000000			3104+	DS	F	debug area
00003BC4	00000000			3105+	DS	F	
				3106+*			
00003BC8				3107+X50	DS	0F	
00003BC8	B29D 85A4		000007A4	3108+	LFPC	FPCINIT	initialize FPC
				3109+*	LFPC	=XL4'08000000'	set trap exception
00003BCC	E320 500C 0014		00003B44	3110+	LGF	R2,V2_50	get v2
00003BD2	E762 0000 0806		00000000	3111+	VL	V22,0(R2)	
00003BD8	E666 0000 1C5D			3112+	VCFN	V22,V22,1,0	test instruction (dest is source)
00003BDE	B29C 5040		00003B78	3113+	STFPC	FPC_R_50	save FPC
00003BE2	E760 5028 080E		00003B60	3114+	VST	V22,V1050	save instruction result
00003BE8	07FB			3115+	BR	R11	return
00003BEC				3116+RE50	DS	0F	expected 16 byte result
00003BEC				3117+	DROP	R5	
00003BEC	00010000 00000000			3118	DC	XL16'0001000000000000000000000000F000'	
00003BF4	00000000 0000F000						
00003BFC	0E000000 00000000			3119	DC	XL16'0E00000000000000000000000000D800'	
00003C04	00000000 0000D800						
				3120			
				3121 *		2 underflows	
				3122	VRR_A	VCFN,1,0,10,14,S	
00003C10				3123+	DS	0FD	
00003C10		00003C10		3124+	USING	*,R5	base for test data and test routine
00003C10	00003CA0			3125+T51	DC	A(X51)	address of test routine
00003C14	0033			3126+	DC	H'51'	test number
00003C16	00			3127+	DC	X'00'	
00003C17	E2			3128+	DC	CL1'S'	Y = skip cross check
00003C18	01			3129+	DC	HL1'1'	m3
00003C19	00			3130+	DC	HL1'0'	m4
00003C1A	10			3131+FLG51	DC	X'10'	expected FPC flags
00003C1B	14			3132+VXC51	DC	X'14'	expected VXC
00003C1C	00003CD4			3133+V2_51	DC	A(RE51+16)	address of v2: 16-byte packed decimal
00003C20	E5C3C6D5 40404040			3134+	DC	CL8'VCFN'	instruction name
00003C28	00000010			3135+	DC	A(16)	result length
00003C2C	00003CC4			3136+	DC	A(RE51)	address of expected resul
00003C30	00000000 00000000			3137+	DS	FD	gap
00003C38	00000000 00000000			3138+V1051	DS	XL16	V1 output
00003C40	00000000 00000000						
00003C48	00000000 00000000			3139+	DS	FD	gap
00003C50	00000000			3140+FPC_R_51	DS	F	FPC after instruction
00003C58	00000000 00000000			3141+	DS	FD	gap

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00003C60	40404040	40404040		3142+	DC	CL8' '	was cross check skipped?
00003C68	00000000	00000000		3143+	DS	FD	gap
00003C70	00000000	00000000		3144+XC051	DS	XL16	Cross check Output
00003C78	00000000	00000000					
00003C80	00000000	00000000		3145+	DS	FD	gap
00003C88	00000000			3146+FPC_XC_51	DS	F	FPC after cross check
00003C90	00000000	00000000		3147+	DS	FD	gap
00003C98	00000000			3148+	DS	F	debug area
00003C9C	00000000			3149+	DS	F	
				3150+*			
00003CA0				3151+X51	DS	0F	
00003CA0	B29D 85A4		000007A4	3152+	LFPC	FPCINIT	initialize FPC
				3153+*	LFPC	=XL4'08000000'	set trap exception
00003CA4	E320 500C 0014		00003C1C	3154+	LGF	R2,V2_51	get v2
00003CAA	E762 0000 0806		00000000	3155+	VL	V22,0(R2)	
00003CB0	E666 0000 1C5D			3156+	VCFN	V22,V22,1,0	test instruction (dest is source)
00003CB6	B29C 5040		00003C50	3157+	STFPC	FPC_R_51	save FPC
00003CBA	E760 5028 080E		00003C38	3158+	VST	V22,V1051	save instruction result
00003CC0	07FB			3159+	BR	R11	return
00003CC4				3160+RE51	DS	0F	expected 16 byte result
00003CC4				3161+	DROP	R5	
00003CC4	00000000	00000000		3162	DC	XL16'0000000000000000000000000000F000'	
00003CCC	00000000	0000F000					
00003CD4	0A000B00	00000000		3163	DC	XL16'0A000B0000000000000000000000D800'	
00003CDC	00000000	0000D800					
				3164			
				3165 * +infinity, -Infinity			
				3166	VRR_A	VCFN,1,0,80,21,S	skip xc - invalid
00003CE8				3167+	DS	0FD	
00003CE8		00003CE8		3168+	USING	*,R5	base for test data and test routine
00003CE8	00003D78			3169+T52	DC	A(X52)	address of test routine
00003CEC	0034			3170+	DC	H'52'	test number
00003CEE	00			3171+	DC	X'00'	
00003CEF	E2			3172+	DC	CL1'S'	Y = skip cross check
00003CF0	01			3173+	DC	HL1'1'	m3
00003CF1	00			3174+	DC	HL1'0'	m4
00003CF2	80			3175+FLG52	DC	X'80'	expected FPC flags
00003CF3	21			3176+VXC52	DC	X'21'	expected VXC
00003CF4	00003DAC			3177+V2_52	DC	A(RE52+16)	address of v2: 16-byte packed decimal
00003CF8	E5C3C6D5	40404040		3178+	DC	CL8'VCFN'	instruction name
00003D00	00000010			3179+	DC	A(16)	result length
00003D04	00003D9C			3180+	DC	A(RE52)	address of expected resul
00003D08	00000000	00000000		3181+	DS	FD	gap
00003D10	00000000	00000000		3182+V1052	DS	XL16	V1 output
00003D18	00000000	00000000					
00003D20	00000000	00000000		3183+	DS	FD	gap
00003D28	00000000			3184+FPC_R_52	DS	F	FPC after instruction
00003D30	00000000	00000000		3185+	DS	FD	gap
00003D38	40404040	40404040		3186+	DC	CL8' '	was cross check skipped?
00003D40	00000000	00000000		3187+	DS	FD	gap
00003D48	00000000	00000000		3188+XC052	DS	XL16	Cross check Output
00003D50	00000000	00000000					
00003D58	00000000	00000000		3189+	DS	FD	gap
00003D60	00000000			3190+FPC_XC_52	DS	F	FPC after cross check
00003D68	00000000	00000000		3191+	DS	FD	gap
00003D70	00000000			3192+	DS	F	debug area

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00003D74	00000000			3193+	DS	F	
				3194+*			
00003D78				3195+X52	DS	0F	
00003D78	B29D 85A4		000007A4	3196+	LFPC	FPCINIT	initialize FPC
				3197+*	LFPC	=XL4'08000000'	set trap exception
00003D7C	E320 500C 0014		00003CF4	3198+	LGF	R2,V2_52	get v2
00003D82	E762 0000 0806		00000000	3199+	VL	V22,0(R2)	
00003D88	E666 0000 1C5D			3200+	VCFN	V22,V22,1,0	test instruction (dest is source)
00003D8E	B29C 5040		00003D28	3201+	STFPC	FPC_R_52	save FPC
00003D92	E760 5028 080E		00003D10	3202+	VST	V22,V1052	save instruction result
00003D98	07FB			3203+	BR	R11	return
00003D9C				3204+RE52	DS	0F	expected 16 byte result
00003D9C				3205+	DROP	R5	
00003D9C	7E000000 FE000000			3206	DC	XL16'7E000000FE0000000000000000000000F000'	
00003DA4	00000000 0000F000						
00003DAC	7FFF0000 FFFF0000			3207	DC	XL16'7FFF0000FFFF00000000000000000000D800'	
00003DB4	00000000 0000D800						
				3208			
				3209 * inexact - bad m3			
				3210	VRR_A	VCFN,5,0,08,05,S	skip xc - inexact
00003DC0				3211+	DS	0FD	
00003DC0		00003DC0		3212+	USING	*,R5	base for test data and test routine
00003DC0	00003E50			3213+T53	DC	A(X53)	address of test routine
00003DC4	0035			3214+	DC	H'53'	test number
00003DC6	00			3215+	DC	X'00'	
00003DC7	E2			3216+	DC	CL1'S'	Y = skip cross check
00003DC8	05			3217+	DC	HL1'5'	m3
00003DC9	00			3218+	DC	HL1'0'	m4
00003DCA	08			3219+FLG53	DC	X'08'	expected FPC flags
00003DCB	05			3220+VXC53	DC	X'05'	expected VXC
00003DCC	00003E84			3221+V2_53	DC	A(RE53+16)	address of v2: 16-byte packed decimal
00003DD0	E5C3C6D5 40404040			3222+	DC	CL8'VCFN'	instruction name
00003DD8	00000010			3223+	DC	A(16)	result length
00003DDC	00003E74			3224+	DC	A(RE53)	address of expected resul
00003DE0	00000000 00000000			3225+	DS	FD	gap
00003DE8	00000000 00000000			3226+V1053	DS	XL16	V1 output
00003DF0	00000000 00000000						
00003DF8	00000000 00000000			3227+	DS	FD	gap
00003E00	00000000			3228+FPC_R_53	DS	F	FPC after instruction
00003E08	00000000 00000000			3229+	DS	FD	gap
00003E10	40404040 40404040			3230+	DC	CL8' '	was cross check skipped?
00003E18	00000000 00000000			3231+	DS	FD	gap
00003E20	00000000 00000000			3232+XC053	DS	XL16	Cross check Output
00003E28	00000000 00000000						
00003E30	00000000 00000000			3233+	DS	FD	gap
00003E38	00000000			3234+FPC_XC_53	DS	F	FPC after cross check
00003E40	00000000 00000000			3235+	DS	FD	gap
00003E48	00000000			3236+	DS	F	debug area
00003E4C	00000000			3237+	DS	F	
				3238+*			
00003E50				3239+X53	DS	0F	
00003E50	B29D 85A4		000007A4	3240+	LFPC	FPCINIT	initialize FPC
				3241+*	LFPC	=XL4'08000000'	set trap exception
00003E54	E320 500C 0014		00003DCC	3242+	LGF	R2,V2_53	get v2
00003E5A	E762 0000 0806		00000000	3243+	VL	V22,0(R2)	
00003E60	E666 0000 5C5D			3244+	VCFN	V22,V22,5,0	test instruction (dest is source)

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00003E66	B29C 5040		00003E00	3245+	STFPC	FPC_R_53	save FPC
00003E6A	E760 5028 080E		00003DE8	3246+	VST	V22,V1053	save instruction result
00003E70	07FB			3247+	BR	R11	return
00003E74				3248+RE53	DS	0F	expected 16 byte result
00003E74				3249+	DROP	R5	
00003E74	00000000 00000000			3250	DC	XL16'00000000000000000000000000000000'	
00003E7C	00000000 00000000						
00003E84	00010000 00000000			3251	DC	XL16'0001000000000000000000000000F000'	
00003E8C	00000000 0000F000						
				3252			
				3253 * inexact - bad m4			
				3254	VRR_A	VCFN,0,5,08,05,S	skip xc - inexact
00003E98				3255+	DS	0FD	
00003E98		00003E98		3256+	USING	*,R5	base for test data and test routine
00003E98	00003F28			3257+T54	DC	A(X54)	address of test routine
00003E9C	0036			3258+	DC	H'54'	test number
00003E9E	00			3259+	DC	X'00'	
00003E9F	E2			3260+	DC	CL1'S'	Y = skip cross check
00003EA0	00			3261+	DC	HL1'0'	m3
00003EA1	05			3262+	DC	HL1'5'	m4
00003EA2	08			3263+FLG54	DC	X'08'	expected FPC flags
00003EA3	05			3264+VXC54	DC	X'05'	expected VXC
00003EA4	00003F5C			3265+V2_54	DC	A(RE54+16)	address of v2: 16-byte packed decimal
00003EA8	E5C3C6D5 40404040			3266+	DC	CL8'VCFN'	instruction name
00003EB0	00000010			3267+	DC	A(16)	result length
00003EB4	00003F4C			3268+	DC	A(RE54)	address of expected resul
00003EB8	00000000 00000000			3269+	DS	FD	gap
00003EC0	00000000 00000000			3270+V1054	DS	XL16	V1 output
00003EC8	00000000 00000000						
00003ED0	00000000 00000000			3271+	DS	FD	gap
00003ED8	00000000			3272+FPC_R_54	DS	F	FPC after instruction
00003EE0	00000000 00000000			3273+	DS	FD	gap
00003EE8	40404040 40404040			3274+	DC	CL8' '	was cross check skipped?
00003EF0	00000000 00000000			3275+	DS	FD	gap
00003EF8	00000000 00000000			3276+XC054	DS	XL16	Cross check Output
00003F00	00000000 00000000						
00003F08	00000000 00000000			3277+	DS	FD	gap
00003F10	00000000			3278+FPC_XC_54	DS	F	FPC after cross check
00003F18	00000000 00000000			3279+	DS	FD	gap
00003F20	00000000			3280+	DS	F	debug area
00003F24	00000000			3281+	DS	F	
				3282+*			
00003F28				3283+X54	DS	0F	
00003F28	B29D 85A4		000007A4	3284+	LFPC	FPCINIT	initialize FPC
				3285+*	LFPC	=XL4'08000000'	set trap exception
00003F2C	E320 500C 0014		00003EA4	3286+	LGF	R2,V2_54	get v2
00003F32	E762 0000 0806		00000000	3287+	VL	V22,0(R2)	
00003F38	E666 0005 0C5D			3288+	VCFN	V22,V22,0,5	test instruction (dest is source)
00003F3E	B29C 5040		00003ED8	3289+	STFPC	FPC_R_54	save FPC
00003F42	E760 5028 080E		00003EC0	3290+	VST	V22,V1054	save instruction result
00003F48	07FB			3291+	BR	R11	return
00003F4C				3292+RE54	DS	0F	expected 16 byte result
00003F4C				3293+	DROP	R5	
00003F4C	00000000 00000000			3294	DC	XL16'00000000000000000000000000000000'	
00003F54	00000000 00000000						
00003F5C	00010000 00000000			3295	DC	XL16'0001000000000000000000000000F000'	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT	
00003F64	00000000 0000F000			3296	
				3297 * more tests	
				3298 * +3.140625, -3.140625	
				3299 VRR_A VCFN,1,0,00,00,N	
00003F70				3300+	DS 0FD
00003F70		00003F70		3301+	USING *,R5 base for test data and test routine
00003F70	00004000			3302+T55	DC A(X55) address of test routine
00003F74	0037			3303+	DC H'55' test number
00003F76	00			3304+	DC X'00'
00003F77	D5			3305+	DC CL1'N' Y = skip cross check
00003F78	01			3306+	DC HL1'1' m3
00003F79	00			3307+	DC HL1'0' m4
00003F7A	00			3308+FLG55	DC X'00' expected FPC flags
00003F7B	00			3309+VXC55	DC X'00' expected VXC
00003F7C	00004034			3310+V2_55	DC A(RE55+16) address of v2: 16-byte packed decimal
00003F80	E5C3C6D5 40404040			3311+	DC CL8'VCFN' instruction name
00003F88	00000010			3312+	DC A(16) result length
00003F8C	00004024			3313+	DC A(RE55) address of expected resul
00003F90	00000000 00000000			3314+	DS FD gap
00003F98	00000000 00000000			3315+V1055	DS XL16 V1 output
00003FA0	00000000 00000000				
00003FA8	00000000 00000000			3316+	DS FD gap
00003FB0	00000000			3317+FPC_R_55	DS F FPC after instruction
00003FB8	00000000 00000000			3318+	DS FD gap
00003FC0	40404040 40404040			3319+	DC CL8' ' was cross check skipped?
00003FC8	00000000 00000000			3320+	DS FD gap
00003FD0	00000000 00000000			3321+XC055	DS XL16 Cross check Output
00003FD8	00000000 00000000				
00003FE0	00000000 00000000			3322+	DS FD gap
00003FE8	00000000			3323+FPC_XC_55	DS F FPC after cross check
00003FF0	00000000 00000000			3324+	DS FD gap
00003FF8	00000000			3325+	DS F debug area
00003FFC	00000000			3326+	DS F
				3327+*	
00004000				3328+X55	DS 0F
00004000	B29D 85A4		000007A4	3329+	LFPC FPCINIT initialize FPC
				3330+*	LFPC =XL4'08000000' set trap exception
00004004	E320 500C 0014		00003F7C	3331+	LGF R2,V2_55 get v2
0000400A	E762 0000 0806		00000000	3332+	VL V22,0(R2)
00004010	E666 0000 1C5D			3333+	VCFN V22,V22,1,0 test instruction (dest is source)
00004016	B29C 5040		00003FB0	3334+	STFPC FPC_R_55 save FPC
0000401A	E760 5028 080E		00003F98	3335+	VST V22,V1055 save instruction result
00004020	07FB			3336+	BR R11 return
00004024				3337+RE55	DS 0F expected 16 byte result
00004024				3338+	DROP R5
00004024	44480000 C4480000			3339	DC XL16'44480000C44800000000000000000000'
0000402C	00000000 00000000				
00004034	42240000 C2240000			3340	DC XL16'42240000C22400000000000000000000'
0000403C	00000000 00000000				
				3341	
				3342	
				3343 * +27.15625, -27.15625	
				3344 VRR_A VCFN,1,0,00,00,N	
00004048				3345+	DS 0FD
00004048		00004048		3346+	USING *,R5 base for test data and test routine

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00004048	000040D8			3347+T56	DC	A(X56)
0000404C	0038			3348+	DC	H'56'
0000404E	00			3349+	DC	X'00'
0000404F	D5			3350+	DC	CL1'N'
00004050	01			3351+	DC	HL1'1'
00004051	00			3352+	DC	HL1'0'
00004052	00			3353+FLG56	DC	X'00'
00004053	00			3354+VXC56	DC	X'00'
00004054	0000410C			3355+V2_56	DC	A(RE56+16)
00004058	E5C3C6D5 40404040			3356+	DC	CL8'VCFN'
00004060	00000010			3357+	DC	A(16)
00004064	000040FC			3358+	DC	A(RE56)
00004068	00000000 00000000			3359+	DS	FD
00004070	00000000 00000000			3360+V1056	DS	XL16
00004078	00000000 00000000					
00004080	00000000 00000000			3361+	DS	FD
00004088	00000000			3362+FPC_R_56	DS	F
00004090	00000000 00000000			3363+	DS	FD
00004098	40404040 40404040			3364+	DC	CL8' '
000040A0	00000000 00000000			3365+	DS	FD
000040A8	00000000 00000000			3366+XC056	DS	XL16
000040B0	00000000 00000000					
000040B8	00000000 00000000			3367+	DS	FD
000040C0	00000000			3368+FPC_XC_56	DS	F
000040C8	00000000 00000000			3369+	DS	FD
000040D0	00000000			3370+	DS	F
000040D4	00000000			3371+	DS	F
				3372+*		
000040D8				3373+X56	DS	0F
000040D8	B29D 85A4		000007A4	3374+	LFPC	FPCINIT
				3375+*	LFPC	=XL4'08000000'
000040DC	E320 500C 0014		00004054	3376+	LGF	R2,V2_56
000040E2	E762 0000 0806		00000000	3377+	VL	V22,0(R2)
000040E8	E666 0000 1C5D			3378+	VCFN	V22,V22,1,0
000040EE	B29C 5040		00004088	3379+	STFPC	FPC_R_56
000040F2	E760 5028 080E		00004070	3380+	VST	V22,V1056
000040F8	07FB			3381+	BR	R11
000040FC				3382+RE56	DS	0F
000040FC				3383+	DROP	R5
000040FC	1ECA0000 9ECA0000			3384	DC	XL16'1ECA00009ECA00000000000000000000'
00004104	00000000 00000000					
0000410C	2F650000 AF650000			3385	DC	XL16'2F650000AF6500000000000000000000'
00004114	00000000 00000000					
				3386		
				3387 * +65536, -65536 (overflow)		
				3388	VRR_A	VCFN,1,0,20,23,S
00004120				3389+	DS	0FD
00004120		00004120		3390+	USING	*,R5
00004120	000041B0			3391+T57	DC	A(X57)
00004124	0039			3392+	DC	H'57'
00004126	00			3393+	DC	X'00'
00004127	E2			3394+	DC	CL1'S'
00004128	01			3395+	DC	HL1'1'
00004129	00			3396+	DC	HL1'0'
0000412A	20			3397+FLG57	DC	X'20'
0000412B	23			3398+VXC57	DC	X'23'

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
0000412C	000041E4			3399+V2_57	DC	A(RE57+16)
00004130	E5C3C6D5 40404040			3400+	DC	CL8'VCFN'
00004138	00000010			3401+	DC	A(16)
0000413C	000041D4			3402+	DC	A(RE57)
00004140	00000000 00000000			3403+	DS	FD
00004148	00000000 00000000			3404+V1057	DS	XL16
00004150	00000000 00000000					
00004158	00000000 00000000			3405+	DS	FD
00004160	00000000			3406+FPC_R_57	DS	F
00004168	00000000 00000000			3407+	DS	FD
00004170	40404040 40404040			3408+	DC	CL8' '
00004178	00000000 00000000			3409+	DS	FD
00004180	00000000 00000000			3410+XC057	DS	XL16
00004188	00000000 00000000					
00004190	00000000 00000000			3411+	DS	FD
00004198	00000000			3412+FPC_XC_57	DS	F
000041A0	00000000 00000000			3413+	DS	FD
000041A8	00000000			3414+	DS	F
000041AC	00000000			3415+	DS	F
				3416+*		
000041B0				3417+X57	DS	0F
000041B0	B29D 85A4		000007A4	3418+	LFPC	FPCINIT
				3419+*	LFPC	=XL4'08000000'
000041B4	E320 500C 0014		0000412C	3420+	LGF	R2,V2_57
000041BA	E762 0000 0806		00000000	3421+	VL	V22,0(R2)
000041C0	E666 0000 1C5D			3422+	VCFN	V22,V22,1,0
000041C6	B29C 5040		00004160	3423+	STFPC	FPC_R_57
000041CA	E760 5028 080E		00004148	3424+	VST	V22,V1057
000041D0	07FB			3425+	BR	R11
000041D4				3426+RE57	DS	0F
000041D4				3427+	DROP	R5
000041D4	7C000000 FC000000			3428	DC	XL16'7C000000FC00000000000000000000000'
000041DC	00000000 00000000					
000041E4	5E000000 DE000000			3429	DC	XL16'5E000000DE00000000000000000000000'
000041EC	00000000 00000000					
				3430		
				3431 * +Normal, -Normal (2.5)		
				3432	VRR_A	VCFN,1,0,00,00,N
000041F8				3433+	DS	0FD
000041F8		000041F8		3434+	USING	*,R5
000041F8	00004288			3435+T58	DC	A(X58)
000041FC	003A			3436+	DC	H'58'
000041FE	00			3437+	DC	X'00'
000041FF	D5			3438+	DC	CL1'N'
00004200	01			3439+	DC	HL1'1'
00004201	00			3440+	DC	HL1'0'
00004202	00			3441+FLG58	DC	X'00'
00004203	00			3442+VXC58	DC	X'00'
00004204	000042BC			3443+V2_58	DC	A(RE58+16)
00004208	E5C3C6D5 40404040			3444+	DC	CL8'VCFN'
00004210	00000010			3445+	DC	A(16)
00004214	000042AC			3446+	DC	A(RE58)
00004218	00000000 00000000			3447+	DS	FD
00004220	00000000 00000000			3448+V1058	DS	XL16
00004228	00000000 00000000					
00004230	00000000 00000000			3449+	DS	FD

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00004238	00000000			3450+FPC_R_58	DS F	FPC after instruction
00004240	00000000 00000000			3451+	DS FD	gap
00004248	40404040 40404040			3452+	DC CL8' '	was cross check skipped?
00004250	00000000 00000000			3453+	DS FD	gap
00004258	00000000 00000000			3454+XC058	DS XL16	Cross check Output
00004260	00000000 00000000					
00004268	00000000 00000000			3455+	DS FD	gap
00004270	00000000			3456+FPC_XC_58	DS F	FPC after cross check
00004278	00000000 00000000			3457+	DS FD	gap
00004280	00000000			3458+	DS F	debug area
00004284	00000000			3459+	DS F	
				3460+*		
00004288				3461+X58	DS 0F	
00004288	B29D 85A4		000007A4	3462+	LFPC FPCINIT	initialize FPC
				3463+*	LFPC =XL4'08000000'	set trap exception
0000428C	E320 500C 0014		00004204	3464+	LGF R2,V2_58	get v2
00004292	E762 0000 0806		00000000	3465+	VL V22,0(R2)	
00004298	E666 0000 1C5D			3466+	VCFN V22,V22,1,0	test instruction (dest is source)
0000429E	B29C 5040		00004238	3467+	STFPC FPC_R_58	save FPC
000042A2	E760 5028 080E		00004220	3468+	VST V22,V1058	save instruction result
000042A8	07FB			3469+	BR R11	return
000042AC				3470+RE58	DS 0F	expected 16 byte result
000042AC				3471+	DROP R5	
000042AC	41000000 C1000000			3472	DC XL16'41000000C1000000000000000000000000'	
000042B4	00000000 00000000					
000042BC	40800000 C0800000			3473	DC XL16'40800000C0800000000000000000000000'	
000042C4	00000000 00000000					
				3474		
				3475 * +0.00006097, -0.00006097 (about)		
				3476	VRR_A VCFN,1,0,00,00,N	
000042D0				3477+	DS 0FD	
000042D0		000042D0		3478+	USING *,R5	base for test data and test routine
000042D0	00004360			3479+T59	DC A(X59)	address of test routine
000042D4	003B			3480+	DC H'59'	test number
000042D6	00			3481+	DC X'00'	
000042D7	D5			3482+	DC CL1'N'	Y = skip cross check
000042D8	01			3483+	DC HL1'1'	m3
000042D9	00			3484+	DC HL1'0'	m4
000042DA	00			3485+FLG59	DC X'00'	expected FPC flags
000042DB	00			3486+VXC59	DC X'00'	expected VXC
000042DC	00004394			3487+V2_59	DC A(RE59+16)	address of v2: 16-byte packed decimal
000042E0	E5C3C6D5 40404040			3488+	DC CL8'VCFN'	instruction name
000042E8	00000010			3489+	DC A(16)	result length
000042EC	00004384			3490+	DC A(RE59)	address of expected resul
000042F0	00000000 00000000			3491+	DS FD	gap
000042F8	00000000 00000000			3492+V1059	DS XL16	V1 output
00004300	00000000 00000000					
00004308	00000000 00000000			3493+	DS FD	gap
00004310	00000000			3494+FPC_R_59	DS F	FPC after instruction
00004318	00000000 00000000			3495+	DS FD	gap
00004320	40404040 40404040			3496+	DC CL8' '	was cross check skipped?
00004328	00000000 00000000			3497+	DS FD	gap
00004330	00000000 00000000			3498+XC059	DS XL16	Cross check Output
00004338	00000000 00000000					
00004340	00000000 00000000			3499+	DS FD	gap
00004348	00000000			3500+FPC_XC_59	DS F	FPC after cross check

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00004350	00000000 00000000			3501+	DS	FD	gap
00004358	00000000			3502+	DS	F	debug area
0000435C	00000000			3503+	DS	F	
				3504+*			
00004360				3505+X59	DS	0F	
00004360	B29D 85A4		000007A4	3506+	LFPC	FPCINIT	initialize FPC
				3507+*	LFPC	=XL4'08000000'	set trap exception
00004364	E320 500C 0014		000042DC	3508+	LGF	R2,V2_59	get v2
0000436A	E762 0000 0806		00000000	3509+	VL	V22,0(R2)	
00004370	E666 0000 1C5D			3510+	VCFN	V22,V22,1,0	test instruction (dest is source)
00004376	B29C 5040		00004310	3511+	STFPC	FPC_R_59	save FPC
0000437A	E760 5028 080E		000042F8	3512+	VST	V22,V1059	save instruction result
00004380	07FB			3513+	BR	R11	return
00004384				3514+RE59	DS	0F	expected 16 byte result
00004384				3515+	DROP	R5	
00004384	03FF0000 83FF0000			3516	DC	XL16'03FF000083FF00000000000000000000'	
0000438C	00000000 00000000						
00004394	21FF0000 A1FF0000			3517	DC	XL16'21FF0000A1FF00000000000000000000'	
0000439C	00000000 00000000						
				3518			
				3519 *			multiple exception types
				3520 *			+Infinity, overflow
				3521	VRR_A	VCFN,1,0,A0,23,S	skip xc - infinity
000043A8				3522+	DS	0FD	
000043A8		000043A8		3523+	USING	*,R5	base for test data and test routine
000043A8	00004438			3524+T60	DC	A(X60)	address of test routine
000043AC	003C			3525+	DC	H'60'	test number
000043AE	00			3526+	DC	X'00'	
000043AF	E2			3527+	DC	CL1'S'	Y = skip cross check
000043B0	01			3528+	DC	HL1'1'	m3
000043B1	00			3529+	DC	HL1'0'	m4
000043B2	A0			3530+FLG60	DC	X'A0'	expected FPC flags
000043B3	23			3531+VXC60	DC	X'23'	expected VXC
000043B4	0000446C			3532+V2_60	DC	A(RE60+16)	address of v2: 16-byte packed decimal
000043B8	E5C3C6D5 40404040			3533+	DC	CL8'VCFN'	instruction name
000043C0	00000010			3534+	DC	A(16)	result length
000043C4	0000445C			3535+	DC	A(RE60)	address of expected resul
000043C8	00000000 00000000			3536+	DS	FD	gap
000043D0	00000000 00000000			3537+V1060	DS	XL16	V1 output
000043D8	00000000 00000000						
000043E0	00000000 00000000			3538+	DS	FD	gap
000043E8	00000000			3539+FPC_R_60	DS	F	FPC after instruction
000043F0	00000000 00000000			3540+	DS	FD	gap
000043F8	40404040 40404040			3541+	DC	CL8' '	was cross check skipped?
00004400	00000000 00000000			3542+	DS	FD	gap
00004408	00000000 00000000			3543+XC060	DS	XL16	Cross check Output
00004410	00000000 00000000						
00004418	00000000 00000000			3544+	DS	FD	gap
00004420	00000000			3545+FPC_XC_60	DS	F	FPC after cross check
00004428	00000000 00000000			3546+	DS	FD	gap
00004430	00000000			3547+	DS	F	debug area
00004434	00000000			3548+	DS	F	
				3549+*			
00004438				3550+X60	DS	0F	
00004438	B29D 85A4		000007A4	3551+	LFPC	FPCINIT	initialize FPC
				3552+*	LFPC	=XL4'08000000'	set trap exception

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
0000443C	E320 500C 0014		000043B4	3553+	LGF	R2,V2_60	get v2
00004442	E762 0000 0806		00000000	3554+	VL	V22,0(R2)	
00004448	E666 0000 1C5D			3555+	VCFN	V22,V22,1,0	test instruction (dest is source)
0000444E	B29C 5040		000043E8	3556+	STFPC	FPC_R_60	save FPC
00004452	E760 5028 080E		000043D0	3557+	VST	V22,V1060	save instruction result
00004458	07FB			3558+	BR	R11	return
0000445C				3559+RE60	DS	0F	expected 16 byte result
0000445C				3560+	DROP	R5	
0000445C	7E000000 7C000000			3561	DC	XL16'7E0000007C000000000000000000000000'	
00004464	00000000 00000000						
0000446C	7FFF0000 5E000000			3562	DC	XL16'7FFF00005E000000000000000000000000'	
00004474	00000000 00000000						
				3563			
				3564 *	Underflow,	-65536 (overflow)	skip xc - dropped digits
				3565	VRR_A	VCFN,1,0,30,23,S	
00004480				3566+	DS	0FD	
00004480		00004480		3567+	USING	*,R5	base for test data and test routine
00004480	00004510			3568+T61	DC	A(X61)	address of test routine
00004484	003D			3569+	DC	H'61'	test number
00004486	00			3570+	DC	X'00'	
00004487	E2			3571+	DC	CL1'S'	Y = skip cross check
00004488	01			3572+	DC	HL1'1'	m3
00004489	00			3573+	DC	HL1'0'	m4
0000448A	30			3574+FLG61	DC	X'30'	expected FPC flags
0000448B	23			3575+VXC61	DC	X'23'	expected VXC
0000448C	00004544			3576+V2_61	DC	A(RE61+16)	address of v2: 16-byte packed decimal
00004490	E5C3C6D5 40404040			3577+	DC	CL8'VCFN'	instruction name
00004498	00000010			3578+	DC	A(16)	result length
0000449C	00004534			3579+	DC	A(RE61)	address of expected resul
000044A0	00000000 00000000			3580+	DS	FD	gap
000044A8	00000000 00000000			3581+V1061	DS	XL16	V1 output
000044B0	00000000 00000000						
000044B8	00000000 00000000			3582+	DS	FD	gap
000044C0	00000000			3583+FPC_R_61	DS	F	FPC after instruction
000044C8	00000000 00000000			3584+	DS	FD	gap
000044D0	40404040 40404040			3585+	DC	CL8' '	was cross check skipped?
000044D8	00000000 00000000			3586+	DS	FD	gap
000044E0	00000000 00000000			3587+XC061	DS	XL16	Cross check Output
000044E8	00000000 00000000						
000044F0	00000000 00000000			3588+	DS	FD	gap
000044F8	00000000			3589+FPC_XC_61	DS	F	FPC after cross check
00004500	00000000 00000000			3590+	DS	FD	gap
00004508	00000000			3591+	DS	F	debug area
0000450C	00000000			3592+	DS	F	
				3593+*			
00004510				3594+X61	DS	0F	
00004510	B29D 85A4		000007A4	3595+	LFPC	FPCINIT	initialize FPC
				3596+*	LFPC	=XL4'08000000'	set trap exception
00004514	E320 500C 0014		0000448C	3597+	LGF	R2,V2_61	get v2
0000451A	E762 0000 0806		00000000	3598+	VL	V22,0(R2)	
00004520	E666 0000 1C5D			3599+	VCFN	V22,V22,1,0	test instruction (dest is source)
00004526	B29C 5040		000044C0	3600+	STFPC	FPC_R_61	save FPC
0000452A	E760 5028 080E		000044A8	3601+	VST	V22,V1061	save instruction result
00004530	07FB			3602+	BR	R11	return
00004534				3603+RE61	DS	0F	expected 16 byte result
00004534				3604+	DROP	R5	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00004534	00000000 7C000000			3605	DC	XL16'000000007C0000000000000000000000'
0000453C	00000000 00000000					
00004544	0A000000 5E000000			3606	DC	XL16'0A0000005E0000000000000000000000'
0000454C	00000000 00000000					
				3607		
				3608	* +Infinity, Underflow, -65536 (overflow) skip xc - infinity	
				3609	VRR_A VCFN,1,0,B0,43,S	
00004558				3610+	DS	0FD
00004558		00004558		3611+	USING	*,R5
00004558	000045E8			3612+T62	DC	A(X62)
0000455C	003E			3613+	DC	H'62'
0000455E	00			3614+	DC	X'00'
0000455F	E2			3615+	DC	CL1'S'
00004560	01			3616+	DC	HL1'1'
00004561	00			3617+	DC	HL1'0'
00004562	B0			3618+FLG62	DC	X'B0'
00004563	43			3619+VXC62	DC	X'43'
00004564	0000461C			3620+V2_62	DC	A(RE62+16)
00004568	E5C3C6D5 40404040			3621+	DC	CL8'VCFN'
00004570	00000010			3622+	DC	A(16)
00004574	0000460C			3623+	DC	A(RE62)
00004578	00000000 00000000			3624+	DS	FD
00004580	00000000 00000000			3625+V1062	DS	XL16
00004588	00000000 00000000					
00004590	00000000 00000000			3626+	DS	FD
00004598	00000000			3627+FPC_R_62	DS	F
000045A0	00000000 00000000			3628+	DS	FD
000045A8	40404040 40404040			3629+	DC	CL8' '
000045B0	00000000 00000000			3630+	DS	FD
000045B8	00000000 00000000			3631+XC062	DS	XL16
000045C0	00000000 00000000					
000045C8	00000000 00000000			3632+	DS	FD
000045D0	00000000			3633+FPC_XC_62	DS	F
000045D8	00000000 00000000			3634+	DS	FD
000045E0	00000000			3635+	DS	F
000045E4	00000000			3636+	DS	F
				3637+*		
000045E8				3638+X62	DS	0F
000045E8	B29D 85A4		000007A4	3639+	LFPC	FPCINIT
				3640+*	LFPC	=XL4'08000000'
000045EC	E320 500C 0014		00004564	3641+	LGF	R2,V2_62
000045F2	E762 0000 0806		00000000	3642+	VL	V22,0(R2)
000045F8	E666 0000 1C5D			3643+	VCFN	V22,V22,1,0
000045FE	B29C 5040		00004598	3644+	STFPC	FPC_R_62
00004602	E760 5028 080E		00004580	3645+	VST	V22,V1062
00004608	07FB			3646+	BR	R11
0000460C				3647+RE62	DS	0F
0000460C				3648+	DROP	R5
0000460C	7E000000 00000000			3649	DC	XL16'7E000000000000007C00000000000000'
00004614	7C000000 00000000					
0000461C	7FFF0000 0A000000			3650	DC	XL16'7FFF00000A0000005E00000000000000'
00004624	5E000000 00000000					
				3651		
				3652	*-----	
				3653	* VCLFNL - VECTOR FP CONVERT AND LENGTHEN FROM NNP LOW	
				3654	*-----	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				3655 * dlfloat -> short float (with cross check: short float -> dlfloat)
				3656 *
				3657
				3658 * +0, -0 simple instruction and 'test' test
				3659 VRR_A VCLFNL,2,0,00,00,N
00004630				3660+ DS 0FD
00004630		00004630		3661+ USING *,R5 base for test data and test routine
00004630	000046C0			3662+T63 DC A(X63) address of test routine
00004634	003F			3663+ DC H'63' test number
00004636	00			3664+ DC X'00'
00004637	D5			3665+ DC CL1'N' Y = skip cross check
00004638	02			3666+ DC HL1'2' m3
00004639	00			3667+ DC HL1'0' m4
0000463A	00			3668+FLG63 DC X'00' expected FPC flags
0000463B	00			3669+VXC63 DC X'00' expected VXC
0000463C	000046F4			3670+V2_63 DC A(RE63+16) address of v2: 16-byte packed decimal
00004640	E5C3D3C6 D5D34040			3671+ DC CL8'VCLFNL' instruction name
00004648	00000010			3672+ DC A(16) result length
0000464C	000046E4			3673+ DC A(RE63) address of expected resul
00004650	00000000 00000000			3674+ DS FD gap
00004658	00000000 00000000			3675+V1063 DS XL16 V1 output
00004660	00000000 00000000			
00004668	00000000 00000000			3676+ DS FD gap
00004670	00000000			3677+FPC_R_63 DS F FPC after instruction
00004678	00000000 00000000			3678+ DS FD gap
00004680	40404040 40404040			3679+ DC CL8' ' was cross check skipped?
00004688	00000000 00000000			3680+ DS FD gap
00004690	00000000 00000000			3681+XC063 DS XL16 Cross check Output
00004698	00000000 00000000			
000046A0	00000000 00000000			3682+ DS FD gap
000046A8	00000000			3683+FPC_XC_63 DS F FPC after cross check
000046B0	00000000 00000000			3684+ DS FD gap
000046B8	00000000			3685+ DS F debug area
000046BC	00000000			3686+ DS F
				3687+*
000046C0				3688+X63 DS 0F
000046C0	B29D 85A4		000007A4	3689+ LFPC FPCINIT initialize FPC
				3690+* LFPC =XL4'08000000' set trap exception
000046C4	E320 500C 0014		0000463C	3691+ LGF R2,V2_63 get v2
000046CA	E762 0000 0806		00000000	3692+ VL V22,0(R2)
000046D0	E666 0000 2C5E			3693+ VCLFNL V22,V22,2,0 test instruction (dest is source)
000046D6	B29C 5040		00004670	3694+ STFPC FPC_R_63 save FPC
000046DA	E760 5028 080E		00004658	3695+ VST V22,V1063 save instruction result
000046E0	07FB			3696+ BR R11 return
000046E4				3697+RE63 DS 0F expected 16 byte result
000046E4				3698+ DROP R5
000046E4	00000000 80000000			3699 DC XL16'00000000800000000000000000000000'
000046EC	00000000 00000000			
000046F4	00000000 00000000			3700 DC XL16'00000000000000000000800000000000'
000046FC	00008000 00000000			
				3701
				3702 * +1, -1
				3703
				VRR_A VCLFNL,2,0,00,00,N
00004708				3704+ DS 0FD
00004708		00004708		3705+ USING *,R5 base for test data and test routine
00004708	00004798			3706+T64 DC A(X64) address of test routine

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
0000470C	0040			3707+	DC	H'64' test number
0000470E	00			3708+	DC	X'00'
0000470F	D5			3709+	DC	CL1'N' Y = skip cross check
00004710	02			3710+	DC	HL1'2' m3
00004711	00			3711+	DC	HL1'0' m4
00004712	00			3712+FLG64	DC	X'00' expected FPC flags
00004713	00			3713+VXC64	DC	X'00' expected VXC
00004714	000047CC			3714+V2_64	DC	A(RE64+16) address of v2: 16-byte packed decimal
00004718	E5C3D3C6 D5D34040			3715+	DC	CL8'VCLFNL' instruction name
00004720	00000010			3716+	DC	A(16) result length
00004724	000047BC			3717+	DC	A(RE64) address of expected resul
00004728	00000000 00000000			3718+	DS	FD gap
00004730	00000000 00000000			3719+V1064	DS	XL16 V1 output
00004738	00000000 00000000					
00004740	00000000 00000000			3720+	DS	FD gap
00004748	00000000			3721+FPC_R_64	DS	F FPC after instruction
00004750	00000000 00000000			3722+	DS	FD gap
00004758	40404040 40404040			3723+	DC	CL8' ' was cross check skipped?
00004760	00000000 00000000			3724+	DS	FD gap
00004768	00000000 00000000			3725+XC064	DS	XL16 Cross check Output
00004770	00000000 00000000					
00004778	00000000 00000000			3726+	DS	FD gap
00004780	00000000			3727+FPC_XC_64	DS	F FPC after cross check
00004788	00000000 00000000			3728+	DS	FD gap
00004790	00000000			3729+	DS	F debug area
00004794	00000000			3730+	DS	F
				3731+*		
00004798				3732+X64	DS	0F
00004798	B29D 85A4		000007A4	3733+	LFPC	FPCINIT initialize FPC
				3734+*	LFPC	=XL4'08000000' set trap exception
0000479C	E320 500C 0014		00004714	3735+	LGF	R2,V2_64 get v2
000047A2	E762 0000 0806		00000000	3736+	VL	V22,0(R2)
000047A8	E666 0000 2C5E			3737+	VCLFNL	V22,V22,2,0 test instruction (dest is source)
000047AE	B29C 5040		00004748	3738+	STFPC	FPC_R_64 save FPC
000047B2	E760 5028 080E		00004730	3739+	VST	V22,V1064 save instruction result
000047B8	07FB			3740+	BR	R11 return
000047BC				3741+RE64	DS	0F expected 16 byte result
000047BC				3742+	DROP	R5
000047BC	3F800000 BF800000			3743	DC	XL16'3F800000BF80000000000000000000000'
000047C4	00000000 00000000					
000047CC	00000000 00000000			3744	DC	XL16'000000000000000003E00BE0000000000'
000047D4	3E00BE00 00000000					
				3745		
				3746 * +0.5, -0.5		
				3747	VRR_A	VCLFNL,2,0,00,00,N
000047E0				3748+	DS	0FD
000047E0		000047E0		3749+	USING	*,R5 base for test data and test routine
000047E0	00004870			3750+T65	DC	A(X65) address of test routine
000047E4	0041			3751+	DC	H'65' test number
000047E6	00			3752+	DC	X'00'
000047E7	D5			3753+	DC	CL1'N' Y = skip cross check
000047E8	02			3754+	DC	HL1'2' m3
000047E9	00			3755+	DC	HL1'0' m4
000047EA	00			3756+FLG65	DC	X'00' expected FPC flags
000047EB	00			3757+VXC65	DC	X'00' expected VXC
000047EC	000048A4			3758+V2_65	DC	A(RE65+16) address of v2: 16-byte packed decimal

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000047F0	E5C3D3C6 D5D34040			3759+	DC	CL8'VCLFNL'	instruction name
000047F8	00000010			3760+	DC	A(16)	result length
000047FC	00004894			3761+	DC	A(RE65)	address of expected resul
00004800	00000000 00000000			3762+	DS	FD	gap
00004808	00000000 00000000			3763+V1065	DS	XL16	V1 output
00004810	00000000 00000000						
00004818	00000000 00000000			3764+	DS	FD	gap
00004820	00000000			3765+FPC_R_65	DS	F	FPC after instruction
00004828	00000000 00000000			3766+	DS	FD	gap
00004830	40404040 40404040			3767+	DC	CL8' '	was cross check skipped?
00004838	00000000 00000000			3768+	DS	FD	gap
00004840	00000000 00000000			3769+XC065	DS	XL16	Cross check Output
00004848	00000000 00000000						
00004850	00000000 00000000			3770+	DS	FD	gap
00004858	00000000			3771+FPC_XC_65	DS	F	FPC after cross check
00004860	00000000 00000000			3772+	DS	FD	gap
00004868	00000000			3773+	DS	F	debug area
0000486C	00000000			3774+	DS	F	
				3775+*			
00004870				3776+X65	DS	0F	
00004870	B29D 85A4		000007A4	3777+	LFPC	FPCINIT	initialize FPC
				3778+*	LFPC	=XL4'08000000'	set trap exception
00004874	E320 500C 0014		000047EC	3779+	LGF	R2,V2_65	get v2
0000487A	E762 0000 0806		00000000	3780+	VL	V22,0(R2)	
00004880	E666 0000 2C5E			3781+	VCLFNL	V22,V22,2,0	test instruction (dest is source)
00004886	B29C 5040		00004820	3782+	STFPC	FPC_R_65	save FPC
0000488A	E760 5028 080E		00004808	3783+	VST	V22,V1065	save instruction result
00004890	07FB			3784+	BR	R11	return
00004894				3785+RE65	DS	0F	expected 16 byte result
00004894				3786+	DROP	R5	
00004894	3F000000 BF000000			3787	DC	XL16'3F000000BF00000000000000000000000'	
0000489C	00000000 00000000						
000048A4	00000000 00000000			3788	DC	XL16'000000000000000000003C00BC00000000000'	
000048AC	3C00BC00 00000000						
				3789			
				3790 * 1/64, -1/64			
				3791	VRR_A	VCLFNL,2,0,00,00,N	
000048B8				3792+	DS	0FD	
000048B8		000048B8		3793+	USING	*,R5	base for test data and test routine
000048B8	00004948			3794+T66	DC	A(X66)	address of test routine
000048BC	0042			3795+	DC	H'66'	test number
000048BE	00			3796+	DC	X'00'	
000048BF	D5			3797+	DC	CL1'N'	Y = skip cross check
000048C0	02			3798+	DC	HL1'2'	m3
000048C1	00			3799+	DC	HL1'0'	m4
000048C2	00			3800+FLG66	DC	X'00'	expected FPC flags
000048C3	00			3801+VXC66	DC	X'00'	expected VXC
000048C4	0000497C			3802+V2_66	DC	A(RE66+16)	address of v2: 16-byte packed decimal
000048C8	E5C3D3C6 D5D34040			3803+	DC	CL8'VCLFNL'	instruction name
000048D0	00000010			3804+	DC	A(16)	result length
000048D4	0000496C			3805+	DC	A(RE66)	address of expected resul
000048D8	00000000 00000000			3806+	DS	FD	gap
000048E0	00000000 00000000			3807+V1066	DS	XL16	V1 output
000048E8	00000000 00000000						
000048F0	00000000 00000000			3808+	DS	FD	gap
000048F8	00000000			3809+FPC_R_66	DS	F	FPC after instruction

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00004A18	00000000			3861+	DS	F	debug area
00004A1C	00000000			3862+	DS	F	
				3863+*			
00004A20				3864+X67	DS	0F	
00004A20	B29D 85A4		000007A4	3865+	LFPC	FPCINIT	initialize FPC
				3866+*	LFPC	=XL4'08000000'	set trap exception
00004A24	E320 500C 0014		0000499C	3867+	LGF	R2,V2_67	get v2
00004A2A	E762 0000 0806		00000000	3868+	VL	V22,0(R2)	
00004A30	E666 0000 2C5E			3869+	VCLFNL	V22,V22,2,0	test instruction (dest is source)
00004A36	B29C 5040		000049D0	3870+	STFPC	FPC_R_67	save FPC
00004A3A	E760 5028 080E		000049B8	3871+	VST	V22,V1067	save instruction result
00004A40	07FB			3872+	BR	R11	return
00004A44				3873+RE67	DS	0F	expected 16 byte result
00004A44				3874+	DROP	R5	
00004A44	41700000 C1700000			3875	DC	XL16'41700000C1700000000000000000000000'	
00004A4C	00000000 00000000						
00004A54	00000000 00000000			3876	DC	XL16'0000000000000000000045C0C5C000000000'	
00004A5C	45C0C5C0 00000000						
				3877			
				3878 * +20/7, -20/7		(about)	
				3879	VRR_A	VCLFNL,2,0,00,00,N	
00004A68				3880+	DS	0FD	
00004A68		00004A68		3881+	USING	*,R5	base for test data and test routine
00004A68	00004AF8			3882+T68	DC	A(X68)	address of test routine
00004A6C	0044			3883+	DC	H'68'	test number
00004A6E	00			3884+	DC	X'00'	
00004A6F	D5			3885+	DC	CL1'N'	Y = skip cross check
00004A70	02			3886+	DC	HL1'2'	m3
00004A71	00			3887+	DC	HL1'0'	m4
00004A72	00			3888+FLG68	DC	X'00'	expected FPC flags
00004A73	00			3889+VXC68	DC	X'00'	expected VXC
00004A74	00004B2C			3890+V2_68	DC	A(RE68+16)	address of v2: 16-byte packed decimal
00004A78	E5C3D3C6 D5D34040			3891+	DC	CL8'VCLFNL'	instruction name
00004A80	00000010			3892+	DC	A(16)	result length
00004A84	00004B1C			3893+	DC	A(RE68)	address of expected resul
00004A88	00000000 00000000			3894+	DS	FD	gap
00004A90	00000000 00000000			3895+V1068	DS	XL16	V1 output
00004A98	00000000 00000000						
00004AA0	00000000 00000000			3896+	DS	FD	gap
00004AA8	00000000			3897+FPC_R_68	DS	F	FPC after instruction
00004AB0	00000000 00000000			3898+	DS	FD	gap
00004AB8	40404040 40404040			3899+	DC	CL8' '	was cross check skipped?
00004AC0	00000000 00000000			3900+	DS	FD	gap
00004AC8	00000000 00000000			3901+XC068	DS	XL16	Cross check Output
00004AD0	00000000 00000000						
00004AD8	00000000 00000000			3902+	DS	FD	gap
00004AE0	00000000			3903+FPC_XC_68	DS	F	FPC after cross check
00004AE8	00000000 00000000			3904+	DS	FD	gap
00004AF0	00000000			3905+	DS	F	debug area
00004AF4	00000000			3906+	DS	F	
				3907+*			
00004AF8				3908+X68	DS	0F	
00004AF8	B29D 85A4		000007A4	3909+	LFPC	FPCINIT	initialize FPC
				3910+*	LFPC	=XL4'08000000'	set trap exception
00004AFC	E320 500C 0014		00004A74	3911+	LGF	R2,V2_68	get v2
00004B02	E762 0000 0806		00000000	3912+	VL	V22,0(R2)	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00004B08	E666 0000 2C5E			3913+	VCLFNL V22,V22,2,0	test instruction (dest is source)
00004B0E	B29C 5040		00004AA8	3914+	STFPC FPC_R_68	save FPC
00004B12	E760 5028 080E		00004A90	3915+	VST V22,V1068	save instruction result
00004B18	07FB			3916+	BR R11	return
00004B1C				3917+RE68	DS 0F	expected 16 byte result
00004B1C				3918+	DROP R5	
00004B1C	4036C000 C036C000			3919	DC XL16'4036C000C036C000000000000000000000'	
00004B24	00000000 00000000					
00004B2C	00000000 00000000			3920	DC XL16'000000000000000040DBC0DB00000000'	
00004B34	40DBC0DB 00000000					
				3921		
				3922	* additional tests	
				3923	* max tiny: (1 - 2^-11) * 2^16 (65504)	
				3924	VRR_A VCLFNL,2,0,00,00,N	
00004B40				3925+	DS 0FD	
00004B40		00004B40		3926+	USING *,R5	base for test data and test routine
00004B40	00004BD0			3927+T69	DC A(X69)	address of test routine
00004B44	0045			3928+	DC H'69'	test number
00004B46	00			3929+	DC X'00'	
00004B47	D5			3930+	DC CL1'N'	Y = skip cross check
00004B48	02			3931+	DC HL1'2'	m3
00004B49	00			3932+	DC HL1'0'	m4
00004B4A	00			3933+FLG69	DC X'00'	expected FPC flags
00004B4B	00			3934+VXC69	DC X'00'	expected VXC
00004B4C	00004C04			3935+V2_69	DC A(RE69+16)	address of v2: 16-byte packed decimal
00004B50	E5C3D3C6 D5D34040			3936+	DC CL8'VCLFNL'	instruction name
00004B58	00000010			3937+	DC A(16)	result length
00004B5C	00004BF4			3938+	DC A(RE69)	address of expected resul
00004B60	00000000 00000000			3939+	DS FD	gap
00004B68	00000000 00000000			3940+V1069	DS XL16	V1 output
00004B70	00000000 00000000					
00004B78	00000000 00000000			3941+	DS FD	gap
00004B80	00000000			3942+FPC_R_69	DS F	FPC after instruction
00004B88	00000000 00000000			3943+	DS FD	gap
00004B90	40404040 40404040			3944+	DC CL8' '	was cross check skipped?
00004B98	00000000 00000000			3945+	DS FD	gap
00004BA0	00000000 00000000			3946+XC069	DS XL16	Cross check Output
00004BA8	00000000 00000000					
00004BB0	00000000 00000000			3947+	DS FD	gap
00004BB8	00000000			3948+FPC_XC_69	DS F	FPC after cross check
00004BC0	00000000 00000000			3949+	DS FD	gap
00004BC8	00000000			3950+	DS F	debug area
00004BCC	00000000			3951+	DS F	
				3952+*		
00004BD0				3953+X69	DS 0F	
00004BD0	B29D 85A4		000007A4	3954+	LFPC FPCINIT	initialize FPC
				3955+*	LFPC =XL4'08000000'	set trap exception
00004BD4	E320 500C 0014		00004B4C	3956+	LGF R2,V2_69	get v2
00004BDA	E762 0000 0806		00000000	3957+	VL V22,0(R2)	
00004BE0	E666 0000 2C5E			3958+	VCLFNL V22,V22,2,0	test instruction (dest is source)
00004BE6	B29C 5040		00004B80	3959+	STFPC FPC_R_69	save FPC
00004BEA	E760 5028 080E		00004B68	3960+	VST V22,V1069	save instruction result
00004BF0	07FB			3961+	BR R11	return
00004BF4				3962+RE69	DS 0F	expected 16 byte result
00004BF4				3963+	DROP R5	
00004BF4	47800000 00000000			3964	DC XL16'47800000000000000000000000000000'	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT	
00004BFC	00000000 00000000				
00004C04	00000000 00000000			3965	DC XL16'000000000000000005E0000000000000'
00004C0C	5E000000 00000000				
				3966	
				3967 * max	dlfloat: 2^(33)-2ulp
00004C18				3968	VRR_A VCLFNL,2,0,00,00,N
00004C18		00004C18		3969+	DS 0FD
00004C18	00004CA8			3970+	USING *,R5
00004C1C	0046			3971+T70	DC A(X70)
00004C1E	00			3972+	DC H'70'
00004C1F	D5			3973+	DC X'00'
00004C20	02			3974+	DC CL1'N'
00004C21	00			3975+	DC HL1'2'
00004C22	00			3976+	DC HL1'0'
00004C22	00			3977+FLG70	DC X'00'
00004C23	00			3978+VXC70	DC X'00'
00004C24	00004CDC			3979+V2_70	DC A(RE70+16)
00004C28	E5C3D3C6 D5D34040			3980+	DC CL8'VCLFNL'
00004C30	00000010			3981+	DC A(16)
00004C34	00004CCC			3982+	DC A(RE70)
00004C38	00000000 00000000			3983+	DS FD
00004C40	00000000 00000000			3984+V1070	DS XL16
00004C48	00000000 00000000				
00004C50	00000000 00000000			3985+	DS FD
00004C58	00000000			3986+FPC_R_70	DS F
00004C60	00000000 00000000			3987+	DS FD
00004C68	40404040 40404040			3988+	DC CL8' '
00004C70	00000000 00000000			3989+	DS FD
00004C78	00000000 00000000			3990+XC070	DS XL16
00004C80	00000000 00000000				
00004C88	00000000 00000000			3991+	DS FD
00004C90	00000000			3992+FPC_XC_70	DS F
00004C98	00000000 00000000			3993+	DS FD
00004CA0	00000000			3994+	DS F
00004CA4	00000000			3995+	DS F
				3996+*	
00004CA8				3997+X70	DS 0F
00004CA8	B29D 85A4		000007A4	3998+	LFPC FPCINIT
				3999+*	LFPC =XL4'08000000'
00004CAC	E320 500C 0014		00004C24	4000+	LGF R2,V2_70
00004CB2	E762 0000 0806		00000000	4001+	VL V22,0(R2)
00004CB8	E666 0000 2C5E			4002+	VCLFNL V22,V22,2,0
00004CBE	B29C 5040		00004C58	4003+	STFPC FPC_R_70
00004CC2	E760 5028 080E		00004C40	4004+	VST V22,V1070
00004CC8	07FB			4005+	BR R11
00004CCC				4006+RE70	DS 0F
00004CCC				4007+	DROP R5
00004CCC	4FFF8000 00000000			4008	DC XL16'4FFF80000000000000000000000000'
00004CD4	00000000 00000000				
00004CDC	00000000 00000000			4009	DC XL16'00000000000000007FFE000000000000'
00004CE4	7FFE0000 00000000				
				4010	
				4011 * min	dlfloat: 2^(-31)*+ulp
				4012	VRR_A VCLFNL,2,0,00,00,N
00004CF0				4013+	DS 0FD
00004CF0		00004CF0		4014+	USING *,R5

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00004CF0	00004D80			4015+T71	DC	A(X71)
00004CF4	0047			4016+	DC	H'71'
00004CF6	00			4017+	DC	X'00'
00004CF7	D5			4018+	DC	CL1'N'
00004CF8	02			4019+	DC	HL1'2'
00004CF9	00			4020+	DC	HL1'0'
00004CFA	00			4021+FLG71	DC	X'00'
00004CFB	00			4022+VXC71	DC	X'00'
00004CFC	00004DB4			4023+V2_71	DC	A(RE71+16)
00004D00	E5C3D3C6 D5D34040			4024+	DC	CL8'VCLFNL'
00004D08	00000010			4025+	DC	A(16)
00004D0C	00004DA4			4026+	DC	A(RE71)
00004D10	00000000 00000000			4027+	DS	FD
00004D18	00000000 00000000			4028+V1071	DS	XL16
00004D20	00000000 00000000					
00004D28	00000000 00000000			4029+	DS	FD
00004D30	00000000			4030+FPC_R_71	DS	F
00004D38	00000000 00000000			4031+	DS	FD
00004D40	40404040 40404040			4032+	DC	CL8' '
00004D48	00000000 00000000			4033+	DS	FD
00004D50	00000000 00000000			4034+XC071	DS	XL16
00004D58	00000000 00000000					
00004D60	00000000 00000000			4035+	DS	FD
00004D68	00000000			4036+FPC_XC_71	DS	F
00004D70	00000000 00000000			4037+	DS	FD
00004D78	00000000			4038+	DS	F
00004D7C	00000000			4039+	DS	F
				4040+*		
00004D80				4041+X71	DS	0F
00004D80	B29D 85A4		000007A4	4042+	LFPC	FPCINIT
				4043+*	LFPC	=XL4'08000000'
00004D84	E320 500C 0014		00004CFC	4044+	LGF	R2,V2_71
00004D8A	E762 0000 0806		00000000	4045+	VL	V22,0(R2)
00004D90	E666 0000 2C5E			4046+	VCLFNL	V22,V22,2,0
00004D96	B29C 5040		00004D30	4047+	STFPC	FPC_R_71
00004D9A	E760 5028 080E		00004D18	4048+	VST	V22,V1071
00004DA0	07FB			4049+	BR	R11
00004DA4				4050+RE71	DS	0F
00004DA4				4051+	DROP	R5
00004DA4	30004000 00000000			4052	DC	XL16'30004000000000000000000000000000'
00004DAC	00000000 00000000					
00004DB4	00000000 00000000			4053	DC	XL16'0000000000000000000010000000000000'
00004DBC	00010000 00000000					
				4054		
				4055 * +Infinity, -Infinity		
				4056	VRR_A	VCLFNL,2,0,80,61,S
00004DC8				4057+	DS	0FD
00004DC8		00004DC8		4058+	USING	*,R5
00004DC8	00004E58			4059+T72	DC	A(X72)
00004DCC	0048			4060+	DC	H'72'
00004DCE	00			4061+	DC	X'00'
00004DCF	E2			4062+	DC	CL1'S'
00004DD0	02			4063+	DC	HL1'2'
00004DD1	00			4064+	DC	HL1'0'
00004DD2	80			4065+FLG72	DC	X'80'
00004DD3	61			4066+VXC72	DC	X'61'

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00004DD4	00004E8C			4067+V2_72	DC	A(RE72+16)
00004DD8	E5C3D3C6 D5D34040			4068+	DC	CL8'VCLFNL'
00004DE0	00000010			4069+	DC	A(16)
00004DE4	00004E7C			4070+	DC	A(RE72)
00004DE8	00000000 00000000			4071+	DS	FD
00004DF0	00000000 00000000			4072+V1072	DS	XL16
00004DF8	00000000 00000000					
00004E00	00000000 00000000			4073+	DS	FD
00004E08	00000000			4074+FPC_R_72	DS	F
00004E10	00000000 00000000			4075+	DS	FD
00004E18	40404040 40404040			4076+	DC	CL8' '
00004E20	00000000 00000000			4077+	DS	FD
00004E28	00000000 00000000			4078+XC072	DS	XL16
00004E30	00000000 00000000					
00004E38	00000000 00000000			4079+	DS	FD
00004E40	00000000			4080+FPC_XC_72	DS	F
00004E48	00000000 00000000			4081+	DS	FD
00004E50	00000000			4082+	DS	F
00004E54	00000000			4083+	DS	F
				4084+*		
00004E58				4085+X72	DS	0F
00004E58	B29D 85A4		000007A4	4086+	LFPC	FPCINIT
				4087+*	LFPC	=XL4'08000000'
00004E5C	E320 500C 0014		00004DD4	4088+	LGF	R2,V2_72
00004E62	E762 0000 0806		00000000	4089+	VL	V22,0(R2)
00004E68	E666 0000 2C5E			4090+	VCLFNL	V22,V22,2,0
00004E6E	B29C 5040		00004E08	4091+	STFPC	FPC_R_72
00004E72	E760 5028 080E		00004DF0	4092+	VST	V22,V1072
00004E78	07FB			4093+	BR	R11
00004E7C				4094+RE72	DS	0F
00004E7C				4095+	DROP	R5
00004E7C	7FC00000 00000000			4096	DC	XL16'7FC0000000000000FFC000000000000'
00004E84	FFC00000 00000000					
00004E8C	00000000 00000000			4097	DC	XL16'00000000000000007FFF0000FFFF0000'
00004E94	7FFF0000 FFFF0000					
				4098		
				4099 * inexact - bad m3		
				4100	VRR_A	VCLFNL,5,0,08,05,S
00004EA0				4101+	DS	0FD
00004EA0		00004EA0		4102+	USING	*,R5
00004EA0	00004F30			4103+T73	DC	A(X73)
00004EA4	0049			4104+	DC	H'73'
00004EA6	00			4105+	DC	X'00'
00004EA7	E2			4106+	DC	CL1'S'
00004EA8	05			4107+	DC	HL1'5'
00004EA9	00			4108+	DC	HL1'0'
00004EAA	08			4109+FLG73	DC	X'08'
00004EAB	05			4110+VXC73	DC	X'05'
00004EAC	00004F64			4111+V2_73	DC	A(RE73+16)
00004EB0	E5C3D3C6 D5D34040			4112+	DC	CL8'VCLFNL'
00004EB8	00000010			4113+	DC	A(16)
00004EBC	00004F54			4114+	DC	A(RE73)
00004EC0	00000000 00000000			4115+	DS	FD
00004EC8	00000000 00000000			4116+V1073	DS	XL16
00004ED0	00000000 00000000					
00004ED8	00000000 00000000			4117+	DS	FD

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00004EE0	00000000			4118+FPC_R_73	DS	F	FPC after instruction
00004EE8	00000000	00000000		4119+	DS	FD	gap
00004EF0	40404040	40404040		4120+	DC	CL8' '	was cross check skipped?
00004EF8	00000000	00000000		4121+	DS	FD	gap
00004F00	00000000	00000000		4122+XC073	DS	XL16	Cross check Output
00004F08	00000000	00000000					
00004F10	00000000	00000000		4123+	DS	FD	gap
00004F18	00000000			4124+FPC_XC_73	DS	F	FPC after cross check
00004F20	00000000	00000000		4125+	DS	FD	gap
00004F28	00000000			4126+	DS	F	debug area
00004F2C	00000000			4127+	DS	F	
				4128+*			
00004F30				4129+X73	DS	0F	
00004F30	B29D 85A4		000007A4	4130+	LFPC	FPCINIT	initialize FPC
				4131+*	LFPC	=XL4'08000000'	set trap exception
00004F34	E320 500C 0014		00004EAC	4132+	LGF	R2,V2_73	get v2
00004F3A	E762 0000 0806		00000000	4133+	VL	V22,0(R2)	
00004F40	E666 0000 5C5E			4134+	VCLFNL	V22,V22,5,0	test instruction (dest is source)
00004F46	B29C 5040		00004EE0	4135+	STFPC	FPC_R_73	save FPC
00004F4A	E760 5028 080E		00004EC8	4136+	VST	V22,V1073	save instruction result
00004F50	07FB			4137+	BR	R11	return
00004F54				4138+RE73	DS	0F	expected 16 byte result
00004F54				4139+	DROP	R5	
00004F54	00000000	00000000		4140	DC	XL16'00000000000000000000000000000000'	
00004F5C	00000000	00000000					
00004F64	00000000	0000F000		4141	DC	XL16'0000000000000F0000000100000000000000'	
00004F6C	00010000	00000000					
				4142			
				4143 * inexact - bad m4			
				4144	VRR_A	VCLFNL,2,5,08,05,S	skip xc - inexact
00004F78				4145+	DS	0FD	
00004F78		00004F78		4146+	USING	*,R5	base for test data and test routine
00004F78	00005008			4147+T74	DC	A(X74)	address of test routine
00004F7C	004A			4148+	DC	H'74'	test number
00004F7E	00			4149+	DC	X'00'	
00004F7F	E2			4150+	DC	CL1'S'	Y = skip cross check
00004F80	02			4151+	DC	HL1'2'	m3
00004F81	05			4152+	DC	HL1'5'	m4
00004F82	08			4153+FLG74	DC	X'08'	expected FPC flags
00004F83	05			4154+VXC74	DC	X'05'	expected VXC
00004F84	0000503C			4155+V2_74	DC	A(RE74+16)	address of v2: 16-byte packed decimal
00004F88	E5C3D3C6 D5D34040			4156+	DC	CL8'VCLFNL'	instruction name
00004F90	00000010			4157+	DC	A(16)	result length
00004F94	0000502C			4158+	DC	A(RE74)	address of expected resul
00004F98	00000000	00000000		4159+	DS	FD	gap
00004FA0	00000000	00000000		4160+V1074	DS	XL16	V1 output
00004FA8	00000000	00000000					
00004FB0	00000000	00000000		4161+	DS	FD	gap
00004FB8	00000000			4162+FPC_R_74	DS	F	FPC after instruction
00004FC0	00000000	00000000		4163+	DS	FD	gap
00004FC8	40404040	40404040		4164+	DC	CL8' '	was cross check skipped?
00004FD0	00000000	00000000		4165+	DS	FD	gap
00004FD8	00000000	00000000		4166+XC074	DS	XL16	Cross check Output
00004FE0	00000000	00000000					
00004FE8	00000000	00000000		4167+	DS	FD	gap
00004FF0	00000000			4168+FPC_XC_74	DS	F	FPC after cross check

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00004FF8	00000000 00000000			4169+	DS	FD	gap
00005000	00000000			4170+	DS	F	debug area
00005004	00000000			4171+	DS	F	
				4172+*			
00005008				4173+X74	DS	0F	
00005008	B29D 85A4		000007A4	4174+	LFPC	FPCINIT	initialize FPC
				4175+*	LFPC	=XL4'08000000'	set trap exception
0000500C	E320 500C 0014		00004F84	4176+	LGF	R2,V2_74	get v2
00005012	E762 0000 0806		00000000	4177+	VL	V22,0(R2)	
00005018	E666 0005 2C5E			4178+	VCLFNL	V22,V22,2,5	test instruction (dest is source)
0000501E	B29C 5040		00004FB8	4179+	STFPC	FPC_R_74	save FPC
00005022	E760 5028 080E		00004FA0	4180+	VST	V22,V1074	save instruction result
00005028	07FB			4181+	BR	R11	return
0000502C				4182+RE74	DS	0F	expected 16 byte result
0000502C				4183+	DROP	R5	
0000502C	00000000 00000000			4184	DC	XL16'00000000000000000000000000000000'	
00005034	00000000 00000000						
0000503C	00000000 0000F000			4185	DC	XL16'0000000000000F0000000100000000000000'	
00005044	00010000 00000000						
				4186			
				4187 * more tests			
				4188 * +3.140625, -3.140625			
				4189	VRR_A	VCLFNL,2,0,00,00,N	
00005050				4190+	DS	0FD	
00005050		00005050		4191+	USING	*,R5	base for test data and test routine
00005050	000050E0			4192+T75	DC	A(X75)	address of test routine
00005054	004B			4193+	DC	H'75'	test number
00005056	00			4194+	DC	X'00'	
00005057	D5			4195+	DC	CL1'N'	Y = skip cross check
00005058	02			4196+	DC	HL1'2'	m3
00005059	00			4197+	DC	HL1'0'	m4
0000505A	00			4198+FLG75	DC	X'00'	expected FPC flags
0000505B	00			4199+VXC75	DC	X'00'	expected VXC
0000505C	00005114			4200+V2_75	DC	A(RE75+16)	address of v2: 16-byte packed decimal
00005060	E5C3D3C6 D5D34040			4201+	DC	CL8'VCLFNL'	instruction name
00005068	00000010			4202+	DC	A(16)	result length
0000506C	00005104			4203+	DC	A(RE75)	address of expected resul
00005070	00000000 00000000			4204+	DS	FD	gap
00005078	00000000 00000000			4205+V1075	DS	XL16	V1 output
00005080	00000000 00000000						
00005088	00000000 00000000			4206+	DS	FD	gap
00005090	00000000			4207+FPC_R_75	DS	F	FPC after instruction
00005098	00000000 00000000			4208+	DS	FD	gap
000050A0	40404040 40404040			4209+	DC	CL8' '	was cross check skipped?
000050A8	00000000 00000000			4210+	DS	FD	gap
000050B0	00000000 00000000			4211+XC075	DS	XL16	Cross check Output
000050B8	00000000 00000000						
000050C0	00000000 00000000			4212+	DS	FD	gap
000050C8	00000000			4213+FPC_XC_75	DS	F	FPC after cross check
000050D0	00000000 00000000			4214+	DS	FD	gap
000050D8	00000000			4215+	DS	F	debug area
000050DC	00000000			4216+	DS	F	
				4217+*			
000050E0				4218+X75	DS	0F	
000050E0	B29D 85A4		000007A4	4219+	LFPC	FPCINIT	initialize FPC
				4220+*	LFPC	=XL4'08000000'	set trap exception

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000050E4	E320 500C 0014		0000505C	4221+	LGF	R2,V2_75	get v2
000050EA	E762 0000 0806		00000000	4222+	VL	V22,0(R2)	
000050F0	E666 0000 2C5E			4223+	VCLFNL	V22,V22,2,0	test instruction (dest is source)
000050F6	B29C 5040		00005090	4224+	STFPC	FPC_R_75	save FPC
000050FA	E760 5028 080E		00005078	4225+	VST	V22,V1075	save instruction result
00005100	07FB			4226+	BR	R11	return
00005104				4227+RE75	DS	0F	expected 16 byte result
00005104				4228+	DROP	R5	
00005104	40890000 00000000			4229	DC	XL16'4089000000000000C089000000000000'	
0000510C	C0890000 00000000						
00005114	00000000 00000000			4230	DC	XL16'000000000000000042240000C2240000'	
0000511C	42240000 C2240000						
				4231			
				4232			
				4233	*	+27.15625, -27.15625	
				4234	VRR_A	VCLFNL,2,0,00,00,N	
00005128				4235+	DS	0FD	
00005128		00005128		4236+	USING	*,R5	base for test data and test routine
00005128	000051B8			4237+T76	DC	A(X76)	address of test routine
0000512C	004C			4238+	DC	H'76'	test number
0000512E	00			4239+	DC	X'00'	
0000512F	D5			4240+	DC	CL1'N'	Y = skip cross check
00005130	02			4241+	DC	HL1'2'	m3
00005131	00			4242+	DC	HL1'0'	m4
00005132	00			4243+FLG76	DC	X'00'	expected FPC flags
00005133	00			4244+VXC76	DC	X'00'	expected VXC
00005134	000051EC			4245+V2_76	DC	A(RE76+16)	address of v2: 16-byte packed decimal
00005138	E5C3D3C6 D5D34040			4246+	DC	CL8'VCLFNL'	instruction name
00005140	00000010			4247+	DC	A(16)	result length
00005144	000051DC			4248+	DC	A(RE76)	address of expected resul
00005148	00000000 00000000			4249+	DS	FD	gap
00005150	00000000 00000000			4250+V1076	DS	XL16	V1 output
00005158	00000000 00000000						
00005160	00000000 00000000			4251+	DS	FD	gap
00005168	00000000			4252+FPC_R_76	DS	F	FPC after instruction
00005170	00000000 00000000			4253+	DS	FD	gap
00005178	40404040 40404040			4254+	DC	CL8' '	was cross check skipped?
00005180	00000000 00000000			4255+	DS	FD	gap
00005188	00000000 00000000			4256+XC076	DS	XL16	Cross check Output
00005190	00000000 00000000						
00005198	00000000 00000000			4257+	DS	FD	gap
000051A0	00000000			4258+FPC_XC_76	DS	F	FPC after cross check
000051A8	00000000 00000000			4259+	DS	FD	gap
000051B0	00000000			4260+	DS	F	debug area
000051B4	00000000			4261+	DS	F	
				4262+*			
000051B8				4263+X76	DS	0F	
000051B8	B29D 85A4		000007A4	4264+	LFPC	FPCINIT	initialize FPC
				4265+*	LFPC	=XL4'08000000'	set trap exception
000051BC	E320 500C 0014		00005134	4266+	LGF	R2,V2_76	get v2
000051C2	E762 0000 0806		00000000	4267+	VL	V22,0(R2)	
000051C8	E666 0000 2C5E			4268+	VCLFNL	V22,V22,2,0	test instruction (dest is source)
000051CE	B29C 5040		00005168	4269+	STFPC	FPC_R_76	save FPC
000051D2	E760 5028 080E		00005150	4270+	VST	V22,V1076	save instruction result
000051D8	07FB			4271+	BR	R11	return
000051DC				4272+RE76	DS	0F	expected 16 byte result

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
000051DC				4273+	DROP R5	
000051DC	3BD94000 00000000			4274	DC	XL16'3BD9400000000000BBD940000000000'
000051E4	BBD94000 00000000					
000051EC	00000000 00000000			4275	DC	XL16'00000000000000002F650000AF650000'
000051F4	2F650000 AF650000					
				4276		
				4277 * +65504, -65504		
				4278	VRR_A	VCLFNL,2,0,00,00,N
00005200				4279+	DS	0FD
00005200		00005200		4280+	USING *	R5 base for test data and test routine
00005200	00005290			4281+T77	DC	A(X77) address of test routine
00005204	004D			4282+	DC	H'77' test number
00005206	00			4283+	DC	X'00'
00005207	D5			4284+	DC	CL1'N' Y = skip cross check
00005208	02			4285+	DC	HL1'2' m3
00005209	00			4286+	DC	HL1'0' m4
0000520A	00			4287+FLG77	DC	X'00' expected FPC flags
0000520B	00			4288+VXC77	DC	X'00' expected VXC
0000520C	000052C4			4289+V2_77	DC	A(RE77+16) address of v2: 16-byte packed decimal
00005210	E5C3D3C6 D5D34040			4290+	DC	CL8'VCLFNL' instruction name
00005218	00000010			4291+	DC	A(16) result length
0000521C	000052B4			4292+	DC	A(RE77) address of expected resul
00005220	00000000 00000000			4293+	DS	FD gap
00005228	00000000 00000000			4294+V1077	DS	XL16 V1 output
00005230	00000000 00000000					
00005238	00000000 00000000			4295+	DS	FD gap
00005240	00000000			4296+FPC_R_77	DS	F FPC after instruction
00005248	00000000 00000000			4297+	DS	FD gap
00005250	40404040 40404040			4298+	DC	CL8' ' was cross check skipped?
00005258	00000000 00000000			4299+	DS	FD gap
00005260	00000000 00000000			4300+XC077	DS	XL16 Cross check Output
00005268	00000000 00000000					
00005270	00000000 00000000			4301+	DS	FD gap
00005278	00000000			4302+FPC_XC_77	DS	F FPC after cross check
00005280	00000000 00000000			4303+	DS	FD gap
00005288	00000000			4304+	DS	F debug area
0000528C	00000000			4305+	DS	F
				4306+*		
00005290				4307+X77	DS	0F
00005290	B29D 85A4		000007A4	4308+	LFPC	FPCINIT initialize FPC
				4309+*	LFPC	=XL4'08000000' set trap exception
00005294	E320 500C 0014		0000520C	4310+	LGF	R2,V2_77 get v2
0000529A	E762 0000 0806		00000000	4311+	VL	V22,0(R2)
000052A0	E666 0000 2C5E			4312+	VCLFNL	V22,V22,2,0 test instruction (dest is source)
000052A6	B29C 5040		00005240	4313+	STFPC	FPC_R_77 save FPC
000052AA	E760 5028 080E		00005228	4314+	VST	V22,V1077 save instruction result
000052B0	07FB			4315+	BR	R11 return
000052B4				4316+RE77	DS	0F expected 16 byte result
000052B4				4317+	DROP	R5
000052B4	47800000 00000000			4318	DC	XL16'4780000000000000C780000000000000'
000052BC	C7800000 00000000					
000052C4	00000000 00000000			4319	DC	XL16'00000000000000005E000000DE000000'
000052CC	5E000000 DE000000					
				4320		
				4321 * +Normal, -Normal (2.5)		
				4322	VRR_A	VCLFNL,2,0,00,00,N

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
000052D8				4323+	DS	0FD
000052D8		000052D8		4324+	USING	*,R5
000052D8	00005368			4325+T78	DC	A(X78)
000052DC	004E			4326+	DC	H'78'
000052DE	00			4327+	DC	X'00'
000052DF	D5			4328+	DC	CL1'N'
000052E0	02			4329+	DC	HL1'2'
000052E1	00			4330+	DC	HL1'0'
000052E2	00			4331+FLG78	DC	X'00'
000052E3	00			4332+VXC78	DC	X'00'
000052E4	0000539C			4333+V2_78	DC	A(RE78+16)
000052E8	E5C3D3C6 D5D34040			4334+	DC	CL8'VCLFNL'
000052F0	00000010			4335+	DC	A(16)
000052F4	0000538C			4336+	DC	A(RE78)
000052F8	00000000 00000000			4337+	DS	FD
00005300	00000000 00000000			4338+V1078	DS	XL16
00005308	00000000 00000000					
00005310	00000000 00000000			4339+	DS	FD
00005318	00000000			4340+FPC_R_78	DS	F
00005320	00000000 00000000			4341+	DS	FD
00005328	40404040 40404040			4342+	DC	CL8' '
00005330	00000000 00000000			4343+	DS	FD
00005338	00000000 00000000			4344+XC078	DS	XL16
00005340	00000000 00000000					
00005348	00000000 00000000			4345+	DS	FD
00005350	00000000			4346+FPC_XC_78	DS	F
00005358	00000000 00000000			4347+	DS	FD
00005360	00000000			4348+	DS	F
00005364	00000000			4349+	DS	F
				4350+*		
00005368				4351+X78	DS	0F
00005368	B29D 85A4		000007A4	4352+	LFPC	FPCINIT
				4353+*	LFPC	=XL4'08000000'
0000536C	E320 500C 0014		000052E4	4354+	LGF	R2,V2_78
00005372	E762 0000 0806		00000000	4355+	VL	V22,0(R2)
00005378	E666 0000 2C5E			4356+	VCLFNL	V22,V22,2,0
0000537E	B29C 5040		00005318	4357+	STFPC	FPC_R_78
00005382	E760 5028 080E		00005300	4358+	VST	V22,V1078
00005388	07FB			4359+	BR	R11
0000538C				4360+RE78	DS	0F
0000538C				4361+	DROP	R5
0000538C	40200000 00000000			4362	DC	XL16'4020000000000000C020000000000000'
00005394	C0200000 00000000					
0000539C	00000000 00000000			4363	DC	XL16'000000000000000040800000C0800000'
000053A4	40800000 C0800000					
				4364		
				4365	*	+0.00006097, -0.00006097 (about)
				4366	VRR_A	VCLFNL,2,0,00,00,N
000053B0				4367+	DS	0FD
000053B0		000053B0		4368+	USING	*,R5
000053B0	00005440			4369+T79	DC	A(X79)
000053B4	004F			4370+	DC	H'79'
000053B6	00			4371+	DC	X'00'
000053B7	D5			4372+	DC	CL1'N'
000053B8	02			4373+	DC	HL1'2'
000053B9	00			4374+	DC	HL1'0'

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000053BA	00			4375+FLG79	DC	X'00'	expected FPC flags
000053BB	00			4376+VXC79	DC	X'00'	expected VXC
000053BC	00005474			4377+V2_79	DC	A(RE79+16)	address of v2: 16-byte packed decimal
000053C0	E5C3D3C6 D5D34040			4378+	DC	CL8'VCLFNL'	instruction name
000053C8	00000010			4379+	DC	A(16)	result length
000053CC	00005464			4380+	DC	A(RE79)	address of expected resul
000053D0	00000000 00000000			4381+	DS	FD	gap
000053D8	00000000 00000000			4382+V1079	DS	XL16	V1 output
000053E0	00000000 00000000						
000053E8	00000000 00000000			4383+	DS	FD	gap
000053F0	00000000			4384+FPC_R_79	DS	F	FPC after instruction
000053F8	00000000 00000000			4385+	DS	FD	gap
00005400	40404040 40404040			4386+	DC	CL8' '	was cross check skipped?
00005408	00000000 00000000			4387+	DS	FD	gap
00005410	00000000 00000000			4388+XC079	DS	XL16	Cross check Output
00005418	00000000 00000000						
00005420	00000000 00000000			4389+	DS	FD	gap
00005428	00000000			4390+FPC_XC_79	DS	F	FPC after cross check
00005430	00000000 00000000			4391+	DS	FD	gap
00005438	00000000			4392+	DS	F	debug area
0000543C	00000000			4393+	DS	F	
				4394+*			
00005440				4395+X79	DS	0F	
00005440	B29D 85A4		000007A4	4396+	LFPC	FPCINIT	initialize FPC
				4397+*	LFPC	=XL4'08000000'	set trap exception
00005444	E320 500C 0014		000053BC	4398+	LGF	R2,V2_79	get v2
0000544A	E762 0000 0806		00000000	4399+	VL	V22,0(R2)	
00005450	E666 0000 2C5E			4400+	VCLFNL	V22,V22,2,0	test instruction (dest is source)
00005456	B29C 5040		000053F0	4401+	STFPC	FPC_R_79	save FPC
0000545A	E760 5028 080E		000053D8	4402+	VST	V22,V1079	save instruction result
00005460	07FB			4403+	BR	R11	return
00005464				4404+RE79	DS	0F	expected 16 byte result
00005464				4405+	DROP	R5	
00005464	387FC000 00000000			4406	DC	XL16'387FC0000000000000B87FC00000000000'	
0000546C	B87FC000 00000000						
00005474	00000000 00000000			4407	DC	XL16'000000000000000021FF0000A1FF0000'	
0000547C	21FF0000 A1FF0000						
				4408			
00005484	00000000			4409	DC	F'0'	END OF TABLE
00005488	00000000			4410	DC	F'0'	
				4411 *			
				4412 *	table of pointers to individual tests		
				4413 *			
0000548C				4414 E6TESTS	DS	0F	
				4415	PTTABLE		
0000548C				4416+TTABLE	DS	0F	
0000548C	000011E0			4417+	DC	A(T1)	TEST &CUR
00005490	000012B8			4418+	DC	A(T2)	TEST &CUR
00005494	00001390			4419+	DC	A(T3)	TEST &CUR
00005498	00001468			4420+	DC	A(T4)	TEST &CUR
0000549C	00001540			4421+	DC	A(T5)	TEST &CUR
000054A0	00001618			4422+	DC	A(T6)	TEST &CUR
000054A4	000016F0			4423+	DC	A(T7)	TEST &CUR
000054A8	000017C8			4424+	DC	A(T8)	TEST &CUR
000054AC	000018A0			4425+	DC	A(T9)	TEST &CUR
000054B0	00001978			4426+	DC	A(T10)	TEST &CUR

LOC	OBJECT CODE	ADDR1	ADDR2	STMT				
000054B4	00001A50			4427+	DC	A(T11)	TEST	&CUR
000054B8	00001B28			4428+	DC	A(T12)	TEST	&CUR
000054BC	00001C00			4429+	DC	A(T13)	TEST	&CUR
000054C0	00001CD8			4430+	DC	A(T14)	TEST	&CUR
000054C4	00001DB0			4431+	DC	A(T15)	TEST	&CUR
000054C8	00001E88			4432+	DC	A(T16)	TEST	&CUR
000054CC	00001F60			4433+	DC	A(T17)	TEST	&CUR
000054D0	00002038			4434+	DC	A(T18)	TEST	&CUR
000054D4	00002110			4435+	DC	A(T19)	TEST	&CUR
000054D8	000021E8			4436+	DC	A(T20)	TEST	&CUR
000054DC	000022C0			4437+	DC	A(T21)	TEST	&CUR
000054E0	00002398			4438+	DC	A(T22)	TEST	&CUR
000054E4	00002470			4439+	DC	A(T23)	TEST	&CUR
000054E8	00002548			4440+	DC	A(T24)	TEST	&CUR
000054EC	00002620			4441+	DC	A(T25)	TEST	&CUR
000054F0	000026F8			4442+	DC	A(T26)	TEST	&CUR
000054F4	000027D0			4443+	DC	A(T27)	TEST	&CUR
000054F8	000028A8			4444+	DC	A(T28)	TEST	&CUR
000054FC	00002980			4445+	DC	A(T29)	TEST	&CUR
00005500	00002A58			4446+	DC	A(T30)	TEST	&CUR
00005504	00002B30			4447+	DC	A(T31)	TEST	&CUR
00005508	00002C08			4448+	DC	A(T32)	TEST	&CUR
0000550C	00002CE0			4449+	DC	A(T33)	TEST	&CUR
00005510	00002DB8			4450+	DC	A(T34)	TEST	&CUR
00005514	00002E90			4451+	DC	A(T35)	TEST	&CUR
00005518	00002F68			4452+	DC	A(T36)	TEST	&CUR
0000551C	00003040			4453+	DC	A(T37)	TEST	&CUR
00005520	00003118			4454+	DC	A(T38)	TEST	&CUR
00005524	000031F0			4455+	DC	A(T39)	TEST	&CUR
00005528	000032C8			4456+	DC	A(T40)	TEST	&CUR
0000552C	000033A0			4457+	DC	A(T41)	TEST	&CUR
00005530	00003478			4458+	DC	A(T42)	TEST	&CUR
00005534	00003550			4459+	DC	A(T43)	TEST	&CUR
00005538	00003628			4460+	DC	A(T44)	TEST	&CUR
0000553C	00003700			4461+	DC	A(T45)	TEST	&CUR
00005540	000037D8			4462+	DC	A(T46)	TEST	&CUR
00005544	000038B0			4463+	DC	A(T47)	TEST	&CUR
00005548	00003988			4464+	DC	A(T48)	TEST	&CUR
0000554C	00003A60			4465+	DC	A(T49)	TEST	&CUR
00005550	00003B38			4466+	DC	A(T50)	TEST	&CUR
00005554	00003C10			4467+	DC	A(T51)	TEST	&CUR
00005558	00003CE8			4468+	DC	A(T52)	TEST	&CUR
0000555C	00003DC0			4469+	DC	A(T53)	TEST	&CUR
00005560	00003E98			4470+	DC	A(T54)	TEST	&CUR
00005564	00003F70			4471+	DC	A(T55)	TEST	&CUR
00005568	00004048			4472+	DC	A(T56)	TEST	&CUR
0000556C	00004120			4473+	DC	A(T57)	TEST	&CUR
00005570	000041F8			4474+	DC	A(T58)	TEST	&CUR
00005574	000042D0			4475+	DC	A(T59)	TEST	&CUR
00005578	000043A8			4476+	DC	A(T60)	TEST	&CUR
0000557C	00004480			4477+	DC	A(T61)	TEST	&CUR
00005580	00004558			4478+	DC	A(T62)	TEST	&CUR
00005584	00004630			4479+	DC	A(T63)	TEST	&CUR
00005588	00004708			4480+	DC	A(T64)	TEST	&CUR
0000558C	000047E0			4481+	DC	A(T65)	TEST	&CUR
00005590	000048B8			4482+	DC	A(T66)	TEST	&CUR

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
				4503	*****		
				4504	* Register equates		
				4505	*****		
		00000000	00000001	4507	R0	EQU	0
		00000001	00000001	4508	R1	EQU	1
		00000002	00000001	4509	R2	EQU	2
		00000003	00000001	4510	R3	EQU	3
		00000004	00000001	4511	R4	EQU	4
		00000005	00000001	4512	R5	EQU	5
		00000006	00000001	4513	R6	EQU	6
		00000007	00000001	4514	R7	EQU	7
		00000008	00000001	4515	R8	EQU	8
		00000009	00000001	4516	R9	EQU	9
		0000000A	00000001	4517	R10	EQU	10
		0000000B	00000001	4518	R11	EQU	11
		0000000C	00000001	4519	R12	EQU	12
		0000000D	00000001	4520	R13	EQU	13
		0000000E	00000001	4521	R14	EQU	14
		0000000F	00000001	4522	R15	EQU	15
				4524	*****		
				4525	* Register equates		
				4526	*****		
		00000000	00000001	4528	FPR0	EQU	0
		00000001	00000001	4529	FPR1	EQU	1
		00000002	00000001	4530	FPR2	EQU	2
		00000003	00000001	4531	FPR3	EQU	3
		00000004	00000001	4532	FPR4	EQU	4
		00000005	00000001	4533	FPR5	EQU	5
		00000006	00000001	4534	FPR6	EQU	6
		00000007	00000001	4535	FPR7	EQU	7
		00000008	00000001	4536	FPR8	EQU	8
		00000009	00000001	4537	FPR9	EQU	9
		0000000A	00000001	4538	FPR10	EQU	10
		0000000B	00000001	4539	FPR11	EQU	11
		0000000C	00000001	4540	FPR12	EQU	12
		0000000D	00000001	4541	FPR13	EQU	13
		0000000E	00000001	4542	FPR14	EQU	14
		0000000F	00000001	4543	FPR15	EQU	15
				4545	*****		
				4546	* Register equates		
				4547	*****		
		00000000	00000001	4549	V0	EQU	0
		00000001	00000001	4550	V1	EQU	1
		00000002	00000001	4551	V2	EQU	2
		00000003	00000001	4552	V3	EQU	3
		00000004	00000001	4553	V4	EQU	4

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES				
FPCPNUM	C	00001015	3	670	430				
FPC_R	F	00000040	4	773	248	253	283	442	478
FPC_R_1	F	00001220	4	922	939				
FPC_R_10	F	000019B8	4	1318	1335				
FPC_R_11	F	00001A90	4	1362	1379				
FPC_R_12	F	00001B68	4	1406	1423				
FPC_R_13	F	00001C40	4	1450	1467				
FPC_R_14	F	00001D18	4	1495	1512				
FPC_R_15	F	00001DF0	4	1539	1556				
FPC_R_16	F	00001EC8	4	1583	1600				
FPC_R_17	F	00001FA0	4	1627	1644				
FPC_R_18	F	00002078	4	1671	1688				
FPC_R_19	F	00002150	4	1715	1732				
FPC_R_2	F	000012F8	4	966	983				
FPC_R_20	F	00002228	4	1759	1776				
FPC_R_21	F	00002300	4	1803	1820				
FPC_R_22	F	000023D8	4	1848	1865				
FPC_R_23	F	000024B0	4	1892	1909				
FPC_R_24	F	00002588	4	1942	1959				
FPC_R_25	F	00002660	4	1986	2003				
FPC_R_26	F	00002738	4	2030	2047				
FPC_R_27	F	00002810	4	2074	2091				
FPC_R_28	F	000028E8	4	2118	2135				
FPC_R_29	F	000029C0	4	2162	2179				
FPC_R_3	F	000013D0	4	1009	1026				
FPC_R_30	F	00002A98	4	2207	2224				
FPC_R_31	F	00002B70	4	2251	2268				
FPC_R_32	F	00002C48	4	2295	2312				
FPC_R_33	F	00002D20	4	2339	2356				
FPC_R_34	F	00002DF8	4	2383	2400				
FPC_R_35	F	00002ED0	4	2427	2444				
FPC_R_36	F	00002FA8	4	2472	2489				
FPC_R_37	F	00003080	4	2517	2534				
FPC_R_38	F	00003158	4	2561	2578				
FPC_R_39	F	00003230	4	2605	2622				
FPC_R_4	F	000014A8	4	1053	1070				
FPC_R_40	F	00003308	4	2649	2666				
FPC_R_41	F	000033E0	4	2699	2716				
FPC_R_42	F	000034B8	4	2743	2760				
FPC_R_43	F	00003590	4	2787	2804				
FPC_R_44	F	00003668	4	2831	2848				
FPC_R_45	F	00003740	4	2875	2892				
FPC_R_46	F	00003818	4	2919	2936				
FPC_R_47	F	000038F0	4	2963	2980				
FPC_R_48	F	000039C8	4	3008	3025				
FPC_R_49	F	00003AA0	4	3052	3069				
FPC_R_5	F	00001580	4	1097	1114				
FPC_R_50	F	00003B78	4	3096	3113				
FPC_R_51	F	00003C50	4	3140	3157				
FPC_R_52	F	00003D28	4	3184	3201				
FPC_R_53	F	00003E00	4	3228	3245				
FPC_R_54	F	00003ED8	4	3272	3289				
FPC_R_55	F	00003FB0	4	3317	3334				
FPC_R_56	F	00004088	4	3362	3379				
FPC_R_57	F	00004160	4	3406	3423				
FPC_R_58	F	00004238	4	3450	3467				

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES							
FPC_R_59	F	00004310	4	3494	3511							
FPC_R_6	F	00001658	4	1141	1158							
FPC_R_60	F	000043E8	4	3539	3556							
FPC_R_61	F	000044C0	4	3583	3600							
FPC_R_62	F	00004598	4	3627	3644							
FPC_R_63	F	00004670	4	3677	3694							
FPC_R_64	F	00004748	4	3721	3738							
FPC_R_65	F	00004820	4	3765	3782							
FPC_R_66	F	000048F8	4	3809	3826							
FPC_R_67	F	000049D0	4	3853	3870							
FPC_R_68	F	00004AA8	4	3897	3914							
FPC_R_69	F	00004B80	4	3942	3959							
FPC_R_7	F	00001730	4	1185	1202							
FPC_R_70	F	00004C58	4	3986	4003							
FPC_R_71	F	00004D30	4	4030	4047							
FPC_R_72	F	00004E08	4	4074	4091							
FPC_R_73	F	00004EE0	4	4118	4135							
FPC_R_74	F	00004FB8	4	4162	4179							
FPC_R_75	F	00005090	4	4207	4224							
FPC_R_76	F	00005168	4	4252	4269							
FPC_R_77	F	00005240	4	4296	4313							
FPC_R_78	F	00005318	4	4340	4357							
FPC_R_79	F	000053F0	4	4384	4401							
FPC_R_8	F	00001808	4	1230	1247							
FPC_R_9	F	000018E0	4	1274	1291							
FPC_XC	F	00000078	4	779	311	314	329	332	347	350	365	368
FPC_XC_1	F	00001258	4	928								
FPC_XC_10	F	000019F0	4	1324								
FPC_XC_11	F	00001AC8	4	1368								
FPC_XC_12	F	00001BA0	4	1412								
FPC_XC_13	F	00001C78	4	1456								
FPC_XC_14	F	00001D50	4	1501								
FPC_XC_15	F	00001E28	4	1545								
FPC_XC_16	F	00001F00	4	1589								
FPC_XC_17	F	00001FD8	4	1633								
FPC_XC_18	F	000020B0	4	1677								
FPC_XC_19	F	00002188	4	1721								
FPC_XC_2	F	00001330	4	972								
FPC_XC_20	F	00002260	4	1765								
FPC_XC_21	F	00002338	4	1809								
FPC_XC_22	F	00002410	4	1854								
FPC_XC_23	F	000024E8	4	1898								
FPC_XC_24	F	000025C0	4	1948								
FPC_XC_25	F	00002698	4	1992								
FPC_XC_26	F	00002770	4	2036								
FPC_XC_27	F	00002848	4	2080								
FPC_XC_28	F	00002920	4	2124								
FPC_XC_29	F	000029F8	4	2168								
FPC_XC_3	F	00001408	4	1015								
FPC_XC_30	F	00002AD0	4	2213								
FPC_XC_31	F	00002BA8	4	2257								
FPC_XC_32	F	00002C80	4	2301								
FPC_XC_33	F	00002D58	4	2345								
FPC_XC_34	F	00002E30	4	2389								
FPC_XC_35	F	00002F08	4	2433								
FPC_XC_36	F	00002FE0	4	2478								

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
FPC_XC_37	F	000030B8	4	2523	
FPC_XC_38	F	00003190	4	2567	
FPC_XC_39	F	00003268	4	2611	
FPC_XC_4	F	000014E0	4	1059	
FPC_XC_40	F	00003340	4	2655	
FPC_XC_41	F	00003418	4	2705	
FPC_XC_42	F	000034F0	4	2749	
FPC_XC_43	F	000035C8	4	2793	
FPC_XC_44	F	000036A0	4	2837	
FPC_XC_45	F	00003778	4	2881	
FPC_XC_46	F	00003850	4	2925	
FPC_XC_47	F	00003928	4	2969	
FPC_XC_48	F	00003A00	4	3014	
FPC_XC_49	F	00003AD8	4	3058	
FPC_XC_5	F	000015B8	4	1103	
FPC_XC_50	F	00003BB0	4	3102	
FPC_XC_51	F	00003C88	4	3146	
FPC_XC_52	F	00003D60	4	3190	
FPC_XC_53	F	00003E38	4	3234	
FPC_XC_54	F	00003F10	4	3278	
FPC_XC_55	F	00003FE8	4	3323	
FPC_XC_56	F	000040C0	4	3368	
FPC_XC_57	F	00004198	4	3412	
FPC_XC_58	F	00004270	4	3456	
FPC_XC_59	F	00004348	4	3500	
FPC_XC_6	F	00001690	4	1147	
FPC_XC_60	F	00004420	4	3545	
FPC_XC_61	F	000044F8	4	3589	
FPC_XC_62	F	000045D0	4	3633	
FPC_XC_63	F	000046A8	4	3683	
FPC_XC_64	F	00004780	4	3727	
FPC_XC_65	F	00004858	4	3771	
FPC_XC_66	F	00004930	4	3815	
FPC_XC_67	F	00004A08	4	3859	
FPC_XC_68	F	00004AE0	4	3903	
FPC_XC_69	F	00004BB8	4	3948	
FPC_XC_7	F	00001768	4	1191	
FPC_XC_70	F	00004C90	4	3992	
FPC_XC_71	F	00004D68	4	4036	
FPC_XC_72	F	00004E40	4	4080	
FPC_XC_73	F	00004F18	4	4124	
FPC_XC_74	F	00004FF0	4	4168	
FPC_XC_75	F	000050C8	4	4213	
FPC_XC_76	F	000051A0	4	4258	
FPC_XC_77	F	00005278	4	4302	
FPC_XC_78	F	00005350	4	4346	
FPC_XC_79	F	00005428	4	4390	
FPC_XC_8	F	00001840	4	1236	
FPC_XC_9	F	00001918	4	1280	
FPR0	U	00000000	1	4528	
FPR1	U	00000001	1	4529	
FPR10	U	0000000A	1	4538	
FPR11	U	0000000B	1	4539	
FPR12	U	0000000C	1	4540	
FPR13	U	0000000D	1	4541	
FPR14	U	0000000E	1	4542	

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES														
FPR15	U	0000000F	1	4543															
FPR2	U	00000002	1	4530															
FPR3	U	00000003	1	4531															
FPR4	U	00000004	1	4532															
FPR5	U	00000005	1	4533															
FPR6	U	00000006	1	4534															
FPR7	U	00000007	1	4535															
FPR8	U	00000008	1	4536															
FPR9	U	00000009	1	4537															
IMAGE	1	00000000	21976	0															
K	U	00000400	1	642	643	644	645												
K64	U	00010000	1	644															
M3	U	00000008	1	762	388	505													
M4	U	00000009	1	763	395	512													
MB	U	00100000	1	645															
MSG	I	000006B0	4	571	221	554													
MSGCMD	C	000006FE	9	601	584	585													
MSGMSG	C	00000707	100	602	578	599	576												
MSGMVC	I	000006F8	6	599	582														
MSGOK	I	000006C6	2	580	577														
MSGRET	I	000006E6	4	595	588	591													
MSGSAVE	F	000006EC	4	598	574	595													
NEXTE6	U	000002EC	1	231	273	530													
OPNAME	C	00000010	8	767	289	292	295	298	385	432	468	502							
PAGE	U	00001000	1	643															
PRT3	C	0000117A	18	741	381	382	383	390	391	392	397	398	399	428	429	430	437		
					438	439	444	445	446	464	465	466	473	474	475	480	481		
					482	498	499	500	507	508	509	514	515	516					
PRTLNE	C	000010CE	13	706	716	519													
PRTLNG	U	00000048	1	716	518														
PRTM3	C	00001107	2	711	509														
PRTM4	C	00001113	2	714	516														
PRTNAME	C	000010F6	8	709	502														
PRTNUM	C	000010DB	3	707	500														
R0	U	00000000	1	4507	135	187	190	210	212	213	214	219	238	239	402	407	408		
					423	424	448	459	460	484	518	526	527	553	555	571	574		
					576	578	580	595											
R1	U	00000001	1	4508	220	267	268	317	318	335	336	353	354	371	372	403	449		
					485	519	536	537	585	599									
R10	U	0000000A	1	4517	175	184	185												
R11	U	0000000B	1	4518	241	242	941	985	1028	1072	1116	1160	1204	1249	1293	1337	1381		
					1425	1469	1514	1558	1602	1646	1690	1734	1778	1822	1867	1911	1961		
					2005	2049	2093	2137	2181	2226	2270	2314	2358	2402	2446	2491	2536		
					2580	2624	2668	2718	2762	2806	2850	2894	2938	2982	3027	3071	3115		
					3159	3203	3247	3291	3336	3381	3425	3469	3513	3558	3602	3646	3696		
					3740	3784	3828	3872	3916	3961	4005	4049	4093	4137	4181	4226	4271		
					4315	4359	4403												
R12	U	0000000C	1	4519	229	232	272	529											
R13	U	0000000D	1	4520															
R14	U	0000000E	1	4521															
R15	U	0000000F	1	4522	244	282	284	302	315	320	333	338	351	356	369	374	401		
					404	405	409	450	486	520	548	558	559						
R2	U	00000002	1	4509	221	379	380	387	388	389	394	395	396	426	427	434	435		
					436	441	442	443	462	463	470	471	472	477	478	479	496		
					497	504	505	506	511	512	513	553	554	555	572	574	580		
					581	582	584	590	595	596	936	937	980	981	1023	1024	1067		

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
RE32	F	00002CBC	4	2315	2288 2291
RE33	F	00002D94	4	2359	2332 2335
RE34	F	00002E6C	4	2403	2376 2379
RE35	F	00002F44	4	2447	2420 2423
RE36	F	0000301C	4	2492	2465 2468
RE37	F	000030F4	4	2537	2510 2513
RE38	F	000031CC	4	2581	2554 2557
RE39	F	000032A4	4	2625	2598 2601
RE4	F	0000151C	4	1073	1046 1049
RE40	F	0000337C	4	2669	2642 2645
RE41	F	00003454	4	2719	2692 2695
RE42	F	0000352C	4	2763	2736 2739
RE43	F	00003604	4	2807	2780 2783
RE44	F	000036DC	4	2851	2824 2827
RE45	F	000037B4	4	2895	2868 2871
RE46	F	0000388C	4	2939	2912 2915
RE47	F	00003964	4	2983	2956 2959
RE48	F	00003A3C	4	3028	3001 3004
RE49	F	00003B14	4	3072	3045 3048
RE5	F	000015F4	4	1117	1090 1093
RE50	F	00003BEC	4	3116	3089 3092
RE51	F	00003CC4	4	3160	3133 3136
RE52	F	00003D9C	4	3204	3177 3180
RE53	F	00003E74	4	3248	3221 3224
RE54	F	00003F4C	4	3292	3265 3268
RE55	F	00004024	4	3337	3310 3313
RE56	F	000040FC	4	3382	3355 3358
RE57	F	000041D4	4	3426	3399 3402
RE58	F	000042AC	4	3470	3443 3446
RE59	F	00004384	4	3514	3487 3490
RE6	F	000016CC	4	1161	1134 1137
RE60	F	0000445C	4	3559	3532 3535
RE61	F	00004534	4	3603	3576 3579
RE62	F	0000460C	4	3647	3620 3623
RE63	F	000046E4	4	3697	3670 3673
RE64	F	000047BC	4	3741	3714 3717
RE65	F	00004894	4	3785	3758 3761
RE66	F	0000496C	4	3829	3802 3805
RE67	F	00004A44	4	3873	3846 3849
RE68	F	00004B1C	4	3917	3890 3893
RE69	F	00004BF4	4	3962	3935 3938
RE7	F	000017A4	4	1205	1178 1181
RE70	F	00004CCC	4	4006	3979 3982
RE71	F	00004DA4	4	4050	4023 4026
RE72	F	00004E7C	4	4094	4067 4070
RE73	F	00004F54	4	4138	4111 4114
RE74	F	0000502C	4	4182	4155 4158
RE75	F	00005104	4	4227	4200 4203
RE76	F	000051DC	4	4272	4245 4248
RE77	F	000052B4	4	4316	4289 4292
RE78	F	0000538C	4	4360	4333 4336
RE79	F	00005464	4	4404	4377 4380
RE8	F	0000187C	4	1250	1223 1226
RE9	F	00001954	4	1294	1267 1270
READDR	A	0000001C	4	769	267
REG2LOW	U	000000DD	1	649	

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES				
REG2PATT	U	AABBCCDD	1	648					
RELEN	A	00000018	4	768					
RPTDWSAV	D	000006A0	8	564	553	555			
RPTERROR	I	0000067A	4	548	404	450	486	520	
RPTSAVE	F	00000698	4	561	548	558			
RPTSVR5	F	0000069C	4	562	549	557			
SKIPXC	C	00000050	8	775	280	288	301		
SKL0001	U	0000005F	1	203	219				
SKT0001	C	0000022A	26	200	203	220			
SVOLDPSW	U	00000140	0	137					
T1	A	000011E0	4	907	4417				
T10	A	00001978	4	1303	4426				
T11	A	00001A50	4	1347	4427				
T12	A	00001B28	4	1391	4428				
T13	A	00001C00	4	1435	4429				
T14	A	00001CD8	4	1480	4430				
T15	A	00001DB0	4	1524	4431				
T16	A	00001E88	4	1568	4432				
T17	A	00001F60	4	1612	4433				
T18	A	00002038	4	1656	4434				
T19	A	00002110	4	1700	4435				
T2	A	000012B8	4	951	4418				
T20	A	000021E8	4	1744	4436				
T21	A	000022C0	4	1788	4437				
T22	A	00002398	4	1833	4438				
T23	A	00002470	4	1877	4439				
T24	A	00002548	4	1927	4440				
T25	A	00002620	4	1971	4441				
T26	A	000026F8	4	2015	4442				
T27	A	000027D0	4	2059	4443				
T28	A	000028A8	4	2103	4444				
T29	A	00002980	4	2147	4445				
T3	A	00001390	4	994	4419				
T30	A	00002A58	4	2192	4446				
T31	A	00002B30	4	2236	4447				
T32	A	00002C08	4	2280	4448				
T33	A	00002CE0	4	2324	4449				
T34	A	00002DB8	4	2368	4450				
T35	A	00002E90	4	2412	4451				
T36	A	00002F68	4	2457	4452				
T37	A	00003040	4	2502	4453				
T38	A	00003118	4	2546	4454				
T39	A	000031F0	4	2590	4455				
T4	A	00001468	4	1038	4420				
T40	A	000032C8	4	2634	4456				
T41	A	000033A0	4	2684	4457				
T42	A	00003478	4	2728	4458				
T43	A	00003550	4	2772	4459				
T44	A	00003628	4	2816	4460				
T45	A	00003700	4	2860	4461				
T46	A	000037D8	4	2904	4462				
T47	A	000038B0	4	2948	4463				
T48	A	00003988	4	2993	4464				
T49	A	00003A60	4	3037	4465				
T5	A	00001540	4	1082	4421				
T50	A	00003B38	4	3081	4466				

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
V1014	X	00001D00	16	1493	1513
V1015	X	00001DD8	16	1537	1557
V1016	X	00001EB0	16	1581	1601
V1017	X	00001F88	16	1625	1645
V1018	X	00002060	16	1669	1689
V1019	X	00002138	16	1713	1733
V102	X	000012E0	16	964	984
V1020	X	00002210	16	1757	1777
V1021	X	000022E8	16	1801	1821
V1022	X	000023C0	16	1846	1866
V1023	X	00002498	16	1890	1910
V1024	X	00002570	16	1940	1960
V1025	X	00002648	16	1984	2004
V1026	X	00002720	16	2028	2048
V1027	X	000027F8	16	2072	2092
V1028	X	000028D0	16	2116	2136
V1029	X	000029A8	16	2160	2180
V103	X	000013B8	16	1007	1027
V1030	X	00002A80	16	2205	2225
V1031	X	00002B58	16	2249	2269
V1032	X	00002C30	16	2293	2313
V1033	X	00002D08	16	2337	2357
V1034	X	00002DE0	16	2381	2401
V1035	X	00002EB8	16	2425	2445
V1036	X	00002F90	16	2470	2490
V1037	X	00003068	16	2515	2535
V1038	X	00003140	16	2559	2579
V1039	X	00003218	16	2603	2623
V104	X	00001490	16	1051	1071
V1040	X	000032F0	16	2647	2667
V1041	X	000033C8	16	2697	2717
V1042	X	000034A0	16	2741	2761
V1043	X	00003578	16	2785	2805
V1044	X	00003650	16	2829	2849
V1045	X	00003728	16	2873	2893
V1046	X	00003800	16	2917	2937
V1047	X	000038D8	16	2961	2981
V1048	X	000039B0	16	3006	3026
V1049	X	00003A88	16	3050	3070
V105	X	00001568	16	1095	1115
V1050	X	00003B60	16	3094	3114
V1051	X	00003C38	16	3138	3158
V1052	X	00003D10	16	3182	3202
V1053	X	00003DE8	16	3226	3246
V1054	X	00003EC0	16	3270	3290
V1055	X	00003F98	16	3315	3335
V1056	X	00004070	16	3360	3380
V1057	X	00004148	16	3404	3424
V1058	X	00004220	16	3448	3468
V1059	X	000042F8	16	3492	3512
V106	X	00001640	16	1139	1159
V1060	X	000043D0	16	3537	3557
V1061	X	000044A8	16	3581	3601
V1062	X	00004580	16	3625	3645
V1063	X	00004658	16	3675	3695
V1064	X	00004730	16	3719	3739

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
V2_16	A	00001E94	4	1576	1597
V2_17	A	00001F6C	4	1620	1641
V2_18	A	00002044	4	1664	1685
V2_19	A	0000211C	4	1708	1729
V2_2	A	000012C4	4	959	980
V2_20	A	000021F4	4	1752	1773
V2_21	A	000022CC	4	1796	1817
V2_22	A	000023A4	4	1841	1862
V2_23	A	0000247C	4	1885	1906
V2_24	A	00002554	4	1935	1956
V2_25	A	0000262C	4	1979	2000
V2_26	A	00002704	4	2023	2044
V2_27	A	000027DC	4	2067	2088
V2_28	A	000028B4	4	2111	2132
V2_29	A	0000298C	4	2155	2176
V2_3	A	0000139C	4	1002	1023
V2_30	A	00002A64	4	2200	2221
V2_31	A	00002B3C	4	2244	2265
V2_32	A	00002C14	4	2288	2309
V2_33	A	00002CEC	4	2332	2353
V2_34	A	00002DC4	4	2376	2397
V2_35	A	00002E9C	4	2420	2441
V2_36	A	00002F74	4	2465	2486
V2_37	A	0000304C	4	2510	2531
V2_38	A	00003124	4	2554	2575
V2_39	A	000031FC	4	2598	2619
V2_4	A	00001474	4	1046	1067
V2_40	A	000032D4	4	2642	2663
V2_41	A	000033AC	4	2692	2713
V2_42	A	00003484	4	2736	2757
V2_43	A	0000355C	4	2780	2801
V2_44	A	00003634	4	2824	2845
V2_45	A	0000370C	4	2868	2889
V2_46	A	000037E4	4	2912	2933
V2_47	A	000038BC	4	2956	2977
V2_48	A	00003994	4	3001	3022
V2_49	A	00003A6C	4	3045	3066
V2_5	A	0000154C	4	1090	1111
V2_50	A	00003B44	4	3089	3110
V2_51	A	00003C1C	4	3133	3154
V2_52	A	00003CF4	4	3177	3198
V2_53	A	00003DCC	4	3221	3242
V2_54	A	00003EA4	4	3265	3286
V2_55	A	00003F7C	4	3310	3331
V2_56	A	00004054	4	3355	3376
V2_57	A	0000412C	4	3399	3420
V2_58	A	00004204	4	3443	3464
V2_59	A	000042DC	4	3487	3508
V2_6	A	00001624	4	1134	1155
V2_60	A	000043B4	4	3532	3553
V2_61	A	0000448C	4	3576	3597
V2_62	A	00004564	4	3620	3641
V2_63	A	0000463C	4	3670	3691
V2_64	A	00004714	4	3714	3735
V2_65	A	000047EC	4	3758	3779
V2_66	A	000048C4	4	3802	3823

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES	
V2_67	A	0000499C	4	3846	3867	
V2_68	A	00004A74	4	3890	3911	
V2_69	A	00004B4C	4	3935	3956	
V2_7	A	000016FC	4	1178	1199	
V2_70	A	00004C24	4	3979	4000	
V2_71	A	00004CFC	4	4023	4044	
V2_72	A	00004DD4	4	4067	4088	
V2_73	A	00004EAC	4	4111	4132	
V2_74	A	00004F84	4	4155	4176	
V2_75	A	0000505C	4	4200	4221	
V2_76	A	00005134	4	4245	4266	
V2_77	A	0000520C	4	4289	4310	
V2_78	A	000052E4	4	4333	4354	
V2_79	A	000053BC	4	4377	4398	
V2_8	A	000017D4	4	1223	1244	
V2_9	A	000018AC	4	1267	1288	
V3	U	00000003	1	4552		
V30	U	0000001E	1	4579		
V31	U	0000001F	1	4580		
V4	U	00000004	1	4553		
V5	U	00000005	1	4554		
V6	U	00000006	1	4555		
V7	U	00000007	1	4556		
V8	U	00000008	1	4557		
V9	U	00000009	1	4558		
VXC	X	0000000B	1	765	253	471
VXC1	X	000011EB	1	914		
VXC10	X	00001983	1	1310		
VXC11	X	00001A5B	1	1354		
VXC12	X	00001B33	1	1398		
VXC13	X	00001C0B	1	1442		
VXC14	X	00001CE3	1	1487		
VXC15	X	00001DBB	1	1531		
VXC16	X	00001E93	1	1575		
VXC17	X	00001F6B	1	1619		
VXC18	X	00002043	1	1663		
VXC19	X	0000211B	1	1707		
VXC2	X	000012C3	1	958		
VXC20	X	000021F3	1	1751		
VXC21	X	000022CB	1	1795		
VXC22	X	000023A3	1	1840		
VXC23	X	0000247B	1	1884		
VXC24	X	00002553	1	1934		
VXC25	X	0000262B	1	1978		
VXC26	X	00002703	1	2022		
VXC27	X	000027DB	1	2066		
VXC28	X	000028B3	1	2110		
VXC29	X	0000298B	1	2154		
VXC3	X	0000139B	1	1001		
VXC30	X	00002A63	1	2199		
VXC31	X	00002B3B	1	2243		
VXC32	X	00002C13	1	2287		
VXC33	X	00002CEB	1	2331		
VXC34	X	00002DC3	1	2375		
VXC35	X	00002E9B	1	2419		
VXC36	X	00002F73	1	2464		

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES	
VXC37	X	0000304B	1	2509		
VXC38	X	00003123	1	2553		
VXC39	X	000031FB	1	2597		
VXC4	X	00001473	1	1045		
VXC40	X	000032D3	1	2641		
VXC41	X	000033AB	1	2691		
VXC42	X	00003483	1	2735		
VXC43	X	0000355B	1	2779		
VXC44	X	00003633	1	2823		
VXC45	X	0000370B	1	2867		
VXC46	X	000037E3	1	2911		
VXC47	X	000038BB	1	2955		
VXC48	X	00003993	1	3000		
VXC49	X	00003A6B	1	3044		
VXC5	X	0000154B	1	1089		
VXC50	X	00003B43	1	3088		
VXC51	X	00003C1B	1	3132		
VXC52	X	00003CF3	1	3176		
VXC53	X	00003DCB	1	3220		
VXC54	X	00003EA3	1	3264		
VXC55	X	00003F7B	1	3309		
VXC56	X	00004053	1	3354		
VXC57	X	0000412B	1	3398		
VXC58	X	00004203	1	3442		
VXC59	X	000042DB	1	3486		
VXC6	X	00001623	1	1133		
VXC60	X	000043B3	1	3531		
VXC61	X	0000448B	1	3575		
VXC62	X	00004563	1	3619		
VXC63	X	0000463B	1	3669		
VXC64	X	00004713	1	3713		
VXC65	X	000047EB	1	3757		
VXC66	X	000048C3	1	3801		
VXC67	X	0000499B	1	3845		
VXC68	X	00004A73	1	3889		
VXC69	X	00004B4B	1	3934		
VXC7	X	000016FB	1	1177		
VXC70	X	00004C23	1	3978		
VXC71	X	00004CFB	1	4022		
VXC72	X	00004DD3	1	4066		
VXC73	X	00004EAB	1	4110		
VXC74	X	00004F83	1	4154		
VXC75	X	0000505B	1	4199		
VXC76	X	00005133	1	4244		
VXC77	X	0000520B	1	4288		
VXC78	X	000052E3	1	4332		
VXC79	X	000053BB	1	4376		
VXC8	X	000017D3	1	1222		
VXC9	X	000018AB	1	1266		
VXCMSG	U	00000582	1	458	255	
VXCPEXP	C	000010B5	3	693	475	
VXCPGOT	C	000010CA	3	696	482	
VXCPLINE	C	0000106B	13	688	698	485
VXCPLNG	U	00000063	1	698	484	
VXCPNAME	C	0000109C	8	691	468	
VXCPNUM	C	00001078	3	689	466	

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES	
WK1	F	00000088	4	781		
WK2	F	0000008C	4	782		
X0001	U	000002C0	1	209	197	210
X1	F	00001270	4	933	907	
X10	F	00001A08	4	1329	1303	
X11	F	00001AE0	4	1373	1347	
X12	F	00001BB8	4	1417	1391	
X13	F	00001C90	4	1461	1435	
X14	F	00001D68	4	1506	1480	
X15	F	00001E40	4	1550	1524	
X16	F	00001F18	4	1594	1568	
X17	F	00001FF0	4	1638	1612	
X18	F	000020C8	4	1682	1656	
X19	F	000021A0	4	1726	1700	
X2	F	00001348	4	977	951	
X20	F	00002278	4	1770	1744	
X21	F	00002350	4	1814	1788	
X22	F	00002428	4	1859	1833	
X23	F	00002500	4	1903	1877	
X24	F	000025D8	4	1953	1927	
X25	F	000026B0	4	1997	1971	
X26	F	00002788	4	2041	2015	
X27	F	00002860	4	2085	2059	
X28	F	00002938	4	2129	2103	
X29	F	00002A10	4	2173	2147	
X3	F	00001420	4	1020	994	
X30	F	00002AE8	4	2218	2192	
X31	F	00002BC0	4	2262	2236	
X32	F	00002C98	4	2306	2280	
X33	F	00002D70	4	2350	2324	
X34	F	00002E48	4	2394	2368	
X35	F	00002F20	4	2438	2412	
X36	F	00002FF8	4	2483	2457	
X37	F	000030D0	4	2528	2502	
X38	F	000031A8	4	2572	2546	
X39	F	00003280	4	2616	2590	
X4	F	000014F8	4	1064	1038	
X40	F	00003358	4	2660	2634	
X41	F	00003430	4	2710	2684	
X42	F	00003508	4	2754	2728	
X43	F	000035E0	4	2798	2772	
X44	F	000036B8	4	2842	2816	
X45	F	00003790	4	2886	2860	
X46	F	00003868	4	2930	2904	
X47	F	00003940	4	2974	2948	
X48	F	00003A18	4	3019	2993	
X49	F	00003AF0	4	3063	3037	
X5	F	000015D0	4	1108	1082	
X50	F	00003BC8	4	3107	3081	
X51	F	00003CA0	4	3151	3125	
X52	F	00003D78	4	3195	3169	
X53	F	00003E50	4	3239	3213	
X54	F	00003F28	4	3283	3257	
X55	F	00004000	4	3328	3302	
X56	F	000040D8	4	3373	3347	
X57	F	000041B0	4	3417	3391	

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES			
X58	F	00004288	4	3461	3435			
X59	F	00004360	4	3505	3479			
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X60	F	00004438	4	3550	3524			
X61	F	00004510	4	3594	3568			
X62	F	000045E8	4	3638	3612			
X63	F	000046C0	4	3688	3662			
X64	F	00004798	4	3732	3706			
X65	F	00004870	4	3776	3750			
X66	F	00004948	4	3820	3794			
X67	F	00004A20	4	3864	3838			
X68	F	00004AF8	4	3908	3882			
X69	F	00004BD0	4	3953	3927			
X7	F	00001780	4	1196	1170			
X70	F	00004CA8	4	3997	3971			
X71	F	00004D80	4	4041	4015			
X72	F	00004E58	4	4085	4059			
X73	F	00004F30	4	4129	4103			
X74	F	00005008	4	4173	4147			
X75	F	000050E0	4	4218	4192			
X76	F	000051B8	4	4263	4237			
X77	F	00005290	4	4307	4281			
X78	F	00005368	4	4351	4325			
X79	F	00005440	4	4395	4369			
X8	F	00001858	4	1241	1215			
X9	F	00001930	4	1285	1259			
XC0001	U	000002E8	1	223	215			
XCFAILMSG	H	0000045C	2	378	319	337	355	373
XCHECK	U	0000033C	1	279	244			
XC01	X	00001240	16	926				
XC010	X	000019D8	16	1322				
XC011	X	00001AB0	16	1366				
XC012	X	00001B88	16	1410				
XC013	X	00001C60	16	1454				
XC014	X	00001D38	16	1499				
XC015	X	00001E10	16	1543				
XC016	X	00001EE8	16	1587				
XC017	X	00001FC0	16	1631				
XC018	X	00002098	16	1675				
XC019	X	00002170	16	1719				
XC02	X	00001318	16	970				
XC020	X	00002248	16	1763				
XC021	X	00002320	16	1807				
XC022	X	000023F8	16	1852				
XC023	X	000024D0	16	1896				
XC024	X	000025A8	16	1946				
XC025	X	00002680	16	1990				
XC026	X	00002758	16	2034				
XC027	X	00002830	16	2078				
XC028	X	00002908	16	2122				
XC029	X	000029E0	16	2166				
XC03	X	000013F0	16	1013				
XC030	X	00002AB8	16	2211				
XC031	X	00002B90	16	2255				
XC032	X	00002C68	16	2299				
XC033	X	00002D40	16	2343				

[illegible]

DESC	SYMBOL	SIZE	POS	ADDR
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Entry: 0

Image	IMAGE	21976	0000-55D7	0000-55D7
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CSECT	ZVE6TST	21976	0000-55D7	0000-55D7

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1 /devstor/sharedvfp/tests/zvector-e6-20-NNPconvert.asm
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**** NO ERRORS FOUND ****